Sur la structure exacte $\mathcal{E}_\diamond$

par

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Cette thèse s'inscrit dans le cadre de la théorie des catégories exactes, avec un accent particulier sur l'étude de structures exactes définies sur des sous-catégories de modules. Elle est présentée sous la forme d'une thèse par articles, chacun des articles étant centré sur l'étude d'une structure exacte spécifique, appelée *structure exacte diamant* et notée $\mathcal{E}_\diamond$. Cette structure est introduite afin de capturer des propriétés particulières des modules, notamment dans le cadre des représentations de carquois de type $\mathbb{A}_n$.

La thèse s'ouvre par une introduction générale qui pose le cadre théorique, expose les motivations du travail et présente les notions clés développées dans les articles. Elle met en évidence l'importance de la structure $\mathcal{E}_\diamond$ dans l'étude des représentations de carquois et des catégories de modules, en soulignant les nouvelles perspectives qu'elle offre dans l'analyse de phénomènes homologiques et combinatoires.

Le premier chapitre présente une synthèse des principaux résultats du premier article annexé, consacré à la définition et à l'analyse de la structure exacte $\mathcal{E}_\diamond$ ainsi qu'à son rôle dans l'étude des modules presque rigides pour les carquois de type $\mathbb{A}_n$. Cette structure est construite à partir d'une classe particulière de suites exactes courtes et il est démontré qu'elle satisfait les axiomes d'une catégorie exacte au sens de Quillen. Dans le cadre des algèbres de chemins de carquois de type $\mathbb{A}_n$, l'article établit des conditions garantissant la pertinence de $\mathcal{E}_\diamond$, en montrant notamment qu'elle permet d'interpréter les modules presque rigides maximaux comme des modules inclinants relativement à cette structure exacte. Cette approche fournit ainsi une reformulation unificatrice de résultats issus de modèles géométriques associés à ces algèbres.

Le deuxième chapitre synthétise les résultats du second article annexé, portant sur l'étude des sous-catégories canoniquement retrouvables de Jordan à l'aide de la structure exacte $\mathcal{E}_\diamond$. Il introduit une nouvelle construction algébrique des sous-catégories canoniquement retrouvables de Jordan (CRJ), reliant explicitement ces sous-catégories à la structure exacte considérée. Ce travail conduit à une caractérisation des sous-catégories CRJ maximales, établissant un lien précis entre structures exactes, invariants de Jordan et propriétés combinatoires des représentations.

La thèse se termine par une conclusion générale qui synthétise les résultats obtenus, discute les apports de la structure $\mathcal{E}_\diamond$ dans le cadre de la théorie des catégories exactes, et propose des pistes pour des travaux futurs. Celles-ci incluent notamment l'extension de la structure $\mathcal{E}_\diamond$ à d'autres types de carquois, l'étude de sous-catégories CRJ dans des contextes algébriques plus généraux, ainsi que l'exploration de liens avec la géométrie combinatoire et certains invariants homologiques.

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INTRODUCTION

Contexte et objectifs

La théorie des catégories exactes, introduite par Quillen en [Qui73], fournit un cadre axiomatique permettant de développer l'algèbre homologique relativement à une classe choisie de suites exactes courtes. Cette approche s'est révélée particulièrement adaptée à l'étude des catégories de modules, notamment en théorie des représentations, où différentes structures exactes peuvent mettre en évidence des phénomènes homologiques distincts. Au-delà de l'introduction d'une structure exacte particulière, la contribution conceptuelle centrale de cette thèse est de montrer que certains phénomènes combinatoires en théorie des représentations tels que la presque rigidité et la retrouvabilité d'invariants deviennent naturellement intelligibles lorsque l'on remplace la structure exacte maximale par un cadre exact relatif adapté. La structure exacte introduite dans ce travail intervient ici comme un *langage minimal* sélectionnant les extensions réellement pertinentes pour ces phénomènes, révélant ainsi leur nature homologique sous-jacente.

Cette thèse s'inscrit dans ce cadre et se concentre sur l'étude d'une structure exacte non standard, appelée *structure exacte diamant* et notée $\mathcal{E}_\diamond$, définie sur des catégories de modules associées à des carquois de type $\mathbb{A}_n$. L'objectif principal est de mettre en évidence que la structure exacte $\mathcal{E}_\diamond$ n'est pas une simple restriction technique de la structure maximale, mais un cadre conceptuellement nécessaire pour formuler et unifier plusieurs notions apparues récemment en théorie des représentations, en particulier celles de modules presque rigides et de sous-catégories canoniquement retrouvables de Jordan.

La thèse est présentée sous la forme d'une thèse par articles. Les articles originaux sont placés en annexe, tandis que le corps du manuscrit en présente le cadre conceptuel, la motivation, ainsi qu'une synthèse cohérente des résultats obtenus.

Motivations

Dans une catégorie abélienne, la structure exacte maximale, constituée de toutes les suites exactes courtes, n'est pas toujours adaptée à l'étude de certaines propriétés combinatoires ou de certains invariants. Plus précisément, dans les situations considérées ici, la profusion d'extensions dans la structure exacte maximale tend à masquer les mécanismes homologiques responsables des contraintes combinatoires observées. L'introduction d'une structure exacte plus fine doit alors être comprise comme une sélection conceptuelle qui filtre les extensions non pertinentes et isole celles qui portent l'information structurante.

Pour les carquois de type $\mathbb{A}_n$, certaines suites exactes courtes non scindées présentent une structure particulière, dite *diamant*, dont le terme central est somme directe de deux indécomposables distincts. La structure exacte $\mathcal{E}_\diamond$ est construite en sélectionnant précisément ces suites. Elle permet ainsi d'ignorer certaines extensions tout en conservant celles qui portent une information combinatoire essentielle.

Ce choix se révèle particulièrement pertinent pour l'étude des modules presque rigides, introduits afin de généraliser la notion classique de rigidité, ainsi que pour l'analyse des sous-catégories canoniquement retrouvables de Jordan, définies à partir d'invariants liés aux endomorphismes nilpotents. La structure $\mathcal{E}_\diamond$ fournit un cadre commun permettant d'approfondir la compréhension de ces deux notions.

Résultats principaux

Le premier article [BHRS24] introduit formellement la structure exacte $\mathcal{E}_\diamond$ sur la catégorie mod KQ, où Q est un carquois de type $\mathbb{A}_n$ et K un corps algébriquement clos. Il est montré que les modules presque rigides coïncident avec les objets $\mathcal{E}_\diamond$-rigides, et que les modules presque rigides maximaux correspondent exactement aux modules $\mathcal{E}_\diamond$-inclinants. Cette correspondance s'inscrit dans le cadre des catégories 0-Auslander et fournit une interprétation structurelle de résultats auparavant obtenus par des méthodes géométriques.

Le deuxième article [DR25] étudie les sous-catégories canoniquement retrouvables de Jordan dans le même contexte. Il introduit un opérateur de type Gen-Sub associé à la structure exacte $\mathcal{E}_\diamond$, permettant de construire des sous-catégories stables par extensions. Le résultat principal établit que les sous-catégories CRJ maximales sont exactement celles de la forme $\mathrm{GS}_{\mathcal{E}_\diamond}(T)$, où T est un module inclinant. Ce résultat met en évidence un lien précis entre structures exactes, invariants de Jordan et propriétés combinatoires des représentations de type $\mathbb{A}_n$.

Organisation de la thèse

Le premier chapitre est consacré à l'étude de la structure exacte $\mathcal{E}_\diamond$ et des modules presque rigides pour les carquois de type $\mathbb{A}_n$. Il en présente le cadre conceptuel et synthétise les principaux résultats du premier article annexé, notamment l'identification des modules presque rigides maximaux aux modules $\mathcal{E}_\diamond$-inclinants. Le deuxième chapitre est dédié aux sous-catégories canoniquement retrouvables de

Jordan et à l'opérateur de clôture homologique Gen–Sub relatif à $\mathcal{E}_\diamond$. Il en expose les constructions fondamentales et la classification des sous-catégories CRJ maximales obtenue dans le second article. La thèse se conclut par une conclusion générale qui met en perspective l'ensemble des résultats et certaines directions de recherche, tandis que les articles originaux sont intégralement en annexe.

CHAPITRE 1

La structure exacte $\mathcal{E}_\diamond$ et les modules presque rigides de type $\mathbb{A}_n$

Ce chapitre présente le cadre conceptuel général de la thèse ainsi qu'une synthèse structurée des résultats obtenus dans le premier article annexé [BHRS24]. L'objectif n'est pas de reproduire l'article, mais d'en dégager les idées centrales, les constructions fondamentales et les conséquences principales, dans une perspective adaptée au format d'une thèse par articles.

Dans ce chapitre, on travaille dans le cadre de la théorie des catégories exactes au sens de Quillen, appliquée à la catégorie $\operatorname{mod} KQ$ des modules de dimension finie sur l'algèbre de chemins KQ d'un carquois Q de type $\mathbb{A}_n$ où K un corps algébriquement clos. On y considère une structure exacte non standard, appelée *structure diamant* et notée $\mathcal{E}_\diamond$, définie de manière à sélectionner une classe spécifique de suites exactes courtes. Cette structure exacte fournit le cadre homologique relatif dans lequel s'interprètent les propriétés de presque rigidité.

1.1 Cadre général

Soit Q un carquois de type $\mathbb{A}_n$ et KQ son algèbre de chemins sur un corps algébriquement clos K. On note $\operatorname{mod} KQ$ la catégorie des modules de dimension finie sur KQ. Cette catégorie est abélienne, héréditaire et de type de représentation fini, ce qui en fait un cadre naturel pour l'étude des structures exactes non standards.

Dans une catégorie additive $\mathscr{A}$, une structure exacte $\mathcal{E}$ est donnée par une classe de suites exactes courtes satisfaisant les axiomes de Quillen. La structure exacte maximale $\mathcal{E}_{\max}$, constituée de toutes les suites exactes courtes, n'est toutefois pas toujours adaptée pour mettre en évidence certains phénomènes combinatoires ou homologiques. L'introduction de structures exactes plus fines permet de sélectionner des extensions pertinentes en fonction des propriétés étudiées.

Dans le cas des carquois de type $\mathbb{A}_n$, l'analyse du carquois d'Auslander–Reiten montre que les suites exactes courtes non scindées entre modules indécomposables se répartissent en deux classes distinctes, selon que leur terme central est indécomposable ou somme directe de deux indécomposables non isomorphes. Ces dernières, appelées *suites exactes diamants*, constituent le point de départ de la construction de la structure exacte $\mathcal{E}_\diamond$.

1.2 La structure exacte $\mathcal{E}_\diamond$

Rappelons qu'une structure exacte $\mathcal{E}$ au sens de Quillen sur une catégorie additive $\mathscr{A}$ est donnée par une classe de suites exactes courtes de $\mathscr{A}$, appelées *suites exactes admissibles*, satisfaisant les conditions suivantes :

(E0) toute suite exacte scindée appartient à $\mathcal{E}$;

(E1) la classe des monomorphismes admissibles est stable par composition, et de même pour les épimorphismes admissibles ;

(E2) $\mathcal{E}$ est stable sur les somme amalgamées et les produits fibrés ;

Ces axiomes assurent que la donnée de $\mathcal{E}$ définit une structure exacte sur $\mathscr{A}$ et permettent de développer une algèbre homologique relative à $\mathcal{E}$. Les définitions formelles et les propriétés générales des structures exactes considérées dans ce travail sont rappelées dans les articles annexés [BHRS24](Définition 2.1) et [DR25] (Définition 3.2).

La structure exacte $\mathcal{E}_\diamond$ est la plus petite structure exacte sur mod KQ contenant toutes les suites exactes scindées et toutes les suites exactes diamants. Elle sélectionne ainsi une classe d'extensions étroitement liée à la combinatoire du carquois d'Auslander–Reiten.

L'un des résultats fondamentaux établis dans l'article [BHRS24] est que $\mathcal{E}_\diamond$ satisfait bien les axiomes d'une structure exacte au sens de Quillen. Cette structure exacte permet alors de reformuler plusieurs notions classiques de la théorie des représentations dans un cadre relatif.

En particulier, la notion de rigidité relative à $\mathcal{E}_\diamond$ coïncide avec la notion de presque rigidité. Plus précisément, un module est presque rigide si et seulement s'il est $\mathcal{E}_\diamond$-rigide. Cette observation constitue un premier lien structurel entre les suites exactes diamants et les propriétés homologiques des modules.

1.3 Modules presque rigides et théorie inclinante relative

L'étude des modules presque rigides maximaux (MAR), introduits dans [BGMS23], constitue l'un des objectifs principaux du premier article. La notion de presque rigidité a été introduite dans le contexte des carquois de type $\mathbb{A}_n$ afin d'ignorer certaines extensions non scindées dont le terme central est indécomposable, et admet une interprétation combinatoire explicite en termes de diagonales non croisées dans un polygone.

Rappelons qu'un module basique $M \in \operatorname{mod} KQ$ est dit *presque rigide* s'il n'existe pas de paires de facteurs directs indécomposables X, Y de M telles que $\operatorname{Ext}^1_{KQ}(X, Y)$ contienne une suite exacte courte non scindée de type diamant. Il est dit *presque rigide maximal* s'il est presque rigide et que $M \oplus X$ n'est pas presque rigide pour tout $X \in \operatorname{mod} KQ$ indécomposable.

Rappelons qu'un module $M \in \operatorname{mod} KQ$ est dit *rigide* au sens classique si $\operatorname{Ext}^1_{KQ}(M, M) = 0$. Il est dit *inclinant* s'il est rigide et maximal pour cette propriété, c'est-à-dire s'il n'existe aucun module indécomposable $X \in \operatorname{mod} KQ$ tel que $M \oplus X$ soit encore rigide. Dans le cas héréditaire des carquois de type $\mathbb{A}_n$, cette caractérisation coïncide avec la définition classique des modules inclinants.

Relativement à la structure exacte $\mathcal{E}_\diamond$, ces modules apparaissent naturellement comme des objets presque rigides maximaux. Dans ce contexte, un module $T \in \operatorname{mod} KQ$ est dit $\mathcal{E}_\diamond$-inclinant s'il est $\mathcal{E}_\diamond$-rigide et maximal pour cette propriété. Cette identification repose sur le fait que la catégorie exacte considérée est une catégorie 0-Auslander, ce qui permet d'appliquer des résultats généraux reliant rigidité maximale et théorie inclinante comme montré dans [GNP23].

Dans ce contexte, il est possible de formuler le résultat central reliant les modules presque rigides maximaux à la théorie inclinante relative.

Théorème 1.3.1 ([BHRS24, Corollaire 4.10]). *Soit Q un carquois de type $\mathbb{A}_n$. Dans la catégorie exacte $(\operatorname{mod} KQ, \mathcal{E}_\diamond)$, les modules presque rigides maximaux coïncident exactement avec les modules $\mathcal{E}_\diamond$-inclinants.*

Ce résultat fournit en particulier une interprétation homologique des propriétés combinatoires observées pour les modules MAR dans [BGMS23], notamment leur bijection avec les triangulations d'un polygone à $n + 1$ sommets. La structure exacte $\mathcal{E}_\diamond$ apparaît ainsi comme le cadre homologique naturel sous-jacent à ces phénomènes.

1.4 Portée et limites

Les résultats obtenus dans ce cadre sont spécifiques aux carquois de type $\mathbb{A}_n$. Des exemples montrent que la construction de $\mathcal{E}_\diamond$ ne se généralise pas directement aux carquois de type $\mathbb{D}$, en raison de l'échec de certaines propriétés de stabilité requises pour une structure exacte. Ces limites soulignent le caractère intrinsèquement combinatoire de la construction et motivent les perspectives développées en conclusion.

1.5 Rôle et contributions de l'auteur

Les résultats présentés dans ce chapitre proviennent de travaux réalisés en collaboration avec T. Brüstle, E. J. Hanson et R. Schiffler. Ces travaux s'inscrivent dans un effort collectif visant à analyser des phénomènes combinatoires et homologiques en théorie des représentations à l'aide de structures exactes non standards.

Dans ce cadre, l'auteur de la présente thèse a apporté une expertise spécifique dans l'étude et l'exploitation des structures exactes. Sa contribution a porté principalement sur l'analyse homologique de la structure exacte $\mathcal{E}_\diamond$, sur la formulation rigoureuse des notions d'objets rigides et d'objets inclinants relatives à cette structure, ainsi que sur leur articulation avec les propriétés générales des catégories exactes au sens de Quillen.

L'auteur a joué un rôle central dans l'identification et l'interprétation des suites exactes diamants comme extensions admissibles pertinentes, ainsi que dans l'étude de leurs conséquences sur la structure homologique de la catégorie $(\operatorname{mod} KQ, \mathcal{E}_\diamond)$. Cette analyse a notamment permis de relier les modules presque rigides aux objets inclinants dans un cadre exact relatif, en faisant intervenir de manière essentielle la théorie des catégories 0-Auslander.

Ces contributions ont consisté à structurer l'approche exacte sous-jacente aux résultats, à en vérifier la cohérence axiomatique et à en dégager les implications conceptuelles, fournissant ainsi le socle homologique sur lequel reposent les interprétations combinatoires développées dans l'article.

CHAPITRE 2

Sous-catégories canoniquement retrouvables de Jordan et opérateur Gen–Sub dans le cadre de la structure exacte $\mathcal{E}_\diamond$

Ce chapitre présente une synthèse structurée des résultats obtenus dans le second article annexé [DR25]. Dans la continuité du chapitre précédent, l'objectif n'est pas de reproduire l'article, mais d'en dégager les constructions fondamentales, les résultats principaux et les interprétations conceptuelles, dans un format adapté à une thèse par articles.

Alors que le chapitre 1 était consacré à l'étude des extensions et de la rigidité relative via la structure exacte $\mathcal{E}_\diamond$, le présent chapitre s'intéresse à un problème complémentaire : la classification de sous-catégories de $\operatorname{mod} KQ$ dont les objets sont entièrement déterminés par un invariant de Jordan générique. Ces sous-catégories, appelées *sous-catégories canoniquement retrouvables de Jordan* (CRJ), sont étudiées à l'aide d'un opérateur de clôture homologique, l'opérateur Gen–Sub, défini relativement à $\mathcal{E}_\diamond$.

2.1 Cadre général

Soit Q un carquois de type $\mathbb{A}_n$ et KQ son algèbre de chemins sur un corps algébriquement clos K. On note $\operatorname{mod} KQ$ la catégorie des modules de dimension finie sur KQ. Comme dans le chapitre précédent, cette catégorie est abélienne, héréditaire et de type de représentation fini.

L'étude des sous-catégories CRJ trouve son origine dans l'analyse des invariants associés aux endomorphismes nilpotents des représentations. À toute représentation $M \in \operatorname{mod} KQ$ est associée une donnée de Jordan générique, qui encode la forme de Jordan maximale apparaissant parmi ses endomorphismes nilpotents. Dans certains cas, cet invariant suffit à déterminer M à isomorphisme près au sein d'une sous-catégorie donnée.

Cette observation conduit naturellement à la définition des sous-catégories CRJ, qui imposent une contrainte forte sur les extensions autorisées entre leurs objets. Comme pour les modules presque rigides, le contrôle de ces extensions constitue un aspect central de l'analyse.

2.2 Sous-catégories canoniquement retrouvables de Jordan

On rappelle la définition centrale utilisée dans [Deq23a], qui constitue un travail précurseur de [DR25].

Définition 2.2.1 ([GPT23]). *Soit Q un carquois lié, fini et connexe. Soit $\mathrm{mod}\, KQ$ la catégorie de ses représentations de dimension finie. Une sous-catégorie additive pleine ($\mathscr{C} \subseteq \mathrm{mod}\, KQ$) est dite canoniquement retrouvable de Jordan (CRJ) si l'invariant de la forme de Jordan générique y est non seulement complet, mais admet de plus une procédure algébrique canonique d'inversion.*

Autrement dit, il existe un procédé explicite et fonctoriel permettant de reconstruire, à isomorphisme près, tout objet de $\mathscr{C}$ à partir de la donnée de Jordan générique associée à ses endomorphismes nilpotents génériques.

Cette condition impose en particulier l'absence de certaines suites exactes non scindées entre objets de $\mathscr{C}$, sous peine de produire plusieurs représentations distinctes partageant le même invariant de Jordan. L'analyse du carquois d'Auslander–Reiten met en évidence que ces situations indésirables sont étroitement liées à des configurations combinatoires spécifiques, notamment des intervalles adjacents.

2.3 Rôle de la structure exacte $\mathcal{E}_\diamond$

Dans ce contexte, la structure exacte $\mathcal{E}_\diamond$, introduite au chapitre 1, fournit un cadre homologique naturel pour contrôler les extensions compatibles avec la retrouvabilité de Jordan. Contrairement à la structure exacte maximale, $\mathcal{E}_\diamond$ exclut précisément les suites exactes non scindées dont le terme central est indécomposable, lesquelles sont à l'origine des ambiguïtés sur les invariants de Jordan.

Ainsi, bien que $\mathcal{E}_\diamond$ ne soit pas l'objet principal de ce chapitre, elle joue un rôle technique essentiel en servant de filtre homologique pour les constructions ultérieures. Elle permet d'isoler une classe d'extensions admissibles compatible avec les contraintes imposées par les sous-catégories CRJ.

2.4 L'opérateur Gen–Sub relatif à $\mathcal{E}_\diamond$

Afin de formaliser la construction de sous-catégories CRJ, l'article [DR25] introduit un opérateur de clôture, noté $\mathrm{GS}_{\mathcal{E}}$, défini relativement à une structure exacte $\mathcal{E}$.

Définition 2.4.1. *Soit $\mathscr{C} \subseteq \operatorname{mod} KQ$ une sous-catégorie additive et $\mathcal{E}$ une structure exacte sur $\operatorname{mod} KQ$. On définit $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ comme la plus petite sous-catégorie additive contenant $\mathscr{C}$ et telle que, pour toute suite exacte courte $\mathcal{E}$-admissible*

$$0 \to A \to B \to C \to 0,$$

si B appartient à $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$, alors A et C y appartiennent également.

Dans le cas particulier de $\mathcal{E}_\diamond$, cet opérateur permet de construire des sous-catégories fermées sous les extensions compatibles avec les invariants de Jordan, en se focalisant sur le rôle des termes centraux des suites exactes diamants.

2.5 Sous-catégories CRJ maximales

Comme dans l'étude des modules presque rigides, une notion de maximalité joue un rôle central.

Définition 2.5.1. *Une sous-catégorie CRJ* $\mathscr{C} \subseteq \mathrm{mod}\, KQ$ *est dite* maximale *si elle n'est contenue strictement dans aucune autre sous-catégorie CRJ.*

Le résultat principal du second article fournit une classification complète de ces sous-catégories.

Théorème 2.5.2 ([DR25, Thm. 4.20])**.** *Une sous-catégorie* $\mathscr{C} \subseteq \mathrm{mod}\, KQ$ *est une sous-catégorie CRJ maximale si et seulement s'il existe une représentation inclinante* $T \in \mathrm{mod}\, KQ$ *telle que*

$$\mathscr{C} = \mathrm{GS}_{\mathcal{E}_\diamond}(T).$$

Ce théorème montre que les représentations inclinantes jouent un rôle structurant analogue à celui observé pour les modules presque rigides maximaux au chapitre 1, l'opérateur Gen–Sub fournissant le mécanisme homologique permettant de passer d'un objet générateur à une sous-catégorie maximale.

2.6 Description combinatoire

Dans le cas des carquois de type $\mathbb{A}_n$, les sous-catégories CRJ admettent une description combinatoire explicite en termes d'ensembles d'intervalles sans adjacence. Cette condition combinatoire traduit algébriquement l'absence de suites exactes incompatibles avec la retrouvabilité de Jordan et fournit un critère effectif pour reconnaître les sous-catégories CRJ.

Cette interprétation complète l'approche homologique fondée sur l'opérateur $\mathrm{GS}_{\mathcal{E}_\diamond}$, en reliant invariants de Jordan et géométrie combinatoire des représentations de carquois.

2.7 Portée et limites

Malgré que les résultats de classification présentés dans ce chapitre soient formulés pour la catégorie $\operatorname{mod} KQ$ associée à un carquois de type $\mathbb{A}_n$, il convient de souligner que les constructions fondamentales introduites dans le second article sur l'opérateur de clôture homologique Gen-Sub défini relativement à une structure exacte sont formulées dans le cadre plus général des catégories abéliennes héréditaires possédant suffisamment de projectifs. C'est d'abord dans ce cadre que les résultats intéressants s'illustrent.

Le choix du type $\mathbb{A}_n$ permet ici d'obtenir une description explicite et combinatoire des sous-catégories CRJ maximales, mais ne reflète pas une restriction du mécanisme homologique sous-jacent.

Cependant, pour la structure exacte $\mathcal{E}_\diamond$ elle-même, les résultats présentés dans ce chapitre sont spécifiques aux carquois de type $\mathbb{A}_n$. Des exemples montrent que l'approche ne se généralise pas directement à d'autres types de carquois, notamment en type $\mathbb{D}$, où $\mathcal{E}_\diamond$ n'est pas une structure exacte.

2.8 Rôle et contributions de l'auteur

Les résultats présentés dans ce chapitre sont issus d'un travail collaboratif avec B. Dequêne portant sur l'étude des sous-catégories canoniquement retrouvables de Jordan dans les catégories de représentations de carquois de type $\mathbb{A}_n$. Dans ce cadre, l'auteur de la présente thèse a joué un rôle structurant dans l'élaboration du cadre mathématique exact et homologique sous-jacent aux résultats principaux.

Plus précisément, l'auteur est à l'origine de l'introduction et de l'exploitation systématique des structures exactes comme outil central pour l'étude des sous-catégories

CRJ. Il a proposé l'utilisation de la structure exacte non standard $\mathcal{E}_\diamond$ comme cadre homologique pertinent pour contrôler les extensions compatibles avec la retrouvabilité des invariants de Jordan, établissant ainsi un lien conceptuel clair entre invariants, extensions et propriétés exactes relatives.

L'auteur a également joué un rôle déterminant dans la définition et la formalisation de l'opérateur homologique $\mathrm{GS}_{\mathcal{E}}$, et en particulier de l'opérateur $\mathrm{GS}_{\mathcal{E}_\diamond}$. Cette contribution comprend la conception de l'opérateur comme un mécanisme de clôture exact relatif, son intégration cohérente dans le cadre des catégories exactes, ainsi que l'analyse de ses propriétés structurelles nécessaires à la classification des sous-catégories CRJ maximales.

La contribution du collaborateur s'inscrit principalement dans l'analyse combinatoire des représentations de carquois de type $\mathbb{A}_n$, notamment dans l'étude fine des intervalles, des configurations sans adjacence et des invariants de Jordan génériques. Ces aspects combinatoires ont joué un rôle essentiel dans l'identification des phénomènes étudiés et dans la validation des résultats, tandis que le cadre exact et homologique développé par l'auteur en fournit la structuration conceptuelle et la justification mathématique.

Ainsi, l'apport principal de l'auteur dans ce travail réside dans la mise en place d'un langage exact unifié permettant d'interpréter, de structurer et de généraliser des phénomènes initialement observés de manière combinatoire, ainsi que dans la démonstration que les sous-catégories CRJ maximales se décrivent naturellement comme des clôtures homologiques relatives associées à des objets générateurs, ce qui permet également de confirmer certaines conjectures posées antérieurement par le collaborateur.

Conclusion

Cette thèse s'inscrit dans le cadre de la théorie des catégories exactes au sens de Quillen, appliquée à l'étude des catégories de représentations de carquois de type $\mathbb{A}_n$. Son objectif principal a été de mettre en évidence le rôle structurant de structures exactes non standards pour l'analyse de phénomènes combinatoires et homologiques apparaissant en théorie des représentations, en particulier dans l'étude des modules presque rigides et des sous-catégories canoniquement retrouvables de Jordan.

Le fil conducteur de ce travail est la structure exacte $\mathcal{E}_\diamond$, introduite comme un outil permettant de sélectionner une classe pertinente d'extensions, étroitement liée à la combinatoire du carquois d'Auslander–Reiten des algèbres de chemins de type $\mathbb{A}_n$. Cette structure exacte non maximale isole précisément les suites exactes non scindées dont le terme central est décomposable en deux indécomposables non isomorphes, appelées suites exactes diamants, et fournit ainsi un cadre homologique adapté à l'étude des propriétés considérées.

Dans le premier chapitre, la structure exacte $\mathcal{E}_\diamond$ permet de reformuler la notion de presque rigidité dans un langage exact relatif. Il est montré que les modules presque rigides coïncident exactement avec les modules $\mathcal{E}_\diamond$-rigides, et que les modules presque rigides maximaux correspondent aux modules inclinants relatifs pour la catégorie exacte $(\operatorname{mod} KQ, \mathcal{E}_\diamond)$. Cette identification fournit une interprétation homologique unifiée de phénomènes combinatoires déjà observés, notamment la bijection avec les triangulations de polygones, et met en évidence le rôle fondamental joué par la structure exacte sous-jacente.

Le second chapitre exploite la même structure exacte dans un contexte différent, celui de la classification des sous-catégories canoniquement retrouvables de Jordan. Dans ce cadre, $\mathcal{E}_\diamond$ intervient comme un outil technique permettant de contrôler les extensions compatibles avec la retrouvabilité des invariants de Jordan. L'introduction de l'opérateur de clôture homologique $\mathrm{GS}_{\mathcal{E}_\diamond}$ permet alors de décrire et de classifier les sous-catégories CRJ maximales comme des clôtures exactes relatives engendrées par des représentations inclinantes. Cette approche fournit un lien conceptuel clair entre invariants de Jordan, structures exactes et combinatoire des carquois.

L'un des apports essentiels de cette thèse réside dans la mise en place d'un cadre exact unifié permettant d'interpréter des phénomènes initialement observés de manière combinatoire dans un langage homologique abstrait. La structure exacte $\mathcal{E}_\diamond$ apparaît ainsi non seulement comme un outil adapté à l'étude de la rigidité relative, mais également comme un cadre pertinent pour la classification de sous-catégories définies par des invariants.

Les résultats obtenus sont toutefois spécifiques aux carquois de type $\mathbb{A}_n$. Des contre-exemples montrent que la construction de $\mathcal{E}_\diamond$ ne se généralise pas directement à d'autres types de carquois, notamment de type $\mathbb{D}$, en raison de l'échec de certaines propriétés de stabilité requises pour une structure exacte. Ces limitations soulignent le caractère intrinsèquement combinatoire de la construction et invitent à rechercher des généralisations adaptées à d'autres contextes algébriques.

Plusieurs perspectives s'ouvrent à la suite de ce travail. Une première direction consiste à explorer des variantes de la structure $\mathcal{E}_\diamond$ dans des classes plus larges d'algèbres, telles que les algèbres aimables ou certaines algèbres de dimension finie, afin de déterminer dans quelle mesure des phénomènes analogues persistent. Une autre direction concerne l'étude d'opérateurs de clôture homologique analogues à $\mathrm{GS}_{\mathcal{E}}$ dans des cadres exacts différents, et leur interaction avec des invariants combinatoires ou géométriques. Enfin, les liens mis en évidence entre structures exactes, théorie inclinante relative et invariants suggèrent des applications potentielles en géométrie combinatoire et en théorie des catégories.

En définitive, cette thèse met en évidence que les structures exactes non standards ne doivent pas être perçues comme de simples raffinements techniques de la structure maximale, mais comme un outil conceptuel permettant d'identifier le bon langage homologique pour des phénomènes combinatoires profonds. Dans le cas des carquois de type $\mathbb{A}_n$, la structure $\mathcal{E}_\diamond$ apparaît ainsi comme un cadre minimal et naturel où rigidité relative, théorie inclinante et retrouvabilité d'invariants se trouvent unifiées.

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Annexe A

Premier article

Référence

T. Brüstle, E. J. Hanson, **S. Roy**, R. Schiffler, *An exact structure approach to almost rigid modules over quivers of type* $\mathbb{A}$, arXiv :2410.04627 (2024).

AN EXACT STRUCTURE APPROACH TO ALMOST RIGID MODULES OVER QUIVERS OF TYPE $\mathbb{A}$

THOMAS BRÜSTLE, ERIC J. HANSON, SUNNY ROY, AND RALF SCHIFFLER

Abstract. Let A be the path algebra of a quiver of Dynkin type $\mathbb{A}_n$. The module category $\operatorname{mod} A$ has a combinatorial model as the category of diagonals in a polygon S with $n+1$ vertices. The recently introduced notion of almost rigid modules is a weakening of the classical notion of rigid modules. The importance of this new notion stems from the fact that maximal almost rigid A-modules are in bijection with the triangulations of the polygon S.

In this article, we give a different realization of maximal almost rigid modules. We introduce a non-standard exact structure $\mathcal{E}_\diamond$ on $\operatorname{mod} A$ such that the maximal almost rigid A-modules in the usual exact structure are exactly the maximal rigid A-modules in the new exact structure. A maximal rigid module in this setting is the same as a tilting module. Thus the tilting theory relative to the exact structure $\mathcal{E}_\diamond$ translates into a theory of maximal almost rigid modules in the usual exact structure.

As an application, we show that with the exact structure $\mathcal{E}_\diamond$, the module category becomes a 0-Auslander category in the sense of Gorsky, Nakaoka and Palu.

We also discuss generalizations to quivers of type $\mathbb{D}$ and gentle algebras.

1. Introduction

Let Q be a quiver of Dynkin type $\mathbb{A}_n$ and K a field. It is a well-known fact that if $\eta = (0 \to X \to E \to Y \to 0)$ is a nonsplit short exact sequence of (right) KQ-modules and both X and Y are indecomposable, then E is either indecomposable or is the direct sum of precisely two nonisomorphic indecomposables. If $E \cong E_1 \oplus E_2$ is not indecomposable, we say that η is a *diamond* short exact sequence.

A module M over KQ is said to be *rigid* if $\operatorname{Ext}^1_{KQ}(M, M) = 0$. In [BGMS21], the notion of rigidity is weakened by "ignoring" those nonsplit short exact sequences whose middle term is indecomposable, and for type $\mathbb{A}$ these are the ones that are not diamonds. More precisely, we can restate their definition as follows. Note that it is assumed that $n \geq 3$ throughout the paper [BGMS21].

Definition 1.1. [BGMS21] Let M be a basic (right) KQ-module. Then M is *almost rigid* if there do not exist indecomposable direct summands X and Y of M such that $\operatorname{Ext}^1_{KQ}(X, Y)$ contains a diamond short exact sequence. We say that M is a *maximal almost rigid (MAR)* module if M is almost rigid and there does not exist a nonzero module N such that $M \oplus N$ is a (basic) almost rigid module.

One of the main results of [BGMS21] shows that the maximal almost rigid modules over KQ are in bijection with the set of triangulations of an $(n+1)$-gon. Moreover, this bijection arises from a geometric realization of the category $\operatorname{mod} KQ$ of finitely-generated (right) modules which associates either an oriented diagonal or an oriented boundary edge of the $(n+1)$-gon to each indecomposable module. As observed in [BGMS21, Remark 4.7], this differs from the realization of the cluster category of type $\mathbb{A}_{n-2}$ established in [CCS06] due to the orientation, the degree shift and the inclusion of boundary edges.

The triangulations of the $(n+1)$-gon in question also appear in [Rea06], where they are used as a model for the "Cambrian lattice" of a quiver of type $\mathbb{A}_{n-2}$. The cover relations of this lattice correspond to "flips" of diagonals. On the level of MAR modules, this manifests as a type of "mutation theory", where any pair of MAR modules related by a cover necessarily differ by only a single direct summand. Implicit in this description are the facts that every MAR module has the same number of indecomposable direct summands and that the modules corresponding to boundary edges are direct summands of every MAR modules, each of which is established in [BGMS21]. These generalize well-known properties from classical tilting theory, namely that every tilting module has the same number of direct summands and that the projective-injective modules are direct summands of every tilting module.

The goal of this paper is to interpret the results described above using the theory of exact categories, and more specifically that of 0-Auslander categories[1] established in the recent paper [GNP].

The paper is outlined as follows. In Section 2, we recall background information about exact categories, 0-Auslander categories, and the associated notions of rigidity and tilting. We then recall technical details regarding the Auslander-Reiten quivers in type $\mathbb{A}_n$ in Section 3. These well-known facts will be necessary in proving our main results later in the paper. In Section 4, we introduce an exact structure $\mathcal{E}_\diamond$. We then prove our main results.

Theorem 1.2. [Proposition 4.3, Corollary 4.4, Corollary 4.5, Theorem 4.9, and Corollary 4.10] *Let Q be a quiver of type $\mathbb{A}_n$. Then the following hold.*

(1) *Let $\eta = (0 \to X \to E \to Y \to 0)$ be a nonsplit short exact sequence with X and Y indecomposable. Then $\eta \in \mathcal{E}_\diamond(Y, X) \subseteq \mathrm{Ext}^1_{KQ}(Y, X)$ if and only if η is a diamond short exact sequence.*
(2) *A basic module $M \in \operatorname{mod} KQ$ is almost rigid if and only if it is $\mathcal{E}_\diamond$-rigid.*
(3) *A basic module $M \in \operatorname{mod} KQ$ is maximal almost rigid if and only if it is $\mathcal{E}_\diamond$-tilting.*
(4) *An indecomposable module $X \in \operatorname{mod} KQ$ is $\mathcal{E}_\diamond$-projective if and only if either X is projective or the Auslander-Reiten sequence $0 \to \tau X \to E \to X \to 0$ has E indecomposable.*
(5) *An indecomposable module $X \in \operatorname{mod} KQ$ is $\mathcal{E}_\diamond$-injective if and only if either X is injective or the Auslander-Reiten sequence $0 \to X \to E \to \tau^{-1}X \to 0$ has E indecomposable.*
(6) *$(\operatorname{mod} KQ, \mathcal{E}_\diamond)$ is a 0-Auslander category.*

As we discuss in Section 2, the theory of tilting in 0-Auslander categories thus gives an alternate explanation for why MAR modules always have the same number of direct summands and always contain the modules corresponding to the boundary edges of the $(n+1)$-gon.

We discuss possible generalizations in Section 5. In Section 5.1, we give examples showing why the theory developed in this paper cannot be readily extended to quivers of Dynkin type $\mathbb{D}$. In Section 5.2 we give an example of a possible generalization of our results to gentle algebras. In that example conditions (1)-(5) of Theorem 1.2 still hold, but condition (6) doesn't.

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2. Exact structures and 0-Auslander categories

In this section, we recall relevant background information on exact structures and 0-Auslander categories. We conclude with a discussion of the mutation theory for maximal rigid objects in 0-Auslander categories as developed in [GNP, Section 4]. Since we are mostly interested in applying these results to an exact structure on the category $\operatorname{mod}\Lambda$ of finitely-generated (right) modules over a finite-dimensional algebra Λ, we restrict our exposition to this case. We note, however, that both exact structures/categories and 0-Auslander categories are defined much more generally.

2.1. **Exact structures.** Exact structures have been introduced by Quillen [Qui72] as an axiomatic framework that allows to use methods from homological algebra in a context relative to a fixed class of short exact sequences. Besides the axiomatic approach, there are a number of ways to describe an exact structure $\mathcal{E}$. We will describe an exact structure as a certain subfunctor $\mathcal{E}$ of the bifunctor Ext^1_Λ, or by specifying what are the projective objects relative to $\mathcal{E}$, or by deciding which Auslander-Reiten sequences belong to $\mathcal{E}$.

Let $\mathcal{E}$ be a class of short exact sequences in $\operatorname{mod}\Lambda$. We assume that $\mathcal{E}$ is closed under isomorphisms. When a short exact sequence $\eta = (0 \to A \xrightarrow{f} E \xrightarrow{g} B \to 0)$ belongs to $\mathcal{E}$, we say that η is an ($\mathcal{E}$-)*admissible short exact sequence*, and also that f is an ($\mathcal{E}$-)*admissible monomorphism* and g is an ($\mathcal{E}$-)*admissible epimorphism.*

[1]Note that 0-Auslander categories are actually defined as certain examples of *extriangulated* categories. Extriangulated categories include both exact and triangulated categories as special cases.

The closure under isomorphisms then implies that the classes of admissible short exact sequences, admissible monomorphisms, and admissible epimorphisms uniquely determine one another. Quillen's definition can then be rephrased as follows:

Definition 2.1. A class $\mathcal{E}$ of short exact sequences in $\operatorname{mod}\Lambda$ is said to be an *exact structure* on $\operatorname{mod}\Lambda$ if all of the following hold:

(1) $\mathcal{E}$ contains all split exact sequences.
(2) $\mathcal{E}$ is closed under compositions of admissible monomorphisms, that is, if $f : X \to Y$ and $g : Y \to Z$ are admissible monomorphisms, then $g \circ f : X \to Z$ is also an admissibe monomorphism. Likewise, $\mathcal{E}$ is closed under compositions of admissible epimorphisms.
(3) $\mathcal{E}$ is closed under pushouts: If $f : X \to Y$ is an admissible monomorphism, and $X \xrightarrow{h} W$ is any morphism in $\operatorname{mod}\Lambda$, then the pushout of f along h yields a short exact sequence in $\mathcal{E}$. Likewise, $\mathcal{E}$ is closed under pullbacks.

Remark 2.2. In the literature, it is common both to refer to $\mathcal{E}$ as an *exact structure* on $\operatorname{mod}\Lambda$ and to refer to the pair $(\operatorname{mod}\Lambda, \mathcal{E})$ as an *exact category*.

Quillen's original definition contained also a fourth (so-called "obscure" axiom) which was shown in [Kel90] to be superfluous. It has been observed in [AS93b] that in the context of module categories, the axioms (1) and (3) together can be rephrased as $\mathcal{E}$ forming an additive subfunctor $\mathcal{E}(-,-)$ of the bifunctor $\operatorname{Ext}^1_\Lambda(-,-)$. Here (1) states that $\mathcal{E}$ contains the zero elements of all groups $\operatorname{Ext}^1(X,Y)$ (under the Baer sum), and axiom (3) corresponds to functoriality. An additive subfunctor of $\operatorname{Ext}^1_\Lambda(-,-)$ satisfying condition (2) has been called a *closed* subfunctor of $\operatorname{Ext}^1_\Lambda(-,-)$, and [DRSS99] show that these correspond precisely to the exact structures on $\operatorname{mod}\Lambda$.

For Λ representation-finite, it has been shown in [Eno18] (see also [BHLR20, Theorem 5.7]) that an exact structure on $\operatorname{mod}\Lambda$ is uniquely determined by the Auslander-Reiten sequences it contains, and in fact there is a bijection between exact structures on $\operatorname{mod}\Lambda$ and sets of (isomorphism classes of) Auslander-Reiten sequences in $\operatorname{mod}\Lambda$. One can also characterize an exact structure by the set of its $\mathcal{E}$-projective (resp. $\mathcal{E}$-injective) objects, defined as follows.

Definition 2.3. Let $\mathcal{E}$ be an exact structure on $\operatorname{mod}\Lambda$.

(1) We say an object $P \in \operatorname{mod}\Lambda$ is *$\mathcal{E}$-projective*, in symbols $P \in \operatorname{proj}(\mathcal{E})$, if $g_* = \operatorname{Hom}_\Lambda(P, g)$ is an epimorphism for all $Y, Z \in \operatorname{mod}\Lambda$ and for all admissible epimorphisms $g : Y \to Z$. Equivalently, $P \in \operatorname{proj}(\mathcal{E})$ if and only if $\operatorname{Hom}_\Lambda(P,-)$ sends admissible exact sequences to exact sequences.
(2) We say an object $I \in \operatorname{mod}\Lambda$ is *$\mathcal{E}$-injective*, in symbols $I \in \operatorname{inj}(\mathcal{E})$, if $f_* = \operatorname{Hom}_\Lambda(f, I)$ is an epimorphism for all $X, Y \in \operatorname{mod}\Lambda$ and for all admissible monomorphisms $f : X \to Y$. Equivalently, $I \in \operatorname{inj}(\mathcal{E})$ if and only if $\operatorname{Hom}_\Lambda(-, I)$ sends admissible exact sequences to exact sequences.

The axiomatic setup of an exact structure allows one to introduce an $\mathcal{E}$-relative version for many concepts from homological algebra, like the existence of a long homology sequence. In particular, the fact that $\operatorname{Hom}_\Lambda(P,-)$ is an exact functor on admissible exact sequences can be rephrased by saying that $\mathcal{E}(P,-)$ is zero. Therefore $P \in \operatorname{mod}\Lambda$ is an $\mathcal{E}$-projective object precisely when there is no non-split short exact sequence $(0 \to A \to B \to P \to 0) \in \mathcal{E}$. The dual characterization likewise holds for $\mathcal{E}$-injective objects.

The following definition from [AS93b] uses this characterization to describe an exact structure in terms of its set of projective objects.

Definition 2.4. Let $\mathcal{X}$ be a subcategory of $\operatorname{mod}\Lambda$. We define

$$\mathcal{F}_\mathcal{X} = \{\eta = (0 \to A \to B \to C \to 0) \mid \operatorname{Hom}_\Lambda(X,-) \text{ is an exact functor on } \eta \text{ for all } X \in \mathcal{X}\}.$$

Remark 2.5. Since covariant hom-functors are always left exact, an exact sequence

$$0 \to A \to B \xrightarrow{f} C \to 0$$

is $\mathcal{F}_\mathcal{X}$-exact if and only if $\operatorname{Hom}(X, f)$ is surjective for all $X \in \mathcal{X}$.

In both the following proposition and the remainder of the paper, we denote by $\operatorname{proj}(\Lambda)$ (resp. $\operatorname{inj}(\Lambda)$) the full subcategory of $\operatorname{mod}\Lambda$ consisting of those objects which satisfy the usual notion of projectivity (resp. injectivity) in $\operatorname{mod}\Lambda$. These can be seen as the "true" projectives and injectives, as opposed to the "relative" projectives and injectives. We also denote by τ the Auslander-Reiten translation. For a subcategory $\mathcal{X}$, we denote by $\tau\mathcal{X}$ the full subcategory of objects of the form τX for $X \in \mathcal{X}$.

Proposition 2.6. [AS93b, Section 1] *Let $\mathcal{X}$ be a subcategory of* $\mathrm{mod}\,\Lambda$. *Then $\mathcal{F}_{\mathcal{X}}$ is an exact structure. Moreover,* $\mathrm{proj}(\mathcal{F}_{\mathcal{X}}) = \mathrm{add}(\mathcal{X} \cup \mathrm{proj}(\Lambda))$ *and* $\mathrm{inj}(\mathcal{F}_{\mathcal{X}}) = \mathrm{add}(\tau\mathcal{X} \cup \mathrm{inj}(\Lambda))$.

2.2. **Relative homological dimensions and relative tilting.** The notions of $\mathcal{E}$-projectives and $\mathcal{E}$-injectives allow one to define relative versions of many of the dimensions studied in homological algebra.

Definition 2.7. Let $\mathcal{E}$ be an exact structure on $\mathrm{mod}\,\Lambda$.

(1) We say that $\mathcal{E}$ *has enough projectives* if for every $M \in \mathrm{mod}\,\Lambda$ there exists an $\mathcal{E}$-admissible epimorphism $f_0 : P_0 \to M$ with $P \in \mathrm{proj}(\mathcal{E})$.
(2) Suppose $\mathcal{E}$ has enough projectives and let $M \in \mathrm{mod}\,\Lambda$. We say an exact sequence
$$\cdots\ P_2 \xrightarrow{f_2} P_1 \xrightarrow{f_1} P_0 \xrightarrow{f_0} M \longrightarrow 0$$
is an $\mathcal{E}$-projective resolution of M if all of P_i are $\mathcal{E}$-projective and each short exact sequence
$$0 \to \ker f_i \to P_i \to \mathrm{im} f_i \to 0$$
is $\mathcal{E}$-admissible. The *length* of a $\mathcal{E}$-projective resolution is the smallest index i for which $P_i = 0$ (or infinity if no such i exists).
(3) Suppose $\mathcal{E}$ has enough projectives and let $M \in \mathrm{mod}\,\Lambda$. The *$\mathcal{E}$-projective dimension* of M, denoted $\mathrm{pd}_{\mathcal{E}}(M)$, is the minimal length among all $\mathcal{E}$-projective resolutions of M.
(4) Suppose $\mathcal{E}$ has enough projectives. The *$\mathcal{E}$-global dimension* of Λ is $\sup_{M \in \mathrm{mod}\,\Lambda} \mathrm{pd}_{\mathcal{E}}(M)$.

Remark 2.8. (1) In later sections, we restrict to the case $\Lambda = KQ$ for Q a quiver of type $\mathbb{A}_n$. This algebra is representation finite, and thus every exact structure of $\mathrm{mod}\,KQ$ has enough projectives.
(2) Note that the $\mathcal{E}$-projective dimension of a module M need not be related to the (classical) projective dimension $\mathrm{pd}(M)$: one may have $\mathrm{pd}_{\mathcal{E}}(M) < \mathrm{pd}(M)$ since there are more $\mathcal{E}$-projective modules than projectives. For example, each $\mathcal{E}$-projective M which is not in $\mathrm{proj}\,\Lambda$ has $0 = \mathrm{pd}_{\mathcal{E}}(M) < \mathrm{pd}(M)$. On the other hand, an $\mathcal{E}$-projective resolution requires all maps be composed of $\mathcal{E}$-admissible maps, so projective resolutions are not necessarily $\mathcal{E}$-projective resolutions. In fact, we give in Section 5.1 an example of a module M over a quiver algebra (which is always of global dimension one) whose $\mathcal{E}$-projective dimension is two.

We now recall the following definition.

Definition 2.9. [AS93a] Let $\mathcal{E}$ be an exact structure on $\mathrm{mod}\,\Lambda$ with enough projectives. A module $T \in \mathrm{mod}\,\Lambda$ is said to be a *$\mathcal{E}$-tilting module* if the following statements all hold:

(1) $\mathrm{pd}_{\mathcal{E}}(T) \leq 1$.
(2) For all $P \in \mathrm{proj}(\mathcal{E})$, there is an $\mathcal{E}$-admissble short exact sequence $0 \to P \to T_1 \to T_2 \to 0$ with $T_1, T_2 \in \mathrm{add}(T)$.
(3) T is *$\mathcal{E}$-rigid*; that is, $\mathcal{E}(T, T) = 0$.

Remark 2.10. Note that [AS93a] more generally defines a notion of $\mathcal{E}$-tilting for modules of global dimension n ($n \in \mathbb{N}$). As is standard in classical tilting theory, this can be clarified by referring to $\mathcal{E}$-tilting modules of projective dimension n as "n-$\mathcal{E}$-tilting" (see e.g. [Wei10]). One can then drop the prefix n when dealing only with $n = 1$, resulting in the convention of the present paper and e.g. [MM22].

Note also that both [GNP] and [Pal24] refer to the notion of $\mathcal{E}$-tilting used in this paper as just "tilting". Both papers also consider a different (but equivalent) notion of "$\mathbb{E}$-tilting".

In Section 4, we will show that the maximal almost rigid modules in $\mathrm{mod}\,KQ$ (Q of type $\mathbb{A}_n$) are precisely the basic $\mathcal{E}$-tilting modules for some exact structure $\mathcal{E}$. In order to do so, it will be useful to relate the notion of $\mathcal{E}$-tilting to that of "maximal $\mathcal{E}$-rigidity". We do this using the theory of 0-Auslander categories from [GNP], the definition of which we recall now. We use the simplified version of the definition appearing as [Pal24, Definition 5.1]. Recall, however, that [Pal24] works with arbitrary extriangulated categories, while we work only with exact structures on $\mathrm{mod}\,\Lambda$.

Definition 2.11. Let $\mathcal{E}$ be an exact structure on $\mathrm{mod}\,\Lambda$ with enough projectives. Then $(\mathrm{mod}\,\Lambda, \mathcal{E})$ is a *0-Auslander category* if the following both hold.

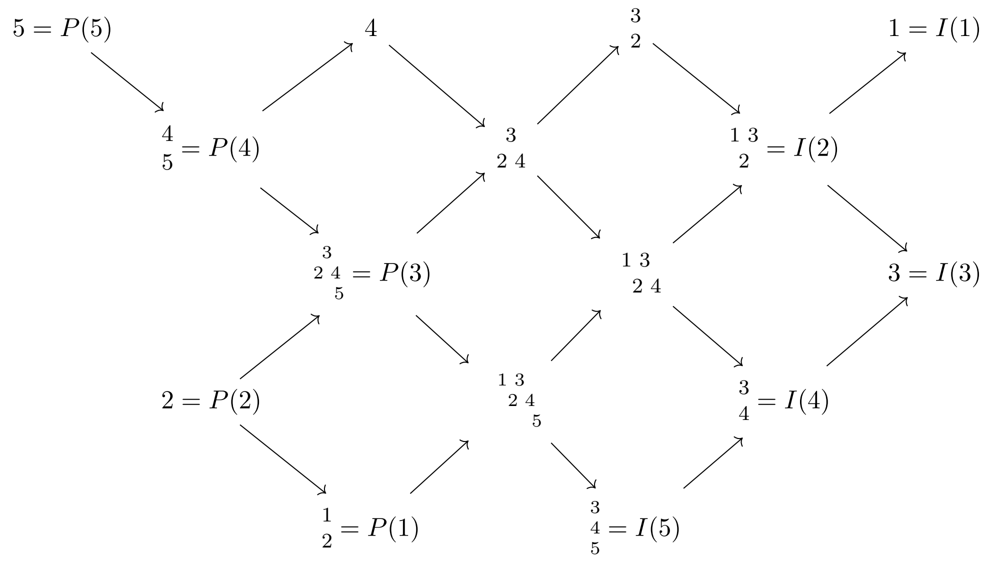

FIGURE 1. The embedded AR quiver $\Gamma(Q)$ for $Q = 1 \to 2 \leftarrow 3 \to 4 \to 5$.

(1) The $\mathcal{E}$-global dimension of Λ is at most 1.
(2) For all $P \in \operatorname{proj}(\mathcal{E})$ there exists an $\mathcal{E}$-admissible short exact sequence $0 \to P \to Q \to I \to 0$ with $Q \in \operatorname{proj}(\mathcal{E}) \cap \operatorname{inj}(\mathcal{E})$ and $I \in \operatorname{inj}(\mathcal{E})$. That is, the *$\mathcal{E}$-dominant dimension* of Λ is at least 1.

We now recall the following from [Pal24, Proposition 5.7] (see also [GNP, Theorem 4.3]). For a module $M \in \operatorname{mod}\Lambda$, we denote by $|M|$ the number of isomorphism classes of indecomposable direct summands of M.

Theorem 2.12. *Let $\mathcal{E}$ be an exact structure on $\operatorname{mod}\Lambda$ with enough projectives, and suppose that $(\operatorname{mod}\Lambda, \mathcal{E})$ is a 0-Auslander category. Suppose further that there exists $P \in \operatorname{mod}(\Lambda)$ such that $\operatorname{proj}(\mathcal{E}) = \operatorname{add}(P)$. Then the following are equivalent for a module $T \in \operatorname{mod}\Lambda$.*

(1) *T is $\mathcal{E}$-tilting.*
(2) *T is* maximal $\mathcal{E}$-rigid*; that is, T is $\mathcal{E}$-rigid and if $T \oplus X$ is $\mathcal{E}$-rigid, then $X \in \operatorname{add}(T)$.*
(3) *T is* complete $\mathcal{E}$-rigid*; that is, T is $\mathcal{E}$-rigid and $|T| = |P|$.*

3. Combinatorics of Auslander-Reiten quivers of type $\mathbb{A}$

In this section, we recall background information on Auslander-Reiten quivers of path algebras of type $\mathbb{A}_n$. The results of this section are well-known, and we refer to [Sch14, Section 3.1] for further details.

Fix for the duration of this section a quiver Q of type $\mathbb{A}_n$. We freely move between the language of (K-)representations of Q and of (right) KQ-modules. We adopt the standard convention of indexing the vertices of Q by $[n] = \{1, \ldots, n\}$ so that the arrows of Q are always of the form $i \to i+1$ or $i \leftarrow i+1$. For $i \in [n]$, we denote by $P(i)$ and $I(i)$ the projective cover and injective envelope of the simple representation $S(i)$ supported at the vertex i. In examples, we denote other representations by their radical filtrations.

The indecomposable representations of Q can be partitioned into sets $\mathcal{L}_1, \ldots, \mathcal{L}_n$ so that a representation M lies in $\mathcal{L}_i$ if and only if $M \cong \tau^{-m} P(i)$ for some m. There is then an embedding of the Auslander-Reiten quiver $\Gamma(Q)$ into $\mathbb{Z} \times [n] \subseteq \mathbb{R}^2$ which satisfies the following:

- The representations in $\mathcal{L}_i$ all lie on the horizontal line $y = i$ (for all $i \in [n]$).
- Every irreducible morphism is represented by an arrow pointing in the direction $(1, 1)$ or $(1, -1)$.

When we refer to $\Gamma(Q)$, we always refer to the Auslander-Reiten quiver with this embedding. An example is shown in Figure 1. Note that the embedding $\Gamma(Q)$ is unique up to horizontal shift.

Let M be an indecomposable representation. We denote by $\Sigma_\nearrow(M)$ and $\Sigma_\searrow(M)$ the rays (in $\mathbb{R}^2$) of slope 1 and -1 with basepoint corresponding to $M \in \Gamma(Q)$. Similarly, we denote by $\Sigma_\swarrow(M)$ and $\Sigma_\nwarrow(M)$ the corays of slope 1 and -1 with basepoint corresponding to $M \in \Gamma(Q)$. We say that an indecomposable representation N (seen as a vertex of $\Gamma(Q)$) lies *at the end of* e.g. $\Sigma_\nearrow(M)$ if N lies on $\Sigma_\nearrow(M)$ and there

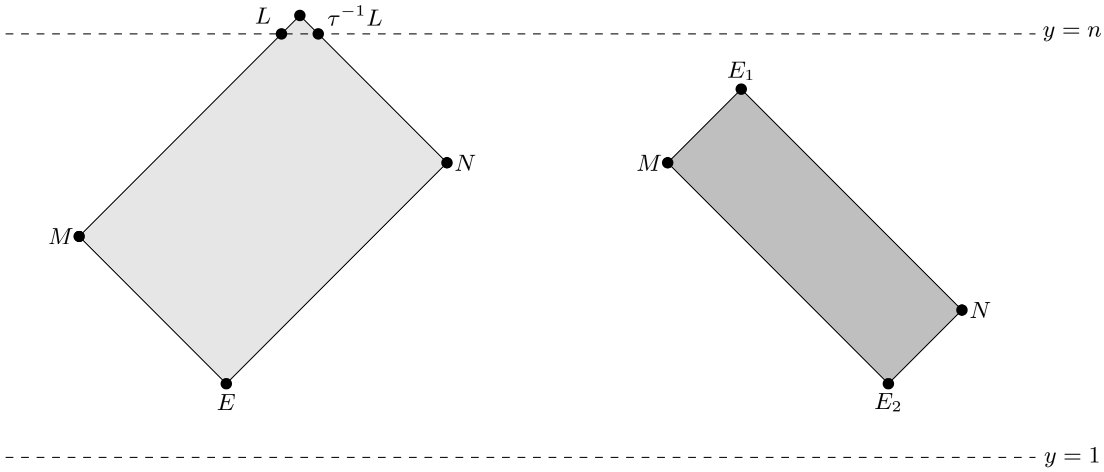

FIGURE 2. Schematic diagrams of an upper deleted (light gray) and of a non-degenerate rectangular pair (dark gray).

does not exist $L \neq N$ such that L lies on $\Sigma_{\nearrow}(N)$. For example, in Figure 1, $I(3), I(4)$, and $I(5)$ lie on $\Sigma_{\nearrow}(I(5))$, with $I(3)$ lying at the end. Note that, in the language of [Sch14, Section 3.1.4.1], we have that there is a sectional path from M to N if and only if N lies on $\Sigma_{\nearrow}(M) \cup \Sigma_{\searrow}(M)$.

We denote by $\mathcal{R}_{\to}(M)$ the set of indecomposable representations whose position in $\Gamma(Q)$ lies in the slanted rectangular region whose left boundary is $\Sigma_{\nearrow}(M) \cup \Sigma_{\searrow}(M)$. For example, in Figure 1, we have

$$\mathcal{R}_{\to}\left(\begin{smallmatrix}1\,3\\ \,2\,4\end{smallmatrix}\right) = \left\{\begin{smallmatrix}1\,3\\ \,2\,4\end{smallmatrix}, I(1), I(2), I(3), I(4)\right\}.$$

We symmetrically denote by $\mathcal{R}_{\leftarrow}(M)$ the set of indecomposable representations whose position in $\Gamma(Q)$ lies in the slanted rectangular region whose right boundary is $\Sigma_{\swarrow}(M) \cup \Sigma_{\nwarrow}(M)$. See [Sch14, Section 3.1.4.1] for additional examples.

The following terminology is useful for characterizing the existence of short exact sequences and nonzero morphisms.

Definition 3.1. Let M and N be indecomposable representations.

(1) We say that (M, N) is a *rectangular pair* if the following both hold.
   (a) The ray $\Sigma_{\nearrow}(M)$ and coray $\Sigma_{\nwarrow}(N)$ intersect and there is a module E_1 at $\Sigma_{\nearrow}(M) \cap \Sigma_{\nwarrow}(N)$.
   (b) The ray $\Sigma_{\searrow}(M)$ and coray $\Sigma_{\swarrow}(N)$ intersect and there is a module E_2 at $\Sigma_{\searrow}(M) \cap \Sigma_{\swarrow}(N)$.

(2) Suppose (M, N) is a rectangular pair. We say that (M, N) is *degenerate* if N lies on $\Sigma_{\nearrow}(M) \cup \Sigma_{\searrow}(M)$. Otherwise, we say that (M, N) is *non-degenerate*.

(3) We say that (M, N) is a *lower deleted pair* if the following both hold.
   (a) The ray $\Sigma_{\nearrow}(M)$ and coray $\Sigma_{\nwarrow}(N)$ intersect and there is a module E at $\Sigma_{\nearrow}(M) \cap \Sigma_{\nwarrow}(N)$.
   (b) Let L be the module at the end of the ray $\Sigma_{\searrow}(M)$. Then $\tau^{-1}L$ is the module at the end of the coray $\Sigma_{\swarrow}(N)$.

(4) We say that (M, N) is a *upper deleted pair* if the following both hold.
   (a) The ray $\Sigma_{\searrow}(M)$ and coray $\Sigma_{\swarrow}(N)$ intersect and there is a module E at $\Sigma_{\searrow}(M) \cap \Sigma_{\swarrow}(N)$.
   (b) Let L be the module at the end of the ray $\Sigma_{\nearrow}(M)$. Then $\tau^{-1}L$ is the module at the end of the coray $\Sigma_{\nwarrow}(N)$.

Schematic diagrams of an upper deleted pair and of a non-degenerate rectangular pair are shown in Figure 2. Note that the shape of an upper (resp. lower) deleted pair is that of a slanted rectangle with a single endpoint deleted. We defer specific examples to Example 3.6.

We now recall how one obtains the structure of morphisms and short exact sequences from the quiver $\Gamma(Q)$.

Lemma 3.2. [Sch14, Sec. 3.1.4.1] *Let M and N be indecomposable representations. Then* $\mathrm{Hom}(M,N) \cong K$ *if and only if the following equivalent conditions hold. Otherwise,* $\mathrm{Hom}(M,N) = 0$.

(1) *(M,N) is a rectangular pair.*
(2) *$N \in \mathcal{R}_{\to}(M)$.*
(3) *$M \in \mathcal{R}_{\leftarrow}(N)$.*

As a consequence of Lemma 3.2, we obtain the following.

Proposition 3.3. *Let M,N and L be indecomposable representations and suppose there are nonzero morphisms $f : M \to N$ and $g : N \to L$. Then every morphism $M \to L$ factors through both f and g.*

The next result explains the structure of short exact sequences.

Lemma 3.4. [Sch14, Section 3.1.4.3] (see also [BGMS21, Proposition 6.5]) *Let M and N be indecomposable representations.*

(1) *Suppose that (M,N) is a non-degenerate rectangular pair. Let E_1 be the module at $\Sigma_{\nearrow}(M) \cap \Sigma_{\nwarrow}(N)$ and let E_2 be the module at $\Sigma_{\searrow}(M) \cap \Sigma_{\swarrow}(N)$. Then* $\mathrm{Ext}^1(N,M) \cong K$ *and is spanned by a (diamond) short exact sequence*
$$0 \to M \to E_1 \oplus E_2 \to N \to 0.$$
(2) *Suppose that (M,N) is an upper or lower deleted pair. Let E be the module at $\Sigma_{\nearrow}(M) \cap \Sigma_{\nwarrow}(N)$ (lower case) or $\Sigma_{\searrow}(M) \cap \Sigma_{\swarrow}(N)$ (upper case). Then* $\mathrm{Ext}^1(N,M) \cong K$ *and is spanned by a short exact sequence*
$$0 \to M \to E \to N \to 0.$$
(3) *Suppose that (M,N) is not a non-degenerate rectangular pair, an upper deleted pair, or a lower deleted pair. Then* $\mathrm{Ext}^1(N,M) = 0$.

Remark 3.5. In the setting of Lemma 3.4(1), Proposition 3.3 implies that the induced compositions $M \to E_1 \to N$ and $M \to E_2 \to N$ must both be nonzero.

Example 3.6. Consider the quiver $\Gamma(Q)$ in Figure 1.

(1) There is a degenerate rectangular pair $\left(P(2), {}^{3}_{2}\right)$. There is a nonzero morphism $P(2) \to {}^{3}_{2}$ and $\mathrm{Ext}^1\left({}^{3}_{2}, P(2)\right) = 0$.
(2) There is a rectangular pair $(P(4), I(4))$. There is a nonzero morphism $P(4) \to I(4)$ and a diamond short exact sequence
$$0 \to P(4) \to I(5) \oplus 4 \to I(4).$$
In this case, we have that $\mathcal{R}_{\to}(P(4)) = \mathcal{R}_{\leftarrow}(I(4))$, and this consists of the eight representations in the rectangle whose vertices are $P(4), I(4)$, $4 = \Sigma_{\nearrow}(P(4)) \cap \Sigma_{\nwarrow}(I(4))$, and $I(5) = \Sigma_{\searrow}(P(4)) \cap \Sigma_{\swarrow}(I(4))$.
(3) There is an upper deleted pair $(4, I(3))$. There is a nonsplit short exact sequence
$$0 \to 4 \to I(4) \to I(3) \to 0$$
and $\mathrm{Hom}(4, I(3)) = 0$.

In Example 3.6(2) above, we have that $\mathcal{R}_{\to}(P(4))$ has the shape of a rectangle in the Auslander-Reiten quiver $\Gamma(Q)$. As observed at the end of [Sch14, Section 3.1.4.1], this happens precisely because $P(4)$ is projective. A consequence of this observation is the following.

Proposition 3.7. *Let $E_1 \in \mathcal{L}_1$ and $E_2 \in \mathcal{L}_n$. Suppose that there is a module M at the intersection $\Sigma_{\nwarrow}(E_1) \cap \Sigma_{\swarrow}(E_2)$ and that there is a module N at the intersection $\Sigma_{\nearrow}(E_1) \cap \Sigma_{\searrow}(E_2)$. Then the following hold.*

(1) *There exists $i \in [n]$ such that $M = P(i)$ and $N = I(i)$.*
(2) *Either $\{M,N\} = \{E_1,E_2\}$ or there is a diamond short exact sequence*
$$0 \to M \to E_1 \oplus E_2 \to N \to 0.$$

In words, Proposition 3.7 above says that any rectangle in $\Gamma(Q)$ which touches both the upper boundary $\mathcal{L}_n$ and lower boundary $\mathcal{L}_1$ must start at a projective and end at an injective. The following shows a sort of converse, namely that the rectangle connecting an indecomposable projective to the corresponding injective will always touch both the upper and lower boundary.

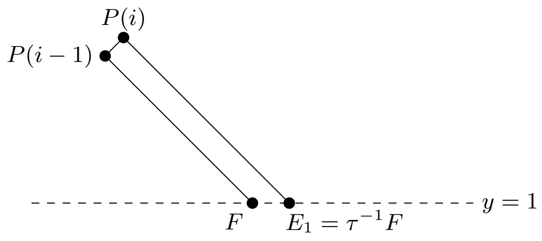

FIGURE 3. Schematic diagram of the proof of Lemma 3.8.

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Lemma 3.8. *Let $i \in [n]$. Let E_1 and E_2 be the modules lying at the ends of $\Sigma_{\searrow}(P(i))$ and $\Sigma_{\nearrow}(P(i))$, respectively. Then $E_1 \in \mathcal{L}_1$ and $E_2 \in \mathcal{L}_n$.*

Proof. We prove only that $E_1 \in \mathcal{L}_1$, the proof that $E_2 \in \mathcal{L}_n$ being similar. We proceed by induction on i. For the base case $i = 1$, we have $E_1 = P(1) \in \mathcal{L}_1$. For the induction step, let F be the module lying at the end of $\Sigma_{\searrow}(P(i-1))$ and assume $F \in \mathcal{L}_1$. We have two cases to consider.

First suppose there is an irreducible morphism $P(i) \hookrightarrow P(i-1)$. Then $P(i-1)$, and thus also F, lies on $\Sigma_{\searrow}(P(i))$. This implies that $E_1 = F \in \mathcal{L}_1$.

Now suppose instead that there is an irreducible morphism $P(i-1) \hookrightarrow P(i)$. Thus $P(i)$ lies in the direction $(1, 1)$ from $P(i-1)$. By Lemma 3.2, this means $\mathrm{Hom}(P(i), F) = 0$. Then since $\mathrm{Hom}(P(i-1), F) \neq 0$ and there is an injection $P(i-1) \hookrightarrow P(i)$, this means F is not injective. It follows that $E_1 = \tau^{-1}F \in \mathcal{L}_1$. See Figure 3 for an illustration. $\square$

Remark 3.9. For $1 < i < n$ and E_1, E_2 as in Lemma 3.8, the well-known description of $\mathcal{R}_{\rightarrow}(P(i))$ implies that there is a diamond short exact sequence

$$0 \to P(i) \to E_1 \oplus E_2 \to I(i) \to 0.$$

See e.g. [Sch14, Section 3.1.4].

We conclude this section by tabulating the following consequence of Lemma 3.4.

Proposition 3.10. *Let M be an indecomposable representation.*

(1) *There exists an indecomposable N such that $\mathrm{Ext}^1(M, N)$ contains a diamond exact sequence if and only if $M \notin \mathrm{proj}(KQ) \cup \mathcal{L}_1 \cup \mathcal{L}_n$.*
(2) *There exists an indecomposable N such that $\mathrm{Ext}^1(N, M)$ contains a diamond exact sequence if and only if $M \notin \mathrm{inj}(KQ) \cup \mathcal{L}_1 \cup \mathcal{L}_n$.*

4. ALMOST RIGID MODULES

In this section, we introduce an exact structure $\mathcal{E}_\diamond$ on $\mathrm{mod}\, KQ$ (Q a quiver of type $\mathbb{A}_n$). We then prove that the classes of almost rigid modules and basic $\mathcal{E}_\diamond$-rigid modules coincide (Corollary 4.4). We further show that $(\mathrm{mod}\,\Lambda, \mathcal{E}_\diamond)$ is a 0-Auslander category (Theorem 4.9), a consequence of which is the fact that the maximal almost rigid modules are precisely the $\mathcal{E}_\diamond$-tilting modules (Corollary 4.10). We conclude by discussing implications for the mutation theory of maximal almost rigid modules in Section 4.3.

4.1. **The diamond exact structure.** Let Q be a quiver of type $\mathbb{A}_n$ with Auslander-Reiten quiver $\Gamma(Q)$. We recall the partition $\mathcal{L}_1, \ldots, \mathcal{L}_n$ of the indecomposable representations introduced in Section 3. We note that the representations in $\mathcal{L}_1 \cup \mathcal{L}_n$ are precisely those for which there is at most one incoming arrow and at most one outgoing arrow in $\Gamma(Q)$. We consider the following definition.

Definition 4.1. The *diamond exact structure* on $\mathrm{mod}\, KQ$ is $\mathcal{E}_\diamond := \mathcal{F}_{\mathcal{L}_1 \cup \mathcal{L}_n}$.

We also need the following, which is an immediate consequence of Lemma 3.2.

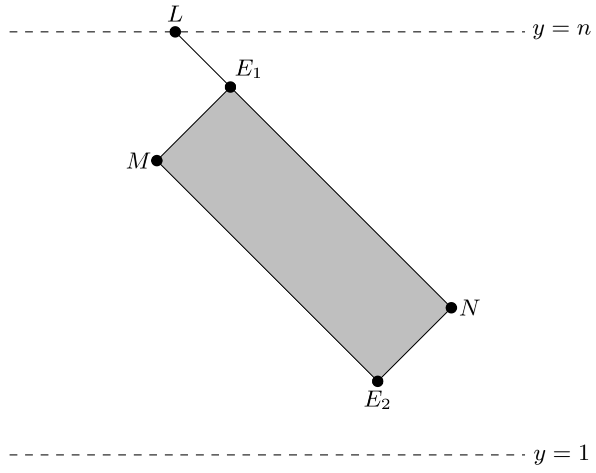


FIGURE 4. Schematic diagrams of the proof of Proposition 4.3.

Lemma 4.2. *Let N be an indecomposable module and let $L \in \mathcal{L}_1 \cup \mathcal{L}_n$. Then $\operatorname{Hom}(L, N) \neq 0$ if and only if L lies at the end of $\Sigma_{\swarrow}(N)$ or $\Sigma_{\nwarrow}(N)$.*

Note in particular that Lemma 4.2 implies that there exist at most two elements of $\mathcal{L}_1 \cup \mathcal{L}_n$ admitting nonzero morphisms to a given X. Moreover, the corresponding rectangular pairs (L, X) will always be degenerate.

We now give a characterization of $\mathcal{E}_\diamond(N, M)$ in the case where M and N are indecomposable. As an additive bifunctor on $\operatorname{mod}\Lambda$, this uniquely determines the exact structure $\mathcal{E}_\diamond$. More precisely, we show that the nonsplit $\mathcal{E}_\diamond$-exact sequences with indecomposable endterms are precisely the diamond exact sequences. The justifies the name diamond exact structure.

Proposition 4.3. *Let M and N be indecomposable. Then*

$$\mathcal{E}_\diamond(N, M) = \{\eta \in \operatorname{Ext}^1(N, M) \mid \eta \textit{ is split or a diamond exact sequence}\}.$$

Proof. Let $\eta = (0 \to M \to E \to N \to 0) \in \operatorname{Ext}^1(N, M)$ be a non-split exact sequence. We suppose first that E is not indecomposable. Then, by Lemma 3.4, we have $E \cong E_1 \oplus E_2$ with E_1 and E_2 both indecomposable. Schematically, the modules E_1 and E_2 lie on the corays $\Sigma_{\swarrow}(N)$ and $\Sigma_{\nwarrow}(N)$, see the right diagram of Figure 2. Now let $L \in \mathcal{L}_1 \cup \mathcal{L}_n$ such that $\operatorname{Hom}(L, N) \neq 0$. By Lemma 4.2, this means L lies at the end of $\Sigma_{\swarrow}(N)$ or $\Sigma_{\nwarrow}(N)$, see Figure 4. Lemma 3.2 and Proposition 3.3 thus imply that every morphism $L \to N$ factors through the map $E_1 \oplus E_2 \to N$ comprising η, see Figure 4. We conclude that $\eta \in \mathcal{F}_{\mathcal{L}_1 \cup \mathcal{L}_n} = \mathcal{E}_\diamond$.

For the converse, suppose that E is indecomposable. By Lemma 3.4, this means (N, M) forms either a lower deleted pair or an upper deleted pair. We consider the case of an upper deleted pair, the other case being similar. Now let X be the module which lies at the end of the coray $\Sigma_{\nwarrow}(N)$. (We have $X = \tau^{-1}L$ in the schematic on the left side of Figure 2.) By Lemma 3.2, we have $\operatorname{Hom}(X, N) \neq 0$ and $\operatorname{Hom}(X, E) = 0$. Thus the sequence

$$0 \to \operatorname{Hom}(X, M) \to \operatorname{Hom}(X, E) \to \operatorname{Hom}(X, N) \to 0$$

is not exact. Since $X \in \mathcal{L}_n$ by construction, we conclude that $\eta \notin \mathcal{E}_\diamond$. $\square$

A consequence of Proposition 4.3 is the following.

Corollary 4.4. *Let $T \in \operatorname{rep} Q$. Then T is almost rigid if and only if T is $\mathcal{E}_\diamond$-rigid.*

Proof. Suppose that T is almost rigid. The only nonzero extensions between two indecomposable summands of T are of the form $\xi : 0 \to M \to E \to N \to 0$ with E indecomposable. Since $\xi \notin \mathcal{E}_\diamond$, then $\mathcal{E}_\diamond(T, T) = 0$.

On the other hand, if $\mathcal{E}_\diamond(T, T) = 0$, then the only non-split extensions between two indecomposable summands of T are of the form $\xi : 0 \to M \to E \to N \to 0$ where E cannot be decomposable. Therefore, T is almost rigid. $\square$

We also obtain the following description of $\mathcal{E}$-projectives and $\mathcal{E}$-injectives.

Corollary 4.5. (a) $\mathrm{proj}(\mathcal{E}_\diamond) = \mathrm{add}(\mathrm{proj}\, KQ \cup \mathcal{L}_1 \cup \mathcal{L}_n)$
(b) $\mathrm{inj}(\mathcal{E}_\diamond) = \mathrm{add}(\mathrm{inj}\, KQ \cup \mathcal{L}_1 \cup \mathcal{L}_n)$
(c) $\mathrm{proj}\text{-}\mathrm{inj}(\mathcal{E}_\diamond) = \mathrm{add}(\mathcal{L}_1 \cup \mathcal{L}_n)$

Proof. Parts (a) and (b) follow directly from Proposition 2.6. Since the set of projective-injectives is the intersection of the two sets in (a) and (b), part (c) follows from the fact that $\mathrm{proj}\, KQ \cap \mathrm{inj}\, KQ$ is empty unless all arrows in Q point in the same direction, and in that case, the only projective-injective KQ-module lies in $\mathcal{L}_1 \cup \mathcal{L}_n$. □

4.2. **0-Auslander property.**
We now turn our attention to proving that the category $(\mathrm{mod}\, KQ, \mathcal{E}_\diamond)$ is 0-Auslander.

Proposition 4.6. *$\mathcal{E}_\diamond$ has dominant dimension 1.*

Proof. Let M be an indecomposable $\mathcal{E}_\diamond$-projective. If $M \in \mathcal{L}_1 \cup \mathcal{L}_n$, then M is $\mathcal{E}_\diamond$-injective by Corollary 4.5, so there is nothing to show. Thus Proposition 2.6 implies that we only need to consider $M = P(i)$ for some $1 < i < n$. Now let E_1 and E_2 be the modules lying at the ends of $\Sigma_\searrow(M)$ and $\Sigma_\nearrow(M)$. By Lemma 3.8, we have that $E_1 \in \mathcal{L}_1$ and $E_2 \in \mathcal{L}_n$, and so $E_1 \oplus E_2$ is $\mathcal{E}_\diamond$-projective-injective by Corollary 4.5. Remark 3.9 and Proposition 4.3 then say that there is an $\mathcal{E}_\diamond$-exact sequence

$$0 \to P(i) \to E_1 \oplus E_2 \to I(i) \to 0.$$

This proves the result. □

Proposition 4.7. *$\mathcal{E}_\diamond$ has global dimension 1.*

Proof. Let M be an indecomposable which is not $\mathcal{E}_\diamond$-projective. Equivalently, $M \notin \mathrm{proj}(KQ) \cup \mathcal{L}_1 \cup \mathcal{L}_2$, by Corollary 4.5. Let E_1 be the module at the end of the coray $\Sigma_\swarrow(M)$ and let E_2 be the module at the end of the coray $\Sigma_\nwarrow(M)$. Since $M \notin \mathrm{proj}(KQ) \cup \mathcal{L}_1 \cup \mathcal{L}_2$, we have that M, E_1, and E_2 are all distinct. We now have two cases to consider.

Suppose first that there is a module N at the intersection $\Sigma_\nwarrow(E_1) \cap \Sigma_\swarrow(E_2)$. In this case, (N, M) is a nondegenerate rectangular pair. By Lemma 3.4 and Proposition 4.3, this means there is an $\mathcal{E}_\diamond$-exact sequence

$$0 \to N \to E_1 \oplus E_2 \to M \to 0.$$

Now if $E_1, E_2 \in \mathcal{L}_1 \cup \mathcal{L}_n$, then $N \in \mathrm{proj}(KQ)$ by Proposition 3.7. Otherwise, without loss of generality we have that $E_1 \in \mathrm{proj}(KQ)$. Now Proposition 3.3 implies that the composite map $N \to E_1 \to M$ is nonzero. Since KQ is hereditary, it follows that $N \in \mathrm{proj}(KQ)$ and thus that N is $\mathcal{E}_\diamond$-projective and hence $\mathrm{pd}_{\mathcal{E}_\diamond} M = 1$.

Now suppose that there is no module at $\Sigma_\nwarrow(E_1) \cap \Sigma_\swarrow(E_2)$. Choose nonzero morphisms $g_1 : E_1 \to M$ and $g_2 : E_2 \to M$ (these exist and are unique up to scalar multiplication by Lemma 3.2), and let $p : P \to M$ be the (standard) projective cover of M. Then there is an exact sequence

$$\eta = (0 \to \ker f \to P \oplus E_1 \oplus E_2 \xrightarrow{f} M \to 0), \tag{1}$$

where $f^\top = \begin{bmatrix} p & g_1 & g_2 \end{bmatrix}$. (See Example 4.8 for an example of why adding the direct summand P is necessary.) We claim that η is $\mathcal{E}_\diamond$-exact. To see this, let $X \in \mathcal{L}_1 \cup \mathcal{L}_n$. By Remark 2.5, it suffices to show that $\mathrm{Hom}(X, f)$ is surjective. We consider the case where $X \in \mathcal{L}_1$, the case where $X \in \mathcal{L}_n$ being similar. Now if $\mathrm{Hom}(X, M) = 0$ there is nothing to show. Thus suppose that $\mathrm{Hom}(X, M) \neq 0$. Then Lemma 4.2 implies that X lies at the end of $\Sigma_\swarrow(M)$; i.e., that $X = E_1$. Then any morphism $X \to M$ must be a scalar multiple of $g_1 = f|_{E_1}$, and thus factors through f. We conclude that $\mathrm{Hom}(X, f)$ is surjective.

Now let N be an indecomposable direct summand of $\ker f$. We will show that N is $\mathcal{E}_\diamond$-projective.

First note that if $\mathrm{Ext}^1(M, N) = 0$, then N is a direct summand of $P \oplus E_1 \oplus E_2$ and we are done. Moreover, if there is a nonzero map from N to P, then N must be projective.

It remains to consider the case where $\mathrm{Ext}^1(M, N) \neq 0$ and $\mathrm{Hom}(N, P) = 0$. We will show that this assumption leads to a contradiction. Suppose that $\mathrm{Hom}(N, E_1) \neq 0$, the case where $\mathrm{Hom}(N, E_2) \neq 0$ being similar. Then Lemmas 3.2 and 3.4 and Proposition 3.3 yield two possibilities.

Suppose first that $\mathrm{Hom}(N, M) \neq 0$. For $i \in \{1, 2\}$, let $h_i : N \to E_i$ be the morphism induced by the inclusion $N \subseteq \ker f \subseteq P \oplus E_1 \oplus E_2$. Since we have supposed $\mathrm{Hom}(N, M) \neq 0$ and $\mathrm{Hom}(N, E_1) \neq 0$,

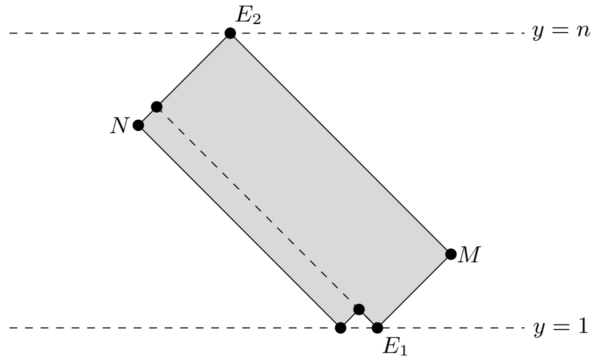


FIGURE 5. Schematic diagram of the proof of Proposition 4.7.

Proposition 3.3 implies that $g_1 \circ h_1 \neq 0$. Moreover, the assumption that $\mathrm{Hom}(N, P) = 0$ implies that $0 = f|_N = g_1 \circ h_1 + g_2 \circ h_2$. We conclude that also $g_2 \circ h_2 \neq 0$, so in particular $\mathrm{Hom}(N, E_2) \neq 0$. But then (N, M) must be a rectangular pair with N lying at the intersection $\Sigma_{\swarrow}(E_1) \cap \Sigma_{\nwarrow}(E_2)$. This is a contradiction to our assumption above.

It remains to consider the case where $\mathrm{Hom}(N, M) = 0$. By Lemmas 3.2 and 3.4, this means that (N, M) is a lower deleted pair, see Figure 5. From this, we observe that there is a module at the intersection $\Sigma_{\swarrow}(E_1) \cap \Sigma_{\nwarrow}(E_2)$. This again is a contradiction to our assumption. □

Example 4.8. Consider the representation $M = \begin{smallmatrix} 1\ 3 \\ 2\ 4 \\ 5 \end{smallmatrix}$ in Figure 1. The representation at the end of $\Sigma_{\swarrow}(M)$ is $P(1)$ and the representation at the end of $\Sigma_{\nwarrow}(M)$ is $P(5)$. In this case, we have that the morphism $P(1) \oplus P(5) \to M$ is not an $\mathcal{E}_\diamond$-admissible epimorphism. (In fact, it is not even an epimorphism.) In the proof of Proposition 4.7, we construct a $\mathcal{E}_\diamond$-admissible epimorphism $P(3) \oplus P(1) \oplus P(1) \oplus P(5) \to M$ by adding the (standard) projective cover as a direct summand. Note that, while this is indeed an admissible epimorphism, it is not a minimal cover by the relative projectives of $\mathcal{E}_\diamond$. That is, it is possible that the exact sequence (1) admits a nonzero direct summand N of $\ker f$ for which the induced inclusion $N \to P \oplus E_1 \oplus E_2$ is split.

Theorem 4.9. $(\mathrm{mod}\, KQ, \mathcal{E}_\diamond)$ *is a 0-Auslander category.*

Proof. This follows from Propositions 4.7 and 4.6. □

Corollary 4.10. *Let $T \in \mathrm{mod}\, KQ$. Then T is maximal almost rigid if and only if T is a basic $\mathcal{E}_\diamond$-tilting module.*

4.3. **Mutation of MAR modules.** It was shown in [BGMS21, Theorem E] that the maximal almost rigid modules form a poset isomorphic to a Cambrian lattice, where the covering relation between two MAR modules T, T' is given as follows. We have $T \lessdot T'$ if there exist indecomposable summands M of T and M' of T' such that $T/M \cong T'/M'$ and there exists a non-split short exact sequence

$$0 \longrightarrow M \xrightarrow{f} E \oplus E' \longrightarrow M' \longrightarrow 0\,,$$

with E, E' indecomposable summands of T/M and f a minimal $\mathrm{add}\, T/M$ approximation. The unique minimal element in this lattice is the unique MAR module that contains every indecomposable projective as a direct summand, and the unique maximal element is the one that contains every indecomposable injective.

In particular, the covering relation corresponds to the usual notion of mutation, and the Hasse diagram is the (oriented) exchange graph.

Because of [BGMS21, Theorem C], the MAR modules are in bijection with the triangulations of a polygon with $n+1$ vertices and mutation has a combinatorial interpretation as the flip of the diagonal corresponding to the indecomposable M that is exchanged in the mutation. The boundary edges of the polygon are not mutable. They correspond to the indecomposable modules in the top and bottom τ-orbits in the Auslander-Reiten quiver. These modules are precisely the indecomposable relative projective-injectives in our exact structure $\mathcal{E}_\diamond$ and therefore required in every MAR module. On the other hand, each of the remaining summands is mutable.

In our setting, we also recover the above mutation result from the theory of 0-Auslander algebras. Indeed, Theorem 1.3 in [GNP] implies that for every MAR module $T = \overline{T} \oplus X$, with X indecomposable and non-projective-injective, there exists a unique indecomposable $Y \ncong X$ such that $\overline{T} \oplus Y$ is an MAR module and there is precisely one exchange triangle

$$\text{either} \quad X \rightarrowtail^{i} F \xrightarrow{p} Y \quad \text{or} \quad Y \rightarrowtail^{i'} F \xrightarrow{p'} X$$

where $F, F' \in \operatorname{add} \overline{T}$ and i, i', p, p' are $\operatorname{add} \overline{T}$-approximations.

The order relation in [BGMS21] is recovered by the uniqueness of the exchange triangle. The fact that the middle term F or F' has at most two indecomposable summands follows from the structure of the Auslander-Reiten quiver in type $\mathbb{A}$. To show that the middle has exactly two indecomposable summands, let us assume that F is indecomposable. Then every non-trivial extension between X and Y is indecomposable, because the space of extensions is at most one-dimensional in type $\mathbb{A}$. On the other hand, $T = \overline{T} \oplus X$ and $\overline{T} \oplus Y$ are almost rigid. However, then $\overline{T} \oplus X \oplus Y$ would be almost rigid, which is a contradiction to the maximality of T.

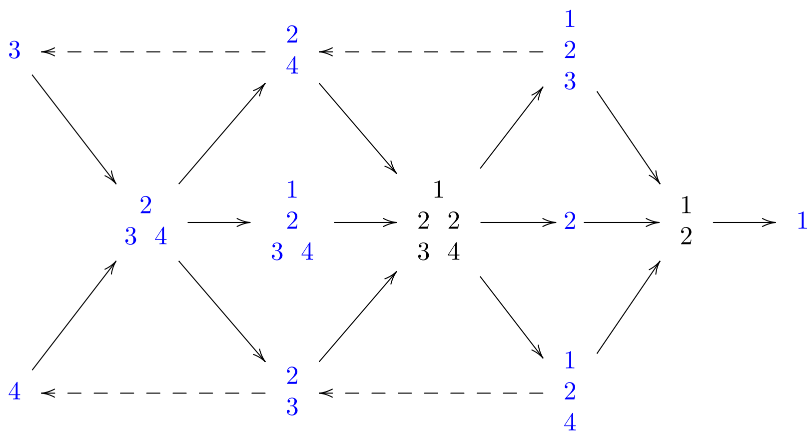
5. Generalizations

5.1. **Type $\mathbb{D}$ quivers.** In Proposition 4.3 we showed that the collection of diamond exact sequences defines additive functor, which describes the exact structure $\mathcal{E}_\diamond$ on indecomposables. We illustrate in this section an example of a quiver of type $\mathbb{D}$ which shows that the diamond exact sequences need not be closed under pushout of morphisms, thus they do *not* define an exact structure for type $\mathbb{D}$ quivers. We propose to replace the diamond condition in this case by requesting three middle terms. This yields an exact structure, which we show to be not hereditary. Note that a different generalization of almost rigid modules to type $\mathbb{D}$ were given in the recent preprint [CZ].

Example 5.1. Let Q be the quiver

$$\begin{tikzcd} & & 3 \\ 1 \arrow[r, "\alpha"] & 2 \arrow[ur, "\beta"] \arrow[dr, "\gamma"'] & \\ & & 4 \end{tikzcd}$$

Its Auslander-Reiten quiver is described as follows:



We consider the diamond exact sequence ξ :

$$0 \longrightarrow {\begin{smallmatrix} & 2 & \\ 3 & & 4 \end{smallmatrix}} \longrightarrow {\begin{smallmatrix} 2 \\ 4 \end{smallmatrix}} \oplus {\begin{smallmatrix} & 1 & \\ & 2 & \\ 3 & & 4 \end{smallmatrix}} \longrightarrow {\begin{smallmatrix} 1 \\ 2 \\ 4 \end{smallmatrix}} \longrightarrow 0$$

By taking the push-out of ξ along $f : \begin{smallmatrix} & 2 & \\ 3 & & 4 \end{smallmatrix} \to \begin{smallmatrix} 2 \\ 3 \end{smallmatrix}$, we obtain the short exact sequence:

$$0 \longrightarrow \begin{smallmatrix} 2 \\ 3 \end{smallmatrix} \longrightarrow \begin{smallmatrix} & 1 & \\ 2 & & 2 \\ 3 & & 4 \end{smallmatrix} \longrightarrow \begin{smallmatrix} 1 \\ 2 \\ 4 \end{smallmatrix} \longrightarrow 0$$

which is not a diamond exact sequence. Therefore, the diamond exact sequences do not form an exact structure for type $\mathbb{D}$ quivers.

In this case, one could consider the exact structure $\mathcal{E}_{\diamondsuit} := \mathcal{F}_{\mathcal{X}}$ where $\mathcal{X} = \left\{ 3, \begin{smallmatrix} & 2 & \\ 3 & & 4 \end{smallmatrix}, 4, \begin{smallmatrix} 2 \\ 4 \end{smallmatrix}, \begin{smallmatrix} 2 \\ 3 \end{smallmatrix}, \begin{smallmatrix} & 1 & \\ & 2 & \\ 3 & & 4 \end{smallmatrix}, \begin{smallmatrix} 1 \\ 2 \\ 3 \end{smallmatrix}, \begin{smallmatrix} 1 \\ 2 \\ 4 \end{smallmatrix}, 2, 1 \right\}$. The only Auslander-Reiten sequences in $\mathcal{E}_{\diamondsuit}$ are the ones having middle terms with three indecomposables:

$$0 \longrightarrow \begin{smallmatrix} & 2 & \\ 3 & & 4 \end{smallmatrix} \longrightarrow \begin{smallmatrix} 2 \\ 4 \end{smallmatrix} \oplus \begin{smallmatrix} & 1 & \\ & 2 & \\ 3 & & 4 \end{smallmatrix} \oplus \begin{smallmatrix} 2 \\ 3 \end{smallmatrix} \longrightarrow \begin{smallmatrix} & 1 & \\ 2 & & 2 \\ 3 & & 4 \end{smallmatrix} \longrightarrow 0$$

and

$$0 \longrightarrow \begin{smallmatrix} & 1 & \\ 2 & & 2 \\ 3 & & 4 \end{smallmatrix} \longrightarrow \begin{smallmatrix} 1 \\ 2 \\ 3 \end{smallmatrix} \oplus 2 \oplus \begin{smallmatrix} 1 \\ 2 \\ 4 \end{smallmatrix} \longrightarrow \begin{smallmatrix} 1 \\ 2 \end{smallmatrix} \longrightarrow 0$$

are in $\mathcal{E}_{\diamondsuit}$.

We observe that one can construct the following $\mathcal{E}_{\diamondsuit}$-projective resolution of the indecomposable $\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}$:

$$0 \longrightarrow \begin{smallmatrix} & 2 & \\ 3 & & 4 \end{smallmatrix} \longrightarrow \begin{smallmatrix} 2 \\ 4 \end{smallmatrix} \oplus \begin{smallmatrix} & 1 & \\ & 2 & \\ 3 & & 4 \end{smallmatrix} \oplus \begin{smallmatrix} 2 \\ 3 \end{smallmatrix} \longrightarrow \begin{smallmatrix} 1 \\ 2 \\ 3 \end{smallmatrix} \oplus 2 \oplus \begin{smallmatrix} 1 \\ 2 \\ 4 \end{smallmatrix} \longrightarrow \begin{smallmatrix} 1 \\ 2 \end{smallmatrix} \longrightarrow 0$$

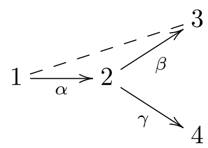
Therefore, $\mathrm{pd}_{\mathcal{E}_{\diamondsuit}}(\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}) = 2$ and so $(\mathrm{rep}\, Q, \mathcal{E}_{\diamondsuit})$ is not a *0-Auslander category*. It is easy to see that $3 \oplus \begin{smallmatrix} & 2 & \\ 3 & & 4 \end{smallmatrix} \oplus 4 \oplus \begin{smallmatrix} 2 \\ 4 \end{smallmatrix} \oplus \begin{smallmatrix} 2 \\ 3 \end{smallmatrix} \oplus \begin{smallmatrix} & 1 & \\ & 2 & \\ 3 & & 4 \end{smallmatrix} \oplus \begin{smallmatrix} 1 \\ 2 \\ 3 \end{smallmatrix} \oplus \begin{smallmatrix} 1 \\ 2 \\ 4 \end{smallmatrix} \oplus 2 \oplus 1 \oplus \begin{smallmatrix} 1 \\ 2 \end{smallmatrix}$ is *maximal almost rigid* in the sense of Definition 1.1 (replacing diamond exact by having three middle terms), but it is *not* $\mathcal{E}_{\diamondsuit}$-tilting since it contains a module of $\mathcal{E}_{\diamondsuit}$-projective dimension 2.

5.2. **Gentle algebras.** In this section, we give an example that illustrates how our results may generalize to the much more general setting of gentle algebras.

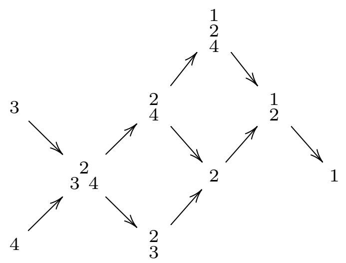
It is shown in [OPS, BCS21] that gentle algebras are in bijection with surface dissections. In [BCSGS], the concept of maximal almost rigid modules is generalized to gentle algebras, and the authors prove that many of the results generalize from type $\mathbb{A}$ to the gentle setting. For example the maximal almost rigid modules over a gentle algebra are in bijection with permissible triangulations of its surface [BCSGS, Theorem 1.2].

For the gentle algebra in the following example, analogs of items (1) - (5) of our Theorem 1.2 hold, but the 0-Auslander property does not.

Example 5.2. Let A be given by the quiver



bound by the relation $\alpha\beta$. Its Auslander-Reiten quiver is



Let $\mathcal{X} = \left\{ 3, 4, \begin{smallmatrix} & 2 & \\ 3 & & 4 \end{smallmatrix}, \begin{smallmatrix} 1 \\ 2 \\ 4 \end{smallmatrix}, \begin{smallmatrix} 2 \\ 4 \end{smallmatrix}, \begin{smallmatrix} 2 \\ 3 \end{smallmatrix}, 1 \right\}$. Then the direct sum of the elements of $\mathcal{X}$ is a maximal almost rigid A-module.

Since the are only finitely many short exact sequences with indecomposable endterms, we can check by hand that $\mathcal{F}_{\mathcal{X}}$ contains only short exact sequences with decomposable middle term. Moreover, it is easy to see that the relative global dimension of $\operatorname{mod} A$ is 1. Indeed the only indecomposables that are not $\mathcal{F}_{\mathcal{X}}$-projective are the modules $\begin{smallmatrix}2\end{smallmatrix}$ and $\begin{smallmatrix}1\\2\end{smallmatrix}$ and their $\mathcal{F}_{\mathcal{X}}$-projective resolutions are

$$0 \longrightarrow \begin{smallmatrix} & 2 & \\ 3 & & 4\end{smallmatrix} \longrightarrow \begin{smallmatrix}2\\4\end{smallmatrix} \oplus \begin{smallmatrix}2\\3\end{smallmatrix} \longrightarrow \begin{smallmatrix}2\end{smallmatrix} \longrightarrow 0 \qquad \text{and} \qquad 0 \longrightarrow \begin{smallmatrix} & 2 & \\ 3 & & 4\end{smallmatrix} \longrightarrow \begin{smallmatrix}1\\2\\4\end{smallmatrix} \oplus \begin{smallmatrix}2\\3\end{smallmatrix} \longrightarrow \begin{smallmatrix}1\\2\end{smallmatrix} \longrightarrow 0\,.$$

Furthermore, the relative projective-injective modules are $\left\{ \begin{smallmatrix}3\end{smallmatrix}, \begin{smallmatrix}4\end{smallmatrix}, \begin{smallmatrix}2\\3\end{smallmatrix}, \begin{smallmatrix}1\\2\\4\end{smallmatrix}, \begin{smallmatrix}1\end{smallmatrix} \right\}$.

However, the category is not 0-Auslander. Indeed the only $\mathcal{F}_{\mathcal{X}}$-projectives that are not $\mathcal{F}_{\mathcal{X}}$-projective-injective are $\begin{smallmatrix} & 2 & \\ 3 & & 4\end{smallmatrix}$ and $\begin{smallmatrix}2\\4\end{smallmatrix}$, and these modules admit the following short exact sequences

$$0 \longrightarrow \begin{smallmatrix} & 2 & \\ 3 & & 4\end{smallmatrix} \longrightarrow \begin{smallmatrix}2\\3\end{smallmatrix} \oplus \begin{smallmatrix}1\\2\\4\end{smallmatrix} \longrightarrow \begin{smallmatrix}1\\2\end{smallmatrix} \longrightarrow 0 \qquad \text{and} \qquad 0 \longrightarrow \begin{smallmatrix}2\\4\end{smallmatrix} \longrightarrow \begin{smallmatrix}1\\2\\4\end{smallmatrix} \longrightarrow \begin{smallmatrix}1\end{smallmatrix} \longrightarrow 0$$

whose middle terms are $\mathcal{F}_{\mathcal{X}}$-projective-injective and whose endterms are $\mathcal{F}_{\mathcal{X}}$-injective. The problem is that the last sequence is not $\mathcal{F}_{\mathcal{X}}$-admissible, because $S(1) \in \mathcal{X}$ but $\operatorname{Hom}(S(1), -)$ is not exact on this sequence. Thus this category is not 0-Auslander.

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Annexe B

Deuxième article

Référence

B. Dequêne, **S. Roy**, *Exact structures and maximal canonically Jordan recoverable subcategories for modules over type $\mathbb{A}$ algebras*, arXiv :2509.25012 (2025).

EXACT STRUCTURES AND MAXIMAL CANONICALLY JORDAN RECOVERABLE SUBCATEGORIES FOR MODULES OVER TYPE A ALGEBRAS

BENJAMIN DEQUÊNE AND SUNNY ROY

Abstract. On one hand, exact structures were introduced by D. Quillen in the '70s. They can be defined as collections of short exact sequences in a fixed abelian category satisfying additional properties.

On the other hand, in a recent work, A. Garver, R. Patrias, and H. Thomas introduced Jordan recoverability. Given a bounded quiver (Q, R), a full additive subcategory of $\mathrm{rep}(Q, R)$ is said to be Jordan recoverable if any $X \in \mathscr{C}$ can be recovered, up to isomorphism, from the Jordan form of its generic nilpotent endomorphisms. Such a subcategory $\mathscr{C}$ is said to be canonically Jordan recoverable if, moreover, there exists a precise algebraic procedure that allows one to get back $X \in \mathscr{C}$ from that same Jordan form data.

We introduce a new family of operators, called Gen-Sub operators $\mathrm{GS}_{\mathcal{E}}$, parametrized by the exact structures $\mathcal{E}$ of abelian categories. After showing some properties of those operators in hereditary abelian categories, by focusing on the setting of modules over type A quivers endowed with the diamond exact structure $\mathcal{E}_\diamond$, we establish that the maximal canonically Jordan recoverable subcategories are precisely of the form $\mathrm{GS}_{\mathcal{E}_\diamond}(T)$ for some tilting object T.

Contents

Date: April 29, 2026.

1. Introduction

1.1. **Exact structures on hereditary abelian categories.** Fix $\mathscr{A}$ an abelian category. An ***exact structure*** on an abelian category $\mathscr{A}$ is a collection $\mathcal{E}$ of short exact sequences of the form

$$0 \longrightarrow A \overset{f}{\rightarrowtail} B \overset{g}{\twoheadrightarrow} C \longrightarrow 0$$

where f is an $\mathcal{E}$-monomorphism, g is an $\mathcal{E}$-epimorphism, and the sequence adheres to axioms ensuring closure under specific categorical operations (see Definition 3.2 for details). Originally introduced by D. Quillen in [Qui73], they provide an axiomatic framework that extends the utility of homological algebra techniques to a context constrained by a predefined class of short exact sequences. This framework is notably versatile, allowing exact structures to be characterized in multiple ways: as subfunctors of the bifunctor $\mathrm{Ext}^1_{\mathscr{A}}$, through projective objects relative to the chosen structure, or via the inclusion of specific Auslander-Reiten sequences.

A category $\mathscr{A}$ is ***hereditary*** if it satisfies $\mathrm{Ext}^i_{\mathscr{A}}(X, Y) = 0$ for all objects $X, Y \in \mathscr{A}$ and all $i \geq 2$. Vanishing of higher extensions simplifies homological analysis, as all extensions are captured by Ext^1. In this paper, we focus on hereditary abelian categories, and, more specifically, on the category $\mathrm{rep}(Q)$ of finite-dimensional representations of some ADE Dynkin type quiver Q. In this setting, exact structures enable us to control which sequences are deemed "admissible", thereby isolating subcategories with distinct algebraic and combinatorial features.

While the standard maximal exact structure on an abelian category encompasses all short exact sequences, alternative structures, such as the ***diamond exact structure*** $\mathcal{E}_\diamond$ on $\mathrm{rep}(Q)$, introduced in [BHRS24] (see Definition 3.4), allows us to refine our focus to sequences that reflect the combinatorial properties of type A quivers.

A pivotal concept in this work is the ***Gen-Sub operator*** $\mathrm{GS}_\mathcal{E}$, which, given an exact structure $\mathcal{E}$ and a subcategory $\mathscr{C} \subseteq \mathscr{A}$, constructs the smallest subcategory containing $\mathscr{C}$ that is closed under operations such as taking $\mathcal{E}$-subobjects and $\mathcal{E}$-quotients (see Section 3.2 for a formal definition). In the context of hereditary abelian categories, $\mathrm{GS}_\mathcal{E}$ iteratively builds subcategories by incorporating terms from $\mathcal{E}$-exact sequences, offering a systematic method to generate subcategories with stability properties.

A subcategory $\mathscr{C} \subseteq \mathscr{A}$ will be called ***$\mathcal{E}$-adapted*** if it behaves well with respect to the exact structure $\mathcal{E}$, allowing us to relate classical constructions to their relative counterparts. More precisely, $\mathscr{C}$ is $\mathcal{E}$-adapted if every short exact sequence in $\mathscr{C}$

is $\mathcal{E}$-exact, that is, it belongs to the exact structure $\mathcal{E}$. A formal definition will be given later (see Definition 3.15).

1.2. **Canonical Jordan recoverability.** A. Garver, R. Patrias, and H. Thomas [GPT23] introduced the notion of Jordan recoverability.

Fix $\mathbb{K}$ an algebraically closed field. Consider (Q, R) a finite connected bounded quiver. Let E be a finite-dimensional representation of Q over $\mathbb{K}$. We study the set $\mathsf{NEnd}(E)$ of nilpotent endomorphisms of E. Any $N \in \mathsf{NEnd}(E)$ induces, at each vertex q of Q, a nilpotent endomorphism N_q of E_q. We can extract from N a sequence of integer partitions $\lambda_q \vdash \dim(E_q)$ from the Jordan block sizes of the Jordan form of each N_q. Write $\mathbf{JF}(N) = (\lambda_q)_{q \in Q_0}$. There exists an open dense set (for the Zariski topology) of $\mathsf{NEnd}(E)$ on which $\mathbf{JF}$ is constant. Denote by $\mathrm{GenJF}(E)$ this constant, and we refer to it as the ***generic Jordan form*** of E.

The map GenJF is an invariant on the representations of (Q, R), but not a complete one. A full subcategory $\mathscr{C}$, closed under sums and summands, is said to be ***Jordan recoverable*** whenever GenJF is complete on representations in $\mathscr{C}$.

Usually, checking whether a subcategory is Jordan recoverable is not an easy task. Under some assumptions, we can define an algebraic inverse to GenJF restricted to a fixed subcategory. We say that a subcategory is ***canonically Jordan recoverable*** if such an algebraic inverse exists. See Section 4.2 for more details.

In the case where Q is an A_n type quiver, and $R = \varnothing$, we have the following characterization of canonically Jordan recoverable subcategories.

Theorem 1.1 ([Deq23a])**.** *Let Q be an A_n type quiver for some $n \in \mathbb{N}^*$ and $\mathscr{C} \subseteq \mathrm{rep}(Q)$ subcategory closed under sums and summands. $\mathscr{C}$ is canonically Jordan recoverable if and only if for any nonsplit short exact sequence*

$$0 \longrightarrow E \rightarrowtail F \twoheadrightarrow G \longrightarrow 0,$$

whenever $E, G \in \mathscr{C}$, the representation F must be decomposable.

There exists a combinatorial characterization of such a subcategory $\mathscr{C}$ based on the behavior of the intervals of $\{1, \ldots, n\}$ corresponding bijectively to the indecomposable representations in $\mathscr{C}$. We even have a precise description of the maximal ones for inclusion. We refer to Section 4.2 for more details.

1.3. **Main results.** First, we study the Gen-Sub operators applied to the tilting objects of $\mathscr{A}$, and we establish the following result.

Theorem 1.2. *Let $T \in \mathscr{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is the unique maximal $\mathcal{E}$-adapted subcategory of $\mathscr{A}$ closed under extensions which contains T.*

Then, by considering Q an ADE Dynkin type quiver, and by fixing an exact structure $\mathcal{E}$ on $\mathrm{rep}(Q)$, we defined $\mathcal{E}$-mutations on tilting objects (see Section 5.2), and an equivalence relation $\approx_{\mathcal{E}}$ on tilting objects of $\mathrm{rep}(Q)$ as follows. Given two tilting objects $T, T' \in \mathrm{rep}(Q)$, we write $T \approx_{\mathcal{E}} T'$ whenever T' can be obtained from T by applying a finite sequence of $\mathcal{E}$-mutations. We write $[T]_{\approx_{\mathcal{E}}}$ for the equivalence class of T for $\approx_{\mathcal{E}}$. We show that, for any tilting object $T \in \mathrm{rep}(Q)$, we can describe $\mathrm{GS}_{\mathcal{E}}(T)$ as the category additively generated by the tilting objects in $[T]_{\approx_{\mathcal{E}}}$ (see Theorem 5.31).

Reducing ourselves to type A quiver representations, we combine the exact structure point of view with the different descriptions known for the maximal canonically Jordan recoverable subcategories to prove conjectures stated in [Deq23a], and hence, strongly link those categories to tilting theory.

Theorem 1.3. *Let Q be an A_n type quiver for some $n \in \mathbb{N}^*$. A subcategory $\mathscr{C} \subseteq \mathrm{rep}(Q)$ is a maximal canonically Jordan recoverable subcategory if and only if there exists a tilting representation $T \in \mathrm{rep}(Q)$ such that $\mathscr{C} = \mathrm{GS}_{\mathcal{E}_\diamond}(T)$. Moreover, for such a tilting representation T, we have*

$$\mathscr{C} = \mathrm{add}\left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}_\diamond}}} T'\right).$$

1.4. **Outline of the paper.** We begin, in Section 2, by laying crucial recalls to set our background framework: quiver representations and, specifically, type A quiver representations. Those recalls enable us to incorporate relevant examples throughout the article, which will gradually lead us to our main results.

In Section 3, given an abelian category $\mathscr{A}$ endowed with an exact structure $\mathcal{E}$, we introduce the Gen-Sub operators which allow one to construct nicely behaved subcategories relative to $\mathcal{E}$ from any subcategory of $\mathscr{A}$. In particular, by assuming that $\mathscr{A}$ is hereditary and Krull–Schmidt, we show that they preserve both $\mathcal{E}$-adaptation and extension-closure properties. We also prove Theorem 1.2.

Afterwards, in Section 4, we recall the notions of Jordan recoverability and canonical Jordan recoverability for subcategories of representations over bounded quivers, and we recall the main results from previous works on type A quivers, namely, a combinatorial and an algebraic characterization of canonically Jordan recoverable subcategories of type A quiver representations.

This brings us to Section 5 where we gather results from previous sections to prove Theorem 1.3. We first study the link between $\mathcal{E}$-mutations and $\mathrm{GS}_{\mathcal{E}}$, for any given exact structure $\mathcal{E}$, on tilting representations of any ADE Dynkin type quiver, before focusing on type A quivers for using results on canonically Jordan recoverable subcategories. Finally, in Section 6, we discuss directions of research that we will likely pursue in the near future.

2. Quiver representations

In this section, we recall the essential definitions that will serve as our main framework throughout this article.

2.1. **Vocabulary.** A ***quiver*** is a quadruple $Q = (Q_0, Q_1, s, t)$ where:

- Q_0 is a set called *vertex set*;
- Q_1 is a set called *arrow set*; and,
- $s, t : Q_0 \longrightarrow Q_1$ are functions called *source* and *target functions*.

Such a quiver Q is said to be ***finite*** whenever Q_0 and Q_1 are finite sets. In the following, we assume that all our quivers are finite.

A ***path*** of Q is either a formal variable e_q for some $q \in Q_0$ called *lazy path at q*, or a finite nonempty word $p = \alpha_k \cdots \alpha_1$ on Q_0 such that, for all $i \in \{1, \ldots, k-1\}$, we have $s(\alpha_{i+1}) = t(\alpha_i)$. Given such a path p, we define:

- its *source* $s(p)$ to be $s(\alpha_1)$ (we set $s(e_q) = q$);

- its *target* $t(p)$ to be $t(\alpha_k)$ (we set $t(e_q) = q$); and,
- its *length* $\ell(p)$ to be k (we set $\ell(e_q) = 0$).

For any arrow $\alpha \in Q_1$, we consider its formal inverse α^{-1} such that $s(\alpha^{-1}) = t(\alpha)$ and $t(\alpha^{-1}) = s(\alpha)$. We denote by $\overline{Q_1}$ the set of formal inverses of arrows in Q_1. A ***walk*** of Q is either a lazy path or a finite nonempty word $w = \beta_k \cdots \beta_1$ on $Q_1 \sqcup \overline{Q_1}$ such that, for all $i \in \{1, \dots, k-1\}$, we have $s(\beta_{i+1}) = t(\beta_i)$. We extend the source, target, and length functions to walks of Q similarly to the case of paths of Q. A quiver Q is said to be ***connected*** whenever for any pair $(a, b) \in Q_0$, there exists a walk w of Q such that $s(w) = a$ and $t(w) = b$. From now on, we assume that all our quivers are connected.

2.2. **Representations.** Let $\mathbb{K}$ be a field. For geometric purposes, we assume that $\mathbb{K}$ is algebraically closed. The ***path algebra*** of Q, denoted by $\mathbb{K}Q$, is the $\mathbb{K}$-vector space generated as a basis by the set of all the paths of Q, together with a multiplication which acts as concatenation on the paths: meaning

$$p_2 \cdot p_1 = \begin{cases} p_2 p_1 & \text{if } s(p_2) = t(p_1); \text{ and,} \\ 0 & \text{otherwise.} \end{cases}$$

For any $m \in \mathbb{N}$, we denote by $\mathbb{K}Q_{\geqslant m}$ the (bi-sided) ideal of $\mathbb{K}Q$ generated by all the paths of length greater than or equal to m. An ideal $I \subseteq \mathbb{K}Q$ is said to be ***admissible*** whenever there exists an integer $m \geqslant 2$ such that $\mathbb{K}Q_{\geqslant m} \subseteq I \subseteq \mathbb{K}Q_{\geqslant 2}$. A ***relation*** on Q is a (finite) $\mathbb{K}$-linear combination of paths of length at least two, and having a common source and a common sink. For any subset of relations R on Q, we denote by $\langle R \rangle$ the ideal of $\mathbb{K}Q$ generated by R. Call ***bounded quiver*** any pair (Q, R) where Q is a quiver and R is a set of relations on Q.

Given a bounded quiver (Q, R), we define a ***representation*** of Q (over $\mathbb{K}$) as a pair $E = ((E_q)_{q \in Q_0}, (E_\alpha)_{\alpha \in Q_1})$ such that:

- E_q is a $\mathbb{K}$-vector space, for all $q \in Q_0$;
- $E_\alpha : E_{s(\alpha)} \longrightarrow E_{t(\alpha)}$ is a $\mathbb{K}$-linear ma, for any $\alpha \in Q_1$; and,
- by setting $E_p = E_{\alpha_k} \circ \cdots \circ E_{\alpha_1}$ for any path $p = \alpha_k \dots \alpha_1$ of Q, whenever we have a relation $k_1 p_1 + \dots + k_s p_s$ on Q in R, then $k_1 E_{p_1} + \dots + k_s E_{p_s} = 0$.

This representation E is ***finite-dimensional*** if, for all $q \in Q_0$, E_q is finite-dimensional. Its ***dimension vector*** $\mathbf{dim}(E)$ is the vector $(\dim(E_q))_{q \in Q_0} \in \mathbb{N}^{\#Q_0}$. Now, we assume that all the representations we consider are finite-dimensional. Given E and F two representations of Q, we define a ***morphism*** $\varphi : E \longrightarrow F$ as a collection $\varphi = (\varphi_q)_{q \in Q_0}$ for linear maps $\varphi_q : E_q \longrightarrow E_q$ such that, for all $\alpha \in Q_1$, we have $\varphi_{t(\alpha)} E_\alpha = F_\alpha \varphi_{s(\alpha)}$. We say that E and F are ***isomorphic*** whenever there exists a bijective morphism $\varphi : E \longrightarrow F$, or equivalently, if there exists a morphism $\varphi : E \longrightarrow F$ such that, for any $q \in Q_0$, φ_q is an isomorphism of $\mathbb{K}$-vector spaces. In such a case, we write $E \cong F$. We obtain the structure of a category by endowing the collection of representations of Q with the morphisms between them. In the following, we denote this category by $\operatorname{rep}(Q, R)$. In the case where $R = \varnothing$, we denote this category by $\operatorname{rep}(Q)$. Note that $\operatorname{rep}(Q, R)$ depends on the field $\mathbb{K}$.

Any (basic) finite-dimensional $\mathbb{K}$-algebra is isomorphic to the quotient algebra $\mathbb{K}Q/\langle R \rangle$ for some bounded quiver (Q, R) where Q is finite, and $\langle R \rangle$ is admissible. Moreover, the category $\operatorname{rep}(Q, R)$ is equivalent to the category of finitely generated left module over $\mathbb{K}Q/\langle R \rangle$. In particular, it implies that $\operatorname{rep}(Q, R)$ is an abelian

Krull–Schmidt category with enough projective objects. We refer the reader to [ASS06] for more details.

Write $E \oplus F$ for the direct sum of E and F in $\mathrm{rep}(Q, R)$. We say that a nonzero representation $X \in \mathrm{rep}(Q, R)$ is ***indecomposable*** if whenever we have $X \cong E \oplus F$ with some $E, F \in \mathrm{rep}(Q, R)$, then $E \cong 0$ or $F \cong 0$. Write $\mathrm{ind}(Q, R)$ for the collection of isomorphism classes of indecomposable representations of (Q, R). Whenever $R = \varnothing$, we denote this collection by $\mathrm{ind}(Q)$. A ***representation-finite type*** bounded quiver is a bounded quiver that has finitely many isomorphism classes of indecomposable representations.

For any $\boldsymbol{d} = (d_q)_{q \in Q_0} \in \mathbb{N}^{\# Q_0}$, we consider the following affine space

$$\mathrm{rep}((Q, R), \boldsymbol{d}) = \prod_{\alpha \in Q_1} \mathrm{Hom}\left(\mathbb{K}^{d_{s(\alpha)}}, \mathbb{K}^{d_{t(\alpha)}}\right) = \prod_{\alpha \in Q_1} \mathrm{Mat}_{d_{t(\alpha)} \times d_{s(\alpha)}}(\mathbb{K}),$$

and refer to it as the ***representation space*** of (Q, R) with dimension vector $\boldsymbol{d}$. Indeed, by choosing a basis for each of the vector spaces, we can identify the representations of (Q, R) with the points of $\mathrm{rep}((Q, R), \boldsymbol{d})$. The isomorphism classes of representations with vector dimension $\boldsymbol{d}$ correspond exactly to the orbits of the group action given by the algebraic group $\mathbf{GL}_{\boldsymbol{d}}(\mathbb{K}) = \prod_{q \in Q_0} \mathrm{GL}_{d_q}(\mathbb{K})$, which acts on $\mathrm{rep}((Q, R), \boldsymbol{d})$ by changing bases at each vertex.

2.3. **ADE Dynkin type quivers.** In this paper, we show examples of our results on representation-finite quivers. We shortly recall a complete characterization of them due to P. Gabriel [Gab72].

Definition 2.1. Let $n \in \mathbb{N}^*$. A quiver Q is said to be:

- of A_n ***type*** if the underlying graph of Q is of the following shape.

$$\begin{tikzcd} 1 \arrow[r, dash] & 2 \arrow[r, dash] & \cdots \arrow[r, dash] & n \end{tikzcd}$$

- of D_n ***type***, for $n \geqslant 4$, if the underlying graph of Q is of the following shape.

$$\begin{tikzcd} & & & & n-1 \\ 1 \arrow[r, dash] & 2 \arrow[r, dash] & \cdots \arrow[r, dash] & n-2 \arrow[ur, dash] \arrow[dr, dash] & \\ & & & & n \end{tikzcd}$$

- of E_6, E_7 or E_8 ***type*** if the shape of the underlying graph of Q is one of the following.

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[u, dash] \arrow[r, dash] & 5 \arrow[r, dash] & 6 \end{tikzcd} \qquad \begin{tikzcd} & & 4 & & & \\ 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[u, dash] \arrow[r, dash] & 5 \arrow[r, dash] & 6 \arrow[r, dash] & 7 \end{tikzcd}$$

$$\begin{tikzcd} & & 4 & & & & \\ 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[u, dash] \arrow[r, dash] & 5 \arrow[r, dash] & 6 \arrow[r, dash] & 7 \arrow[r, dash] & 8 \end{tikzcd}$$

We say that Q is an ADE ***Dynkin type*** quiver whenever Q is one of the above types.

We have the following main result.

Theorem 2.2 ([Gab72])**.** *Let Q be a quiver. Then Q is a representation-finite type quiver if and only if Q is an ADE Dynkin type quiver.*

Remark 2.3. This result holds among bounded quivers (Q, R) with $R = \varnothing$.

In the following, we recall the precise description of $\mathrm{rep}(Q)$ for Q a type A quiver. The reader can find those results in [Sch14].

Let $n \in \mathbb{N}^*$. The ***intervals*** of $\{1, \dots, n\}$ are the sets $[\![i, j]\!] := \{i, i+1, \dots, j\}$ given by all $1 \leqslant i \leqslant j \leqslant n$. If $i = j$, we write $[\![i, i]\!] = [\![i]\!]$. Denote by $\mathcal{I}_n$ the set of intervals in $\{1, \dots n\}$. For $K = [\![i, j]\!] \in \mathcal{I}_n$, set $b(K) = i$ and $e(K) = j$. Call ***interval set*** any subset of $\mathcal{I}_n$.

Definition 2.4. Let Q be an A_n type quiver. Consider $K, L \in \mathcal{I}_n$ such that $K \subseteq L$. We say that:

- K is ***above*** L (***relative to*** Q) if the following two assertions are satisfied:
  - $b(K) = b(L)$ or we have the arrow $b(K) - 1 \longleftarrow b(K)$ in Q;
  - $e(K) = e(L)$ or we have the arrow $e(K) \longrightarrow e(K) + 1$ in Q.
- K is ***below*** L (***relative to*** Q) if the following two assertions are satisfied:
  - $b(K) = b(L)$ or we have the arrow $b(K) - 1 \longrightarrow b(K)$ in Q;
  - $e(K) = e(L)$ or we have the arrow $e(K) \longleftarrow e(K) + 1$ in Q.

Example 2.5. Consider the following quiver.

$$Q = \quad 1 \longrightarrow 2 \longrightarrow 3$$

Then $[\![2]\!]$ is above $[\![2, 3]\!]$ and below $[\![1, 2]\!]$.

Note that any interval is above and below itself, relative to all A_n type quivers. Let Q be a quiver of A_n type. To any interval $K \in \mathcal{I}_n$, we consider X_K the representation of Q defined as it follows:

- $(X_K)_q = \mathbb{K}$ if $q \in K$, $(X_K)_q = 0$ otherwise;
- $(X_K)_\alpha = \mathrm{Id}_{\mathbb{K}}$ if α is such that $\{s(\alpha), t(\alpha)\} \subseteq K$, $(X_K)_\alpha = 0$ otherwise;

Note that X_K is an indecomposable representation of Q for all $K \in \mathcal{I}_n$.

Theorem 2.6. *Let Q be an A_n type quiver.*

(*a*) *The isomorphism classes of indecomposable representations of Q are in bijection with $\mathcal{I}_n$; more precisely, they are described by indecomposable representations X_K for $K \in \mathcal{I}_n$;*

(*b*) *The homomorphism space between two indecomposable representations of Q is of dimension at most one; more precisely,* $\mathrm{Hom}(X_K, X_L)$ *is nonzero if and only if there exists an interval J such that J is above K and below L relative to Q; if such an interval exists, it is unique and* $\mathrm{Hom}(X_K, X_L)$ *consists of scalar multiples of the morphism $\phi = (\phi_q)_{q \in Q_0}$ such that $\phi_q =$* $\mathrm{Id}_{\mathbb{K}}$ *if $q \in J$ and $\phi_q = 0$ otherwise.*

In the light of the previous result :

- for all interval sets $\mathscr{J} \subseteq \mathcal{I}_n$ and all quivers Q of A_n type, write $\mathrm{Cat}_Q(\mathscr{J})$ for the subcategory of $\mathrm{rep}(Q)$ additively generated by X_K for $K \in \mathscr{J}$;
- for all quivers Q of A_n type and for all nonzero subcategories $\mathscr{C}$ of $\mathrm{rep}(Q)$, let $\mathrm{Int}(\mathscr{C})$ be the interval set of $\mathcal{I}_n$ consisting of intervals K such that $X_K \in \mathscr{C}$.

Under Convention 3.1, the subcategories we are considering are additively generated by X_K for $K \in \mathscr{J}$ for some $\mathscr{J} \subset \mathcal{I}_n$.

Hence, for any A_n type quiver Q, for all $\mathscr{J} \subseteq \mathcal{I}_n$ and for all subcategories $\mathscr{C} \subseteq \operatorname{rep}(Q)$, we have $\mathscr{J} = \operatorname{Int}(\operatorname{Cat}_Q(\mathscr{J}))$ and $\mathscr{C} = \operatorname{Cat}_Q(\operatorname{Int}(\mathscr{C}))$.

Now we recall the description of all the short exact sequences between indecomposable representations.

Theorem 2.7. *Let Q be an A_n type quiver. Let $D, F \in \operatorname{ind}(Q)$ and consider a non-split exact sequence*

$$0 \longrightarrow D \rightarrowtail E \twoheadrightarrow F \longrightarrow 0.$$

Then exactly one of the following assertions holds:

(a) *We have $E \in \operatorname{ind}(Q)$, and if we consider $K, L \in \mathcal{I}_n$ such that $D \cong X_K$ and $F \cong X_L$, then $K \cup L \in \mathcal{I}_n$, $K \cap L = \varnothing$ and $E \cong X_{K\cup L}$; or,*

(b) *We have $E \cong E_1 \oplus E_2$ with $E_1, E_2 \in \operatorname{ind}(Q)$, and there exist $K, L \in \mathcal{I}_n$ with $K \cap L \neq \varnothing$ such that both assertions hold:*
- *$K, L, K\cap L$ and $K \cup L$ are four different intervals in $\mathcal{I}_n$; and,*
- *either $\{D, F\} = \{X_K, X_L\}$ and $\{E_1, E_2\} = \{X_{K\cap L}, X_{K\cup L}\}$, or $\{D, F\} = \{X_{K\cap L}, X_{K\cup L}\}$ and $\{E_1, E_2\} = \{X_K, X_L\}$.*

In this case, we call it a ***diamond exact sequence.***

Definition 2.8 (Diamond exact sequence)**.** Let Q be an A_n type quiver. A non-split short exact sequence

$$0 \longrightarrow D \longrightarrow E \longrightarrow F \longrightarrow 0$$

in $\operatorname{rep}(Q)$ is called a *diamond exact sequence* if

(i) $D, F \in \operatorname{ind}(Q)$;

(ii) E is decomposable and,

$$E \cong E_1 \oplus E_2 \quad \text{with } E_1, E_2 \in \operatorname{ind}(Q).$$

3. Gen-Sub operators on exact categories

Let $\mathscr{A}$ be a hereditary abelian category with enough projectives.

Convention 3.1. Our subcategories are *full*, *additive* and closed under isomorphisms. By additive subcategories, we mean subcategories that are closed under sums and summands.

In this section, after a few meaningful recalls on exact structures, we define the *Gen-Sub operator*. By fixing $\mathcal{E}$ as an exact structure on $\mathscr{A}$, it allows one to construct a subcategory $\mathscr{D} \supseteq \mathscr{C}$ satisfying crucial properties relative to $\mathcal{E}$ from any subcategory $\mathscr{C}$. We illustrate the following definitions with examples within the framework of $\operatorname{rep}(Q)$, where Q is a type A quiver.

3.1. **Exact structures.** Let $\mathscr{A}$ be an abelian category. Recall that two short exact sequences in $\mathscr{A}$

$$\xi : \qquad 0 \longrightarrow D \overset{f}{\rightarrowtail} E \overset{g}{\twoheadrightarrow} F \longrightarrow 0$$

$$\varsigma : \qquad 0 \longrightarrow X \overset{i}{\rightarrowtail} Y \overset{j}{\twoheadrightarrow} Z \longrightarrow 0$$

are said to be *isomorphic* whenever there exists a triplet of isomorphisms

$$\left(D \xrightarrow{\phi} X \ , \ E \xrightarrow{\psi} Y \ , \ F \xrightarrow{\mu} Z \right)$$

such that every square in the following diagram commutes.

$$\begin{tikzcd}
\xi : & 0 \arrow[r] & D \arrow[r, tail, "f"] \arrow[d, "\phi"] & E \arrow[r, two heads, "g"] \arrow[d, "\psi"] & F \arrow[r] \arrow[d, "\mu"] & 0 \\
\varsigma : & 0 \arrow[r] & X \arrow[r, tail, "i"] & Y \arrow[r, two heads, "j"] & Z \arrow[r] & 0
\end{tikzcd}$$

From now on, by abusing notations, we identify a short exact sequence with its isomorphism class of short exact sequences. Fix a subcollection $\mathcal{E}$ of short exact sequences in $\mathscr{A}$. A short exact sequence

$$\xi : \qquad 0 \longrightarrow A \overset{f}{\rightarrowtail} E \overset{g}{\twoheadrightarrow} B \longrightarrow 0$$

is said to be ***$\mathcal{E}$-exact*** (or ***$\mathcal{E}$-admissible***) if $\xi \in \mathcal{E}$. In such a case, we say that the map f is said to be an ***$\mathcal{E}$-monomorphism***, and the map g is said to be an ***$\mathcal{E}$-epimorphism***. Considering the short exact sequences up to isomorphism, the definitions of $\mathcal{E}$-exact sequences, $\mathcal{E}$-monomorphisms, and short $\mathcal{E}$-epimorphisms are interdependent, uniquely determining each other within this framework.

Quillen's original formulation of exact structures thus encapsulates these relationships, providing a robust foundation for applying homological constructs in settings that deviate from traditional abelian categories. Quillen's definition can be rephrased as follows.

Definition 3.2. A collection $\mathcal{E}$ of short exact sequence in $\mathscr{A}$ is an *exact structure* on $\mathscr{A}$ whenever $\mathcal{E}$ satisfies the following axioms:

(ES1) All the split short exact sequences are in $\mathcal{E}$;

(ES2) The collection of $\mathcal{E}$-monomorphisms and the one of $\mathcal{E}$-epimorphisms are both closed under compositions;

(ES3) $\mathcal{E}$ is closed under the pushouts along any morphism: that is, any short exact sequence

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r, tail, "f"] \arrow[d, "h"] & Y \arrow[r, two heads] \arrow[d] & Z \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & D \arrow[r, tail] & PO \arrow[r, two heads] & Z \arrow[r] & 0,
\end{tikzcd}$$

where f is an $\mathcal{E}$-monomorphism gives a short exact sequence is in $\mathcal{E}$ whenever taking the pushout along any morphism h;

(ES4) $\mathcal{E}$ is closed under the pullbacks along any morphism: that is, any short exact sequence

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r, tail] & Y \arrow[r, two heads, "g"] & Z \arrow[r] & 0 \\
0 \arrow[r] & X \arrow[r, tail] \arrow[u, equal] & PB \arrow[r, two heads] \arrow[u] & F \arrow[r] \arrow[u, "h"] & 0,
\end{tikzcd}$$

where g is an $\mathcal{E}$-epimorphism gives a short exact sequence in $\mathcal{E}$ whenever taking the pullback along any morphism h.

If $\mathcal{E}$ is an exact structure on $\mathscr{A}$, then we refer to the pair $(\mathscr{A}, \mathcal{E})$ as an ***exact category***. We also refer to the exact structure containing all short exact sequences as $\mathcal{E}_{\max}$, and the one containing only the split short exact sequences as $\mathcal{E}_{\min}$. We denote by $\mathcal{E}(-,-)$ the additive subbifunctor of $\mathrm{Ext}^1_{\mathscr{A}}(-,-)$ that consists of extensions belonging to $\mathcal{E}$.

Definition 3.3. Let $(\mathscr{A}, \mathcal{E})$ be an exact category.

- An object P in $\mathscr{A}$ is ***$\mathcal{E}$-projective*** if for every $\mathcal{E}$-epimorphism $g : Y \to Z$, the map $g_* = \mathrm{Hom}_{\mathscr{A}}(P, g)$ is an epimorphism across all Y, Z in $\mathscr{A}$. Equivalently, P is $\mathcal{E}$-projective if the functor $\mathrm{Hom}_{\mathscr{A}}(P, -)$ preserves the exactness of $\mathcal{E}$-admissible sequences. We write $\mathrm{proj}_{\mathcal{E}}(\mathscr{A})$ for the collection of $\mathcal{E}$-projective objects in $\mathscr{A}$. We observe that $\mathrm{proj}_{\mathcal{E}_{\max}}(\mathscr{A}) = \mathrm{proj}(\mathscr{A})$ where $\mathcal{E}_{\max}$ is the exact structure containing all the short exact sequences.
- An object I in $\mathscr{A}$ is ***$\mathcal{E}$-injective*** if for every $\mathcal{E}$-monomorphism $f : X \to Y$, the map $f^* = \mathrm{Hom}_{\mathscr{A}}(f, I)$ is an epimorphism across all X, Y in $\mathscr{A}$. Equivalently, I is $\mathcal{E}$-injective if the functor $\mathrm{Hom}_{\mathscr{A}}(-, I)$ preserves the exactness of $\mathcal{E}$-admissible sequences. We write $\mathrm{inj}_{\mathcal{E}}(\mathscr{A})$ for the collection of $\mathcal{E}$-injective objects in $\mathscr{A}$. We observe that $\mathrm{inj}_{\mathcal{E}_{\max}}(\mathscr{A}) = \mathrm{inj}(\mathscr{A})$.

In the following, we focus on a relevant example of exact structure for $\mathscr{A} = \mathrm{rep}(Q)$ for Q a type A quiver, called the *diamond exact structure*.

Definition 3.4. Let Q be an A_n type quiver. We define the collection of short exact sequences $\mathcal{E}_\diamond$ given by the additive bifunctor $\mathcal{E}_\diamond(-,-)$ that is uniquely determined by

$$\mathcal{E}_\diamond(N, M) = \{\eta \in \mathrm{Ext}^1(N, M) \mid \eta \text{ is split or a diamond exact sequence}\}.$$

with $M, N \in \mathrm{rep}\, Q$ indecomposable.

The following result ensures that $\mathcal{E}_\diamond$ is indeed an exact structure of $\mathrm{rep}(Q)$.

Proposition 3.5 ([BHRS24])**.** *Let $n \in \mathbb{N}^*$, and Q be an A_n type quiver. The collection $\mathcal{E}_\diamond$ is an exact structure on $\mathscr{A} = \mathrm{rep}(Q)$.*

Before defining the *Gen-Sub operators*, we recall and restate some useful results on diagrams where the exact structure $\mathcal{E}$ intervenes, based on the work of T. Bühler [Bü10].

Lemma 3.6 (3×3-lemma)**.** *Let $(\mathscr{A}, \mathcal{E})$ be an exact category. Consider the following commutative diagram.*

$$\begin{tikzcd}
0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] \\
A \arrow[r] \arrow[d, tail] & B \arrow[r] \arrow[d, tail] & C \arrow[d, two heads] \\
A' \arrow[r, "f"] \arrow[d, two heads] & B' \arrow[r, "g"] \arrow[d, two heads] & C' \arrow[d, two heads] \\
A'' \arrow[r] \arrow[d] & B'' \arrow[r] \arrow[d] & C'' \arrow[d] \\
0 & 0 & 0
\end{tikzcd}$$

Assume that:

- *all the short sequences given by the columns are $\mathcal{E}$-exact; and,*
- *one of the following assertions holds:*
  - *two out of the three rows, including the middle one, give short $\mathcal{E}$-exact sequences; or,*
  - *the short sequences given by the top and the bottom rows are $\mathcal{E}$-exact, and $gf = 0$.*

Then the short sequence given by the remaining row is $\mathcal{E}$-exact.

Remark 3.7. We can exchange the roles of the rows and the columns to obtain a dual result applied to the columns of such a diagram.

Lemma 3.8 (Obscure axiom)**.** *Let $(\mathscr{A}, \mathcal{E})$ be an exact category. Let $X, Y \in \mathscr{A}$ and $i : X \longrightarrow Y$ admitting a cokernel. If there exist $Z \in \mathscr{A}$, and $j : Y \longrightarrow Z$ such that $j \circ i$ is an $\mathcal{E}$-monomorphism, then i is an $\mathcal{E}$-monomorphism. Dually, it holds for $\mathcal{E}$-epimorphisms.*

Remark 3.9. This lemma was initially stated as an axiom from D. Quillen's definition of an exact category [Qui73]. N. Yoneda proved in his thesis [Yon61] that this axiom is a consequence of the other axioms. Later on, B. Keller [Kel90] rediscovered this redundancy. The name "obscure axiom" was given by R. W. Thomason [TT07].

3.2. **Gen-Sub operators.** Fix $(\mathscr{A}, \mathcal{E})$ an exact category. This section introduces a closure operator on subcategories of $\mathscr{A}$. More precisely, given a subcategory $\mathscr{C}$, we will construct the smallest full subcategory $\mathscr{D} \supseteq \mathscr{C}$ such that:

- $\mathscr{D}$ is closed under cokernels of $\mathcal{E}$-monomorphisms; and,
- $\mathscr{D}$ is closed under kernels of $\mathcal{E}$-epimorphisms.

The $\mathrm{Gen}_{\mathcal{E}}$***-operator*** on subcategories of $\mathscr{A}$ is defined as follows: given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define $\mathrm{Gen}_{\mathcal{E}}(\mathscr{C})$ as the full subcategory, closed under sums and summands, of $\mathscr{A}$ containing the cokernel of any $\mathcal{E}$-monomorphism $f : X \longrightarrow Y$ such that $Y \in \mathscr{C}$. The $\mathrm{Sub}_{\mathcal{E}}$***-operator*** on subcategories of $\mathscr{A}$ is defined analogously: given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define $\mathrm{Sub}_{\mathcal{E}}(\mathscr{C})$ as the full subcategory, closed by sums and summands, of $\mathscr{A}$ containing the kernel of any $\mathcal{E}$-epimorphism $g : Y \longrightarrow X$ such that $Y \in \mathscr{C}$. Given an object $M \in \mathscr{C}$, by abusing of notation, we set $\mathrm{Gen}_{\mathcal{E}}(M) = \mathrm{Gen}_{\mathcal{E}}(\mathrm{add}(M))$ and $\mathrm{Sub}_{\mathcal{E}}(M) = \mathrm{Sub}_{\mathcal{E}}(\mathrm{add}(M))$.

Note that $\mathrm{Gen}_{\mathcal{E}_{\max}}(\mathscr{C}) = \mathrm{Gen}(\mathscr{C})$ and $\mathrm{Sub}_{\mathcal{E}_{\max}}(\mathscr{C}) = \mathrm{Sub}(\mathscr{C})$. As $\mathcal{E}$ is an exact structure on $\mathscr{A}$, then $\mathscr{C}$ is a subcategory of both $\mathrm{Gen}_{\mathcal{E}}(\mathscr{C})$ and $\mathrm{Sub}_{\mathcal{E}}(\mathscr{C})$.

Given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define a sequence of subcategories $(\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C}))_{i \in \mathbb{N}}$ as it follows:

- we set $\mathrm{GS}^0_{\mathcal{E}}(\mathscr{C}) = \mathscr{C}$; and,
- for all $i \geqslant 1$, we define $\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C})$ to be the subcategory of $\mathscr{A}$ additively generated by objects in both $\mathrm{Gen}_{\mathcal{E}}(\mathrm{GS}^{i-1}_{\mathcal{E}}(\mathscr{C}))$ and $\mathrm{Sub}_{\mathcal{E}}(\mathrm{GS}^{i-1}_{\mathcal{E}}(\mathscr{C}))$.

By the previous construction, $(\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C}))_{i \in \mathbb{N}}$ is an increasing sequence of additive subcategories of $\mathscr{A}$. Therefore, as $\mathscr{A}$ is abelian, this sequence of subcategories admits a colimit.

Definition 3.10. The $\mathrm{GS}_{\mathcal{E}}$***-operator*** on subcategories of $\mathscr{A}$ is defined as follows: given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ as the colimit of the sequence of subcategories $(\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C}))_{i\in\mathbb{N}}$. As before, we set $\mathrm{GS}_{\mathcal{E}}(M) = \mathrm{GS}_{\mathcal{E}}(\mathrm{add}(M))$ by abuse of notations.

Here are some apparent results coming directly from the definition.

Lemma 3.11. *Let $\mathscr{C} \subseteq \mathscr{A}$. The following assertions hold:*

(a) *The category $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is the (full additive) subcategory of $\mathscr{A}$ whose objects are the ones appearing in $\mathrm{GS}^i(\mathscr{C})$ for some $i \in \mathbb{N}$.*

(b) *The category $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is the smallest subcategory $\mathscr{A}$ containing $\mathscr{C}$, closed under taking $\mathcal{E}$-subobjects and $\mathcal{E}$-quotients.*

Remark 3.12. A subcategory $\mathscr{D} \subseteq \mathscr{A}$ satisfying (b) must be closed under extensions to be a so-called $\mathcal{E}$-*Serre* category. We recall that $\mathscr{D}$ is ***$\mathcal{E}$-Serre*** whenever for any short $\mathcal{E}$-exact sequence

$$\xi : \qquad 0 \longrightarrow E \rightarrowtail F \twoheadrightarrow G \longrightarrow 0,$$

we have $F \in \mathscr{D}$ if and only if $E, G \in \mathscr{D}$.

Example 3.13. In Figure 1, we calculate $\mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C})$ for a subcategory $\mathscr{C}$ of $\mathrm{rep}(Q)$ and a fixed quiver Q of A_7 type.

$$Q = \ 1 \rightarrow 2 \leftarrow 3 \rightarrow 4 \rightarrow 5 \leftarrow 6 \rightarrow 7$$

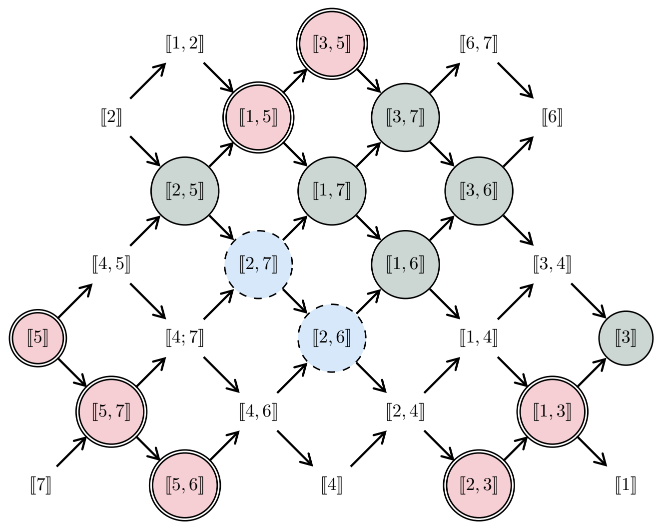

FIGURE 1. Calculation of $\mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C})$ where $\mathscr{C} = \mathrm{add}\,(\ )$. We get $\mathrm{GS}^1_{\mathcal{E}_\diamond}(\mathscr{C}) = \mathrm{add}\,(\ + \ )$ and $\mathrm{GS}^2_{\mathcal{E}_\diamond}(\mathscr{C}) = \mathrm{add}\,(\ + \ + \ )$. Here, we obtain that $\mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C}) = \mathrm{GS}^2_{\mathcal{E}_\diamond}(\mathscr{C})$.

3.3. **Subcategories adapted to an exact structure.** Let $(\mathscr{A}, \mathcal{E})$ be an exact category. We recall that $\mathscr{A}$ is *hereditary* if, for all $X, Y \in \mathscr{A}$, and for all $i \in \mathbb{N}_{\geqslant 2}$, we have $\mathrm{Ext}^i_{\mathscr{A}}(X, Y) = 0$.

Convention 3.14. From now on, $\mathscr{A}$ is a hereditary abelian category.

We define the following useful notion on subcategories of $\mathscr{A}$.

Definition 3.15. A subcategory $\mathscr{D} \subseteq \mathscr{A}$ is ***$\mathcal{E}$-adapted*** if for any pair (X, Y) of objects in $\mathscr{D}$, we have $\mathrm{Ext}^1(X, Y) \subseteq \mathcal{E}$

Example 3.16.

- Any subcategory of $\mathscr{A}$ is $\mathcal{E}_{\max}$-adapted.
- A subcategory $\mathscr{C} \subseteq \mathscr{A}$ is $\mathcal{E}_{\min}$-adapted only if for any $(X, Y) \in \mathscr{C}^2$, we have $\mathrm{Ext}^1_{\mathscr{A}}(X, Y) = 0$.

In this section, we will show that for any $\mathcal{E}$-adapted subcategory $\mathscr{C} \subseteq \mathscr{A}$, the category $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is also $\mathcal{E}$-adapted.

Lemma 3.17. *Consider a subcategory $\mathscr{C} \subseteq \mathscr{A}$. Let $D, E \in \mathscr{C}$ such that $\mathrm{Ext}^1(D, E) \subseteq \mathcal{E}$. Then, for any $X \in \mathrm{Gen}(E)$, the following assertions hold:*

(*a*) *Any $\xi \in \mathrm{Ext}^1(D, X)$ is obtained by a pushout of some $\eta \in \mathrm{Ext}^1(D, E^{\oplus r})$ along an epimorphism $g : E^{\oplus r} \twoheadrightarrow X$; and,*

(*b*) $\mathrm{Ext}^1(D, X) \subseteq \mathcal{E}$.

Furthermore, if $X \in \mathrm{Gen}_{\mathcal{E}}(E)$, then g can be chosen as an $\mathcal{E}$-epimorphism.

Proof. Assume that $X \in \mathrm{Gen}(E)$. Then there is a short exact sequence

$$0 \longrightarrow K \rightarrowtail E^{\oplus r} \xrightarrow{g} X \longrightarrow 0$$

By applying the $\mathrm{Hom}(D, -)$ functor on this short exact sequence, we obtain the following long exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{Hom}(D,K) \arrow[r, tail] & \mathrm{Hom}(D,E^{\oplus r}) \arrow[r] & \mathrm{Hom}(D,X) \arrow[dll] & \\ & \mathrm{Ext}^1(D,K) \arrow[r] & \mathrm{Ext}^1(D,E^{\oplus r}) \arrow[r, two heads, "g_\bullet"] & \mathrm{Ext}^1(D,X) \arrow[r] & 0, \end{tikzcd}$$

where $g_\bullet = \mathrm{Ext}^1(D, g)$. As $\mathscr{A}$ is hereditary, we have that $\mathrm{Ext}^2(D, K) = 0$. So $g_\bullet$ is surjective, and therefore, for any short exact sequence $\xi \in \mathrm{Ext}^1(D, X)$, there exists $\eta \in \mathrm{Ext}^1(D, E^{\oplus r})$ such that $g_\bullet(\eta) = \xi$. Since $g_\bullet$ is the pushout along g, we proved the assertion (a).

By hypothesis, $\mathrm{Ext}^1(D, E) \subseteq \mathcal{E}$. So we have $\mathrm{Ext}^1(D, X) \subseteq \mathcal{E}$ as $\mathcal{E}$ is an exact structure. If moreover $X \in \mathrm{Gen}_{\mathcal{E}}(E)$, then g can obviously be chosen as an $\mathcal{E}$-epimorphism. ■

Lemma 3.18. *Let $\mathscr{C} \subseteq \mathscr{A}$ be a subcategory. Let $E, F \in \mathscr{C}$ with $\mathrm{Ext}^1(E, F) \subseteq \mathcal{E}$. Then, for any object X in $\mathrm{Gen}_{\mathcal{E}}(E)$, we have $\mathrm{Ext}^1(X, F) \subseteq \mathcal{E}$.*

Proof. Suppose we have the following short exact sequence

$$\xi : \qquad 0 \longrightarrow F \rightarrowtail E' \twoheadrightarrow X \longrightarrow 0.$$

Since $X \in \mathrm{Gen}_{\mathcal{E}}(E)$, there is a short exact sequence

$$0 \longrightarrow K \rightarrowtail E^{\oplus r} \xrightarrow{\varphi} X \longrightarrow 0 \qquad \in \mathcal{E}.$$

We can construct the following diagram.

$$
\begin{tikzcd}
 & & & 0 & \\
0 \arrow[r] & F \arrow[r, tail] & E' \arrow[r, two heads] & X \arrow[r] \arrow[u] & 0 \\
 & & & E^{\oplus r} \arrow[u, two heads, "\varphi"] & \\
 & & & K \arrow[u, tail] & \\
 & & & 0 \arrow[u] &
\end{tikzcd}
$$

We complete the diagram by taking the pullback along φ and by using the kernel lifting property.

$$
\begin{tikzcd}
 & 0 & 0 & 0 & \\
0 \arrow[r] & F \arrow[r, tail] \arrow[u] & E' \arrow[r, two heads] \arrow[u] & X \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & F \arrow[r, tail] \arrow[u, equal] & PB \arrow[r, two heads] \arrow[u, two heads] & E^{\oplus r} \arrow[r] \arrow[u, two heads, "\varphi"] & 0 \\
0 \arrow[r] & 0 \arrow[r, tail] \arrow[u, tail] & K' \arrow[r, two heads] \arrow[u, tail] & K \arrow[r] \arrow[u, tail] & 0 \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}
$$

As the short sequences given by the three columns and those given by the first two rows of the above diagram are exact, by Lemma 3.6, the short sequence given by the last row is exact too. So $K' \cong K$.

Moreover, the short exact sequences given by the third column and the middle row are in $\mathcal{E}$ by hypothesis. As $\mathcal{E}$ is an exact structure, the short exact sequence given by the middle column is in $\mathcal{E}$. As the short exact sequence given by the first column and the one provided by the last row are split, they are in $\mathcal{E}$.

By Lemma 3.6, we have that $\xi \in \mathcal{E}$. ■

Lemma 3.19. *Consider a subcategory $\mathscr{C} \subseteq \mathscr{A}$, and $E, F \in \mathscr{C}$ such that $\operatorname{Ext}^1(E,F) \subseteq \mathcal{E}$. Then, for any $X \in \operatorname{Sub}(E)$, the following assertions hold:*

- (a) *Any $\xi \in \operatorname{Ext}^1(X,F)$ is obtained by a pullback of some $\eta \in \operatorname{Ext}^1(E^{\oplus r},F)$ along a monomorphism $f : X \rightarrowtail E^{\oplus r}$; and,*
- (b) $\operatorname{Ext}^1(X,F) \subseteq \mathcal{E}$.

Furthermore, if $X \in \operatorname{Sub}_{\mathcal{E}}(E)$, then f can be chosen as an $\mathcal{E}$-monomorphism.

Proof. This is the dual result of Lemma 3.17. ■

Lemma 3.20. *Let $\mathscr{C} \subseteq \mathscr{A}$ be a subcategory. Let D, E be in $\mathscr{C}$ such that $\operatorname{Ext}^1(D,E) \subseteq \mathcal{E}$. Then, for any object X in $\operatorname{Sub}_{\mathcal{E}}(E)$, we have $\operatorname{Ext}^1(D,X) \subseteq \mathcal{E}$.*

Proof. This is the dual result of Lemma 3.18. ■

Proposition 3.21. *Consider an exact category $(\mathscr{A}, \mathcal{E})$ and let $\mathscr{C} \subseteq \mathscr{A}$ be a subcategory. If $\mathscr{C}$ is $\mathcal{E}$-adapted, then $\operatorname{Ext}^1\left(\mathscr{C}, \mathrm{GS}^1_{\mathcal{E}}(\mathscr{C})\right)$ and $\operatorname{Ext}^1\left(\mathrm{GS}^1_{\mathcal{E}}(\mathscr{C}), \mathscr{C}\right)$ are contained in $\mathcal{E}$.*

Proof. Consider X an object of $\mathrm{GS}^1_{\mathcal{E}}(\mathscr{C})$. Then X is in $\mathrm{Gen}_{\mathcal{E}}(\mathscr{C})$ or in $\mathrm{Sub}_{\mathcal{E}}(\mathscr{C})$.

Assume that $X \in \mathrm{Gen}_{\mathcal{E}}(\mathscr{C})$. We can then suppose that $X \in \mathrm{Gen}_{\mathcal{E}}(E) \subseteq \mathrm{Gen}(E)$ for $E \in \mathscr{C}$. By hypothesis, we know for any $D, F \in \mathscr{C}$, $\operatorname{Ext}^1(D, E), \operatorname{Ext}^1(E, F) \subseteq \mathcal{E}$. By Lemma 3.17, $\operatorname{Ext}^1(D, X) \subseteq \mathcal{E}$ for any $D \in \mathrm{GS}^1_{\mathcal{E}}(\mathscr{C})$. By Lemma 3.18, $\operatorname{Ext}^1(X, F) \subseteq \mathcal{E}$ for any $F \in \mathscr{C}$.

Assume that $X \in \mathrm{Sub}_{\mathcal{E}}(\mathscr{C})$. Similarly, we have that $X \in \mathrm{Sub}_{\mathcal{E}}(E) \subseteq \mathrm{Sub}(E)$ for some $E \in \mathscr{C}$. By hypothesis, we know for any $D, F \in \mathscr{C}$, $\operatorname{Ext}^1(D, E), \operatorname{Ext}^1(E, F) \subseteq \mathcal{E}$. By Lemma 3.19, $\operatorname{Ext}^1(X, F) \subseteq \mathcal{E}$ for any $F \in \mathscr{C}$. By Lemma 3.20, $\operatorname{Ext}^1(D, X) \subseteq \mathcal{E}$ for any $D \in \mathscr{C}$.

Therefore, we get to the conclusion that, for any $X \in \mathrm{GS}^1_{\mathcal{E}}(\mathscr{C})$ and $E' \in \mathscr{C}$, we have $\operatorname{Ext}^1(X, E'), \operatorname{Ext}^1(E', X) \subseteq \mathcal{E}$. ■

We can now finally prove and state the fact that the $\mathrm{GS}_{\mathcal{E}}$-operator preserves $\mathcal{E}$-adaptability.

Proposition 3.22. *Let $\mathscr{C} \subseteq \mathscr{A}$ be a subcategory. If $\mathscr{C}$ is $\mathcal{E}$-adapted, then $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is $\mathcal{E}$-adapted too.*

Proof. We prove by induction that $\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C})$ is $\mathcal{E}$-adpated, for all $i \geq 0$. For $i = 0$, the statement holds as $\mathcal{C}$ is $\mathcal{E}$-adapted.

Fix $k \geqslant 0$. Assume that $\mathrm{GS}^k_{\mathcal{E}}(\mathscr{C})$ is $\mathcal{E}$-adapted. Suppose that $X, Y \in \mathrm{GS}^{k+1}_{\mathcal{E}}(\mathscr{C})$. Then $X \in \mathrm{Gen}_{\mathcal{E}}(\mathrm{GS}^k_{\mathcal{E}}(\mathscr{C}))$ or $X \in \mathrm{Sub}_{\mathcal{E}}(\mathrm{GS}^k_{\mathcal{E}}(\mathscr{C}))$.

If $X \in \mathrm{Gen}_{\mathcal{E}}(\mathrm{GS}^k_{\mathcal{E}}(\mathscr{C}))$. So $X \in \mathrm{Gen}_{\mathcal{E}}(E) \subseteq \mathrm{Gen}(E)$ for some $E \in \mathrm{GS}^k_{\mathcal{E}}(\mathscr{C})$. By induction hypothesis and Proposition 3.21, we get that $\operatorname{Ext}^1(Y, E), \operatorname{Ext}^1(E, Y) \subseteq \mathcal{E}$. Therefore, by Lemma 3.17, $\operatorname{Ext}^1(Y, X) \subseteq \mathcal{E}$. Using Lemma 3.18, we also have $\operatorname{Ext}^1(X, Y) \subseteq \mathcal{E}$.

If $X \in \mathrm{Sub}_{\mathcal{E}}(\mathrm{GS}^k_{\mathcal{E}}(\mathscr{C}))$. Let $E \in \mathrm{GS}^k_{\mathcal{E}}(\mathscr{C})$ such that $X \in \mathrm{Sub}_{\mathcal{E}}(E) \subseteq \mathrm{Sub}(E)$. By induction hypothesis and Proposition 3.21, we get that both $\operatorname{Ext}^1(Y, E) \subseteq \mathcal{E}$ and $\operatorname{Ext}^1(E, Y) \subseteq \mathcal{E}$. Therefore, by Lemma 3.20, we also have that $\operatorname{Ext}^1(Y, X) \subseteq \mathcal{E}$.

In either case, we prove that $\operatorname{Ext}^1(Y, X) \subseteq \mathcal{E}$. So $\mathrm{GS}^{k+1}_{\mathcal{E}}(\mathscr{C})$ is $\mathcal{E}$-adapted. We get the desired result on $\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C})$ for all $i \geq 0$. So, $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is $\mathcal{E}$-adapted. ■

3.4. **Preserving extension closure.** In the following, we will show that $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is extension-closed whenever $\mathscr{C} \subseteq \mathscr{A}$ is an $\mathcal{E}$-adapted subcategory closed under extensions. To this end, we prove the following lemma.

Lemma 3.23. *Let $\mathscr{C} \subset \mathscr{A}$ be an $\mathcal{E}$-adapted subcategory closed under extensions. If there is a short exact sequence*

$$0 \longrightarrow X \rightarrowtail E \twoheadrightarrow U \longrightarrow 0,$$

with $X \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ and $U \in \mathscr{C}$, then $E \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$.

Proof. It is enough to show that, for any $m \in \mathbb{N}$, if we have a short exact sequence

$$0 \longrightarrow X \rightarrowtail E \twoheadrightarrow U \longrightarrow 0,$$

with $X \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$ and $U \in \mathscr{C}$, then $E \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$. We proceed by induction on $m \in \mathbb{N}$.

If $m = 0$, then $X \in \mathrm{GS}^0_{\mathcal{E}}(T) = \mathscr{C}$. As $\mathscr{C}$ is closed under extensions, the result follows. Fix $m \in \mathbb{N}$. Assume that if there is a short exact sequence

$$0 \longrightarrow X \rightarrowtail E \twoheadrightarrow U \longrightarrow 0,$$

with $X \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$ and $U \in \mathscr{C}$, then $E \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$.

Consider a short exact sequence

$$0 \longrightarrow Y \rightarrowtail E' \twoheadrightarrow U' \longrightarrow 0,$$

with $Y \in \mathrm{GS}^{m+1}_{\mathcal{E}}(\mathscr{C})$ and $U' \in \mathscr{C}$. Then we have that $Y \in \mathrm{Sub}_{\mathcal{E}}(\mathrm{GS}^m_{\mathcal{E}}(\mathscr{C}))$ or $Y \in \mathrm{Gen}_{\mathcal{E}}(\mathrm{GS}^m_{\mathcal{E}}(\mathscr{C}))$.

Suppose $Y \in \mathrm{Gen}_{\mathcal{E}}(\mathrm{GS}^m_{\mathcal{E}}(\mathscr{C}))$. Let $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$ such that $Y \in \mathrm{Gen}_{\mathcal{E}}(F)$. Note that $\mathrm{Gen}_{\mathcal{E}}(F) \subseteq \mathrm{Gen}(F)$. By Proposition 3.22, we have that $\mathrm{Ext}^1(U, F) \subseteq \mathcal{E}$. Using Lemma 3.17, we have:

- any $\xi \in \mathrm{Ext}^1(U', Y)$ is obtained by a pushout of some $\eta \in \mathrm{Ext}^1(U', F)$ along an $\mathcal{E}$-epimorphism $g : F \twoheadrightarrow Y$; and,
- $\mathrm{Ext}^1(U', Y) \subseteq \mathcal{E}$.

We obtain the following commutative diagram.

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & K' \arrow[r, tail] \arrow[d, tail] & K'' \arrow[r, two heads] \arrow[d, tail] & 0 \arrow[r] \arrow[d, tail] & 0 \\
0 \arrow[r] & F \arrow[r, tail] \arrow[d, two heads] & E'' \arrow[r, two heads] \arrow[d, two heads] & U' \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & Y \arrow[r, tail] \arrow[d] & PO \arrow[r, two heads] \arrow[d] & U' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

The short sequences given by the three columns and those given by the last two rows are exact. By Lemma 3.6, the short sequence given by the first row is exact, too. Therefore $K' \cong K''$.

As $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$, we have that $E'' \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ by induction hypothesis. So, in the diagram above, the short sequences given by the three rows and those given by the first and third columns are $\mathcal{E}$-exact. By Lemma 3.6, the short sequence given by the middle column is $\mathcal{E}$-exact. Thus $PO \in \mathrm{Gen}_{\mathcal{E}}(E'') \subseteq \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$.

As every short exact sequence in $\mathrm{Ext}^1(U', Y)$ is obtained by a pushout of a short exact sequence in $\mathrm{Ext}^1(U', F)$, we get that $PO \cong E' \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$.

Assume that $Y \in \mathrm{Sub}_{\mathcal{E}}(\mathrm{GS}^m_{\mathcal{E}}(\mathscr{C}))$. There exists a short exact sequence

$$0 \longrightarrow Y \rightarrowtail F \twoheadrightarrow C \longrightarrow 0,$$

with $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$. We obtain the following pushout diagram.

$$\begin{tikzcd}
& 0 & 0 & 0 & \\
0 \arrow[r] & C' \arrow[r, tail] \arrow[u] & C'' \arrow[r, two heads] \arrow[u] & 0 \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & F \arrow[r, tail] \arrow[u, two heads] & PO \arrow[r, two heads] \arrow[u, two heads] & U' \arrow[r] \arrow[u, two heads] & 0 \\
0 \arrow[r] & Y \arrow[r, tail] \arrow[u, tail] & E' \arrow[r, two heads] \arrow[u, tail] & U' \arrow[r] \arrow[u, equal] & 0 \\
& 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

The short sequences given by the three columns and those given by the last two rows are exact. By Lemma 3.6, the short sequence given by the first row is exact, too. So $C' \cong C''$.

As $F, Y, T \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$, the two last rows are in $\mathcal{E}$ by Proposition 3.22. Moreover we know that $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$, which implies that $PO \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ by induction hypothesis.

The short sequences given by the three rows and those given by the first and third columns are $\mathcal{E}$-exact. By Lemma 3.6, the short sequence given by the middle column is also $\mathcal{E}$-exact. Then, $E' \in \mathrm{Sub}_{\mathcal{E}}(PO) \subseteq \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$.

In either case, we get $E' \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$. This completes the proof by induction. ∎

Proposition 3.24. *Let $\mathscr{C} \subseteq \mathscr{A}$ be an $\mathcal{E}$-adapted subcategory closed under extensions. Then $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is closed under extensions.*

Proof. Let us prove by induction that, for any $m \in \mathbb{N}$, if we have a short exact sequence

$$0 \longrightarrow X \rightarrowtail E \twoheadrightarrow Y \longrightarrow 0,$$

with $X \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ and $Y \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$, then $E \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$. The $m = 0$ case follows Lemma 3.23.

Fix $m \in \mathbb{N}$. Assume that if we have the short exact sequence

$$0 \longrightarrow U \rightarrowtail E' \twoheadrightarrow V \longrightarrow 0,$$

with $U \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ and $V \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$, then $E' \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$. Consider a short exact sequence

$$0 \longrightarrow X \rightarrowtail E \twoheadrightarrow Y \longrightarrow 0,$$

with $X \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ and $Y \in \mathrm{GS}^{m+1}_{\mathcal{E}}(\mathscr{C})$. As before, we must consider two cases.

Suppose $Y \in \mathrm{Sub}_{\mathcal{E}}(F)$ for some $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$. By Proposition 3.22, we have that $\mathrm{Ext}^1(F, X) \subseteq \mathcal{E}$. By Lemma 3.19, we have:

- any $\xi \in \mathrm{Ext}^1(Y, X)$ is obtained by a pullback of some $\eta \in \mathrm{Ext}^1(F, X)$ along an $\mathcal{E}$-monomorphism $f : Y \rightarrowtail F$; and,
- $\mathrm{Ext}^1(Y, X) \subseteq \mathcal{E}$.

We obtain the following commutative diagram.

$$
\begin{tikzcd}
 & 0 & 0 & 0 & \\
0 \arrow[r] & 0 \arrow[r, tail] \arrow[u] & C'' \arrow[r, two heads] \arrow[u] & C' \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & X \arrow[r, tail] \arrow[u, two heads] & E'' \arrow[r, two heads] \arrow[u, two heads] & F \arrow[r] \arrow[u, two heads] & 0 \\
0 \arrow[r] & X \arrow[r, tail] \arrow[u, equal] & PB \arrow[r, two heads] \arrow[u, tail] & Y \arrow[r] \arrow[u, tail] & 0 \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}
$$

The short sequences given by the three columns and those given by the last two rows are exact. By Lemma 3.6, the short sequence given by the first row is exact, too. Then $C'' \cong C$. Moreover, as $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$, we have that $E'' \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ by induction hypothesis.

The short exact sequences given by the three rows and those given by the first and third columns are in $\mathcal{E}$. By Lemma 3.6, the short sequence given by the middle column is also $\mathcal{E}$-exact. So $PB \in \mathrm{Sub}_{\mathcal{E}}(E'') \subseteq \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$. By the fact that every short exact sequence in $\mathrm{Ext}^1(Y, X)$ is obtained by a pullback of a short exact sequence in $\mathrm{Ext}^1(E', X)$, we get that $PB \cong E \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$.

Now, assume that $Y \in \mathrm{Gen}_{\mathcal{E}}(F)$ for some $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$. Therefore, there exists a short exact sequence

$$
\begin{tikzcd}
0 \arrow[r] & K' \arrow[r, tail] & F \arrow[r, two heads] & Y \arrow[r] & 0
\end{tikzcd} \qquad \in \mathcal{E}.
$$

We obtain the following pullback diagram.

$$
\begin{tikzcd}
 & 0 & 0 & 0 & \\
0 \arrow[r] & X \arrow[r, tail] \arrow[u] & E \arrow[r, two heads] \arrow[u] & Y \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & X \arrow[r, tail] \arrow[u, equal] & PB \arrow[r, two heads] \arrow[u, two heads] & F \arrow[r] \arrow[u, two heads] & 0 \\
0 \arrow[r] & 0 \arrow[r, tail] \arrow[u, tail] & K \arrow[r, two heads] \arrow[u, tail] & K' \arrow[r] \arrow[u, tail] & 0 \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}
$$

Once again, by Lemma 3.6, the short sequence given by the last row is exact, and so $K \cong K'$. Furthermore, we have that $F, X, Y \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$. So the first two rows are in $\mathcal{E}$ by Proposition 3.22. Moreover $F \in \mathrm{GS}^m_{\mathcal{E}}(\mathscr{C})$. So we have that $PB \in \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ by induction hypothesis. Using Lemma 3.6, the short sequence given by the middle column is $\mathcal{E}$-exact. Therefore $E \in \mathrm{Gen}_{\mathcal{E}}(PB) \subseteq \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$.

This completes the induction proof, and allows us to get that $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is closed under extensions. ■

We can state the following corollary, which is a direct consequence of Lemma 3.11 (b) and the last proposition.

Corollary 3.25. *Let $\mathscr{C} \subseteq \mathscr{A}$ be an $\mathcal{E}$-adapted subcategory closed under extensions. Then the following statements hold:*

(a) $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ *is an $\mathcal{E}$-torsion class;*

(b) $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ *is an $\mathcal{E}$-cotorsion class;*

(c) $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ *is $\mathcal{E}$-Serre; and,*

(d) $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ *is the smallest $\mathcal{E}$-Serre category containing $\mathscr{C}$.*

Note the extension-closed assumption on $\mathscr{C}$ is essential.

Example 3.26. Consider the exact category $(\operatorname{rep} Q, \mathcal{E}_\diamond)$, where $Q = 1 \longrightarrow 2 \longrightarrow 3$. Let $\mathscr{C} = \operatorname{add}(X_{[\![2,3]\!]}, X_{[\![1,2]\!]})$ (See Figure 2). We can check that $\mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C}) = \mathscr{C}$, and $\mathscr{C}$ is $\mathcal{E}_\diamond$-adapted but not closed under extensions.

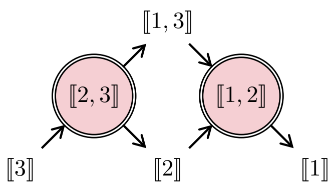

FIGURE 2. The Auslander–Reiten quiver of Q in Example 3.26. We have $\mathscr{C} = \operatorname{add}(\circledcirc) = \mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C})$, and we can check that $\mathscr{C}$ is not closed under extensions even though it is $\mathcal{E}_\diamond$-adapted.

3.5. **Tilting objects in hereditary subcategories.** In this section, we highlight some interesting properties of the $\mathrm{GS}_{\mathcal{E}}$-operator with tilting objects in $\mathscr{A}$. Recall that we assume $\mathscr{A}$ hereditary (Convention 3.14).

Definition 3.27. An object $T \in \mathscr{A}$ is said to be ***rigid*** whenever $\operatorname{Ext}^1_{\mathscr{A}}(T, T) = 0$. An object $T \in \mathscr{A}$ is said to be ***tilting*** if the following statements hold:

- T is *rigid*.
- For any projective object P, there is a short exact sequence

$$\varsigma : \qquad 0 \longrightarrow P \rightarrowtail T_0 \twoheadrightarrow T_1 \longrightarrow 0.$$

with $T_0, T_1 \in \operatorname{add}(T)$.

Denote by $\operatorname{Tilt}(\mathscr{A})$ the collection of tilting objects in $\mathscr{A}$, considered up to isomorphisms.

We recall the following crucial result, which gives us an easier criterion for tilting objects of $\mathscr{A}$.

Proposition 3.28. *Assume that $\mathscr{A}$ admits n nonisomorphic indecomposable projective objects. Then, for any $T \in \mathscr{A}$, we have that $T \in \operatorname{Tilt}(\mathscr{A})$ if and only if T is both rigid and made of n nonisomorphic indecomposable summands.*

Example 3.29. In Figure 3, we enumerate the fourteen tilting modules in $\operatorname{rep}(Q)$ for some A_4 type quiver Q.

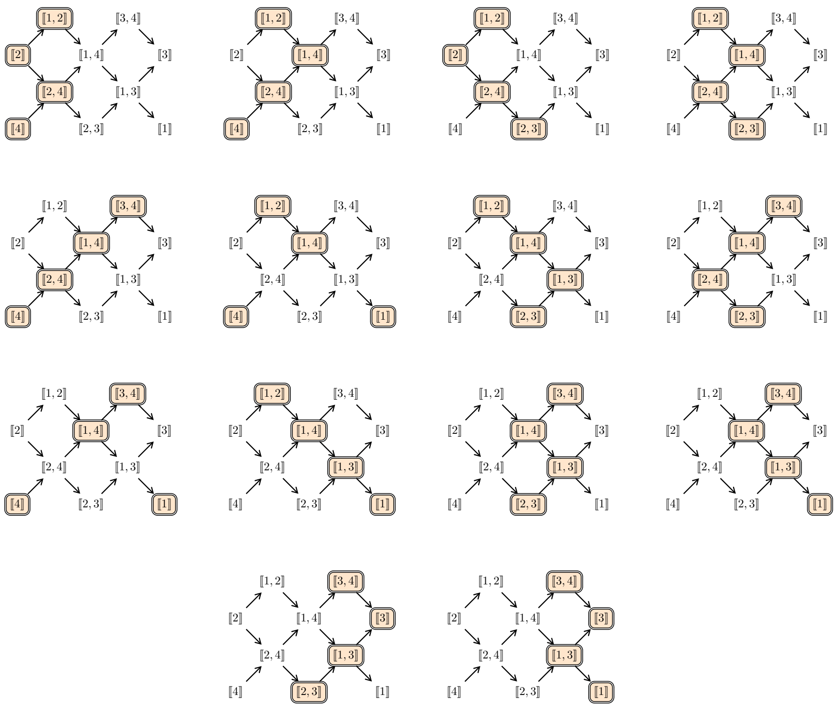

FIGURE 3. Tilting representations of rep(Q) for some A_4 type quiver Q. A tilting object in this figure is the direct sum of indecomposable objects in for one copy of the Auslander–Reiten quiver of Q

The following result allows us to handle objects in $\mathrm{Gen}_{\mathcal{E}}(T)$ and those in $\mathrm{Sub}_{\mathcal{E}}(T)$ for $T \in \mathrm{Tilt}(\mathscr{A})$. See [ASS06] for more details.

Proposition 3.30. *Let $T \in \mathrm{Tilt}(\mathscr{A})$. The following assertions hold:*

(*i*) *for any $M \in \mathrm{Gen}_{\mathcal{E}}(T)$, there exists a short $\mathcal{E}$-exact sequence*

$$0 \longrightarrow T_1 \rightarrowtail T_0 \twoheadrightarrow M \longrightarrow 0.$$

where $T_0, T_1 \in \mathrm{add}(T)$; and,

(*ii*) *for any $M \in \mathrm{Sub}_{\mathcal{E}}(T)$, there exists a short $\mathcal{E}$-exact sequence*

$$0 \longrightarrow M \rightarrowtail \Theta_0 \twoheadrightarrow \Theta_1 \longrightarrow 0.$$

where $\Theta_0, \Theta_1 \in \mathrm{add}(T)$.

Before being further in the study of $\mathrm{GS}_{\mathcal{E}}(T)$ for any rigid object $T \in \mathscr{A}$, we can state a few noticeable results related to $\mathcal{E}$-adaptability.

Lemma 3.31. *For any rigid object $T \in \mathscr{A}$, the category $\mathrm{add}(T)$ is $\mathcal{E}$-adapted and closed under extension.*

Thanks to what we proved previously, we have the following result.

Corollary 3.32. *If $T \in \mathscr{A}$ is rigid, then $\mathrm{GS}_{\mathcal{E}}(T)$ is $\mathcal{E}$-adapted and closed under extension.*

Proof. It follows from Proposition 3.24 and Lemma 3.31. ■

Theorem 3.33. *Let $T \in \mathrm{Tilt}(\mathscr{A})$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is a maximal $\mathcal{E}$-adapted subcategory of $\mathscr{A}$ closed under extensions which contains T.*

Proof. Let $\mathscr{C} \subseteq \mathscr{A}$ be an $\mathcal{E}$-adapted subcategory closed under extensions that contains T. By Convention 3.1, to prove that $\mathscr{C} \subseteq \mathrm{GS}_{\mathcal{E}}(T)$, it is enough to show that all the objects in $\mathscr{C}$ are in $\mathrm{GS}_{\mathcal{E}}(T)$.

Consider $V \in \mathscr{C}$. Suppose $V \in \mathrm{proj}(\mathscr{A})$. Since T is tilting, by Definition 3.27, there is a short exact sequence

$$\xi: \qquad 0 \longrightarrow V \rightarrowtail \Theta_0 \twoheadrightarrow \Theta_1 \longrightarrow 0,$$

with $\Theta_0, \Theta_1 \in \mathrm{add}(T)$. Since $V, \Theta_1 \in \mathscr{C}$ which is $\mathcal{E}$-adapted, then $\xi \in \mathcal{E}$. Thus $V \in \mathrm{Sub}_{\mathcal{E}}(T_0) \subseteq \mathrm{GS}_{\mathcal{E}}(T)$.

Assume that $V \notin \mathrm{proj}(\mathscr{A})$. Consider the projective resolution of V

$$0 \longrightarrow P_1 \rightarrowtail P_0 \twoheadrightarrow V \longrightarrow 0$$

with $P_0, P_1 \in \mathrm{proj}(\mathscr{A})$. Since T is tilting, by Definition 3.27, there is a short exact sequence

$$0 \longrightarrow P_1 \overset{\psi}{\rightarrowtail} \Theta_0 \twoheadrightarrow \Theta_1 \longrightarrow 0,$$

with $\Theta_0, \Theta_1 \in \mathrm{add}(T)$. We obtain the following diagram by taking the pushout along ψ and by using the cokernel lifting property.

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & P_1 \arrow[r, tail] \arrow[d, tail, "\psi"] & P_0 \arrow[r, two heads] \arrow[d, tail] & V \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \Theta_0 \arrow[r, tail] \arrow[d, two heads] & PO \arrow[r, two heads] \arrow[d, two heads] & V \arrow[r] \arrow[d, two heads] & 0 \\
0 \arrow[r] & \Theta_1 \arrow[r, tail] \arrow[d] & C \arrow[r, two heads] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

As the short sequences given by the three columns and those given by the first two rows of the above diagram are exact, by Lemma 3.6, the short sequence given by the last row is exact too. So $C \cong \Theta_1$. Since $V, \Theta_0 \in \mathscr{C}$, by assumption on $\mathscr{C}$, then the middle row lies in $\mathcal{E}$, and $W = PO \in \mathscr{C}$.

Since T is tilting, by Definition 3.27, there is also a short exact sequence

$$0 \longrightarrow P_0 \overset{\phi}{\rightarrowtail} \Theta_2 \twoheadrightarrow \Theta_3 \longrightarrow 0,$$

with $\Theta_2, \Theta_3 \in \mathrm{add}(T)$. Given the short exact sequence and the middle column of the diagram above, we obtain the following diagram by taking the pushout along

ϕ and applying the cokernel lifting property again.

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & P_0 \arrow[r, tail, "\phi"] \arrow[d, tail] & \Theta_2 \arrow[r, two heads] \arrow[d, tail] & \Theta_3 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & W \arrow[r, tail] \arrow[d, two heads] & PO' \arrow[r, two heads] \arrow[d, two heads] & C' \arrow[r] \arrow[d, two heads] & 0 \\
0 \arrow[r] & \Theta_1 \arrow[r, equal] \arrow[d] & \Theta_1 \arrow[r, two heads] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

As the short sequences given by the three rows and those given by the first two columns of the above diagram are exact, by Lemma 3.6, the short sequence given by the last column is exact too. So $C' \cong \Theta_3$. Since T is tilting, the middle column splits, hence $PO' \cong \Theta_1 \oplus \Theta_2$.

We get the following short exact sequence.

$$\varsigma : \qquad 0 \longrightarrow W \rightarrowtail \Theta_1 \oplus \Theta_2 \twoheadrightarrow \Theta_3 \longrightarrow 0$$

As $W, \Theta_3 \in \mathscr{C}$, as $\mathscr{C}$ is $\mathcal{E}$-adapted, then $\varsigma \in \mathcal{E}$ and $W \in \mathrm{Sub}_{\mathcal{E}}(\Theta_1 \oplus \Theta_2) \subseteq \mathrm{GS}_{\mathcal{E}}(T)$.

We finally obtain the following short exact sequence from the middle row of the first diagram

$$\mu : \qquad 0 \longrightarrow T_0 \rightarrowtail W \twoheadrightarrow V \longrightarrow 0$$

As $\mu \in \mathcal{E}$ and $W \in \mathrm{GS}_{\mathcal{E}}(T)$, we get that $V \in \mathrm{Gen}_{\mathcal{E}}(W) \subseteq \mathrm{GS}_{\mathcal{E}}(T)$.

So all the objects in $\mathscr{C}$ are in $\mathrm{GS}_{\mathcal{E}}(T)$, and therefore, we get the desired result. ∎

Following the proof, the corollary below occurs.

Corollary 3.34. *Let $T \in \mathrm{Tilt}(\mathscr{A})$ be a tilting object. The following assertions hold:*

- (a) *we have $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}^2_{\mathcal{E}}(T)$, and moreover, if T is an $\mathcal{E}$-projective object, then $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}^1_{\mathcal{E}}(T)$; and,*
- (b) *$\mathrm{GS}_{\mathcal{E}}(T)$ is the unique $\mathcal{E}$-adapted $\mathcal{E}$-Serre subcategory of $\mathscr{A}$ containing T.*

Proof. The assertion (a) follows directly from the proof of Theorem 3.33. The assertion (b) is a direct consequence of both Theorem 3.33 and Corollary 3.25 (d). ∎

Note that, in general, a maximal $\mathcal{E}$-adapted subcategory is not extension-closed.

Example 3.35. Consider $\mathscr{A} = \mathrm{rep}\, Q$ with $Q = 1 \longrightarrow 2 \longrightarrow 3$. Fix the exact structure $\mathcal{E}'$ generated by the two following Auslander–Reiten sequences.

$$\xi_1 : \qquad 0 \longrightarrow X_{[\![3]\!]} \rightarrowtail X_{[\![2,3]\!]} \twoheadrightarrow X_{[\![2]\!]} \longrightarrow 0$$

$$\xi_2 : \qquad 0 \longrightarrow X_{[\![2]\!]} \rightarrowtail X_{[\![1,2]\!]} \twoheadrightarrow X_{[\![1]\!]} \longrightarrow 0$$

Let $\mathscr{C} = \mathrm{add}(X_{[\![1]\!]}, X_{[\![2]\!]}, X_{[\![3]\!]}, X_{[\![1;3]\!]})$ (see Figure 4). We can check that $\mathscr{C}$ is a maximal $\mathcal{E}'$-adapted subcategory of $\mathscr{A}$, but $\mathscr{C}$ is not closed under extension.

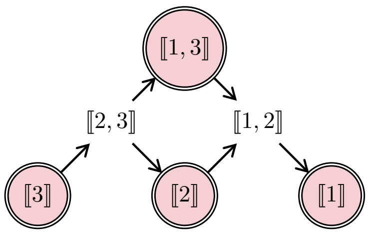

FIGURE 4. The Auslander–Reiten quiver of Q in Example 3.26. We have $\mathscr{C} = \operatorname{add}\left(\circledcirc\right) = \mathrm{GS}_{\mathcal{E}'}(\mathscr{C})$, and we can check that $\mathscr{C}$ is not closed under extensions even though it is a maximal $\mathcal{E}'$-adapted subcategory of $\mathscr{A}$.

4. Jordan recoverability

Let (Q, R) be a bounded quiver. In this section, we recall the notion of Jordan recoverability on subcategories of $\operatorname{rep}(Q, R)$, and a few results relative to this subject. We refer the reader to [GPT23, Deq23b, Deq23a].

4.1. Jordan recoverability.

Let $E \in \operatorname{rep}(Q, R)$. A ***nilpotent endomorphism*** of E is a morphism $N : E \longrightarrow E$ such that there exists a $k \in \mathbb{N}^*$ such that $N^k = 0$, or equivalently, for all $q \in Q_0$, the linear map N_q is nilpotent. We denote the set of nilpotent endomorphisms of E by $\mathsf{NEnd}(E)$. [GPT23] shows that $\mathsf{NEnd}(E)$ is an irreducible algebraic variety.

An ***integer partition*** is a finite weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_k)$ of positive integers. We also consider, by convention, that (0) is an integer partition called *zero partition*. The ***parts*** of λ are the values, counted with multiplicities, taken by λ. The ***size*** of λ, denoted by $|\lambda|$, is the sum of all its parts; meaning $|\lambda| = \lambda_1 + \ldots + \lambda_k$. More precisely, if $|\lambda| = m$ for some $m \in \mathbb{N}$, we say that λ is an integer partition of m, and we write $\lambda \vdash m$. The ***length*** of λ, denoted by $\ell(\lambda)$, is the number of parts of λ; meaning $\ell(\lambda) = k$. Let $m \in \mathbb{N}$. We define the ***dominance order*** $\leqslant$ on integer partitions of m as follows: for any $\lambda, \mu \vdash m$, we say that $\lambda \leqslant \mu$ whenever for all $1 \leqslant i \leqslant \min(\ell(\lambda), \ell(\mu))$, we have $\lambda_1 + \ldots + \lambda_i \leqslant \mu_1 + \ldots + \mu_i$.

Let $p \in \mathbb{N}^*$. We extend some notions to p-tuples of integer partitions, which are called ***p-integer partitions***. Let $\boldsymbol{\lambda} = (\lambda^1, \dots, \lambda^p)$ be a p-integer partition. The ***size vector*** of $\boldsymbol{\lambda}$ is $|\boldsymbol{\lambda}| = (|\lambda^1|, \dots, |\lambda^p|)$. Given $\boldsymbol{d} = (d_1, \dots, d_p) \in \mathbb{N}^p$, we say that $\boldsymbol{\lambda}$ is a p-integer partition of $\boldsymbol{d}$ whenever $|\boldsymbol{\lambda}| = \boldsymbol{d}$. In such case, we write $\boldsymbol{\lambda} \vdash \boldsymbol{d}$. We define the *dominance order* on p-integer partitions of $\boldsymbol{d}$ as follows: for any $\boldsymbol{\lambda}, \boldsymbol{\mu} \vdash \boldsymbol{d}$, we say that $\boldsymbol{\lambda} \trianglelefteq \boldsymbol{\mu}$ if, for all $j \in \{1, \dots, p\}$, we have $\lambda^j \leqslant \mu^j$.

Let $N \in \mathsf{NEnd}(E)$. For any $q \in Q_0$, we denote $\mathrm{JF}(N_q)$ the *Jordan form* of the nilpotent linear map N_q, understood as an integer partition whose parts are the sizes of the Jordan blocks. So $\mathrm{JF}(N_q) \vdash d_q$ for any $q \in Q_0$. Write $\mathbf{JF}(N)$ for the ***Jordan form*** of N, defined as the $\#Q_0$-integer partitions $(\mathrm{JF}(N_q))_{q \in Q_0}$. Note that $\mathbf{JF}(N) \vdash \mathbf{dim}(E)$.

Theorem 4.1 ([GPT23])**.** *Let (Q, R) be a bounded quiver, $\mathbb{K}$ be an algebraically closed field, and $E \in \operatorname{rep}(Q, R)$. Then $\mathbf{JF}$ admits a maximum value on $\mathsf{NEnd}(E)$, and it is attained in a dense open (for Zariski topology) subset of $\mathsf{NEnd}(E)$.*

We define $\mathrm{GenJF}(E)$ as the maximal value of $\mathbf{JF}$ on $\mathsf{NEnd}(E)$. We call it ***generic Jordan form*** of E. Note that GenJF defines an invariant on isomorphism classes of representations of (Q, R). Still, it is not a complete invariant in general: that is, we can find $E, F \in \mathrm{rep}(Q, R)$ such that $E \ncong F$ and $\mathrm{GenJF}(E) = \mathrm{GenJF}(F)$. However, we can focus on the subcategories of $\mathrm{rep}(Q, R)$ such that GenJF is complete.

Definition 4.2. Let (Q, R) be a bounded quiver. A subcategory $\mathscr{C}$ or $\mathrm{rep}(Q, R)$ is said to be ***Jordan recoverable*** if, for any $\#Q_0$-integer partition $\boldsymbol{\lambda}$, there exists at most one $E \in \mathrm{rep}(Q, R)$, up to isomorphism, such that $\mathrm{GenJF}(E) = \boldsymbol{\lambda}$.

To determine whether a subcategory of $\mathrm{rep}(Q, R)$ is Jordan recoverable, we are led to solve a system of tropical equations, which can be a challenging task. The following section introduces an algebraic way to build a representation from any $\#Q_0$-integer partition. Under some more restrictions, it allows one to return a representation of (Q, R) from its generic Jordan form.

4.2. **Canonical Jordan recoverability.** Let $\boldsymbol{\lambda}$ be a $\#Q_0$-integer partition, and set $|\boldsymbol{\lambda}| = \boldsymbol{d}$. Consider $N = \left(N_q : \mathbb{K}^{d_q} \longrightarrow \mathbb{K}^{d_q}\right)_{q \in Q_0}$ to be a $\#Q_0$-tuple of nilpotent linear maps such that $\mathrm{JF}(N_q) = \lambda^q$. We say that a representation E is ***compatible with*** N whenever there exists $N' \in \mathsf{NEnd}(E)$ such that, for all $q \in Q_0$, the linear maps N'_q and N_q are similar. We denote by $\mathrm{rep}((Q, R), N)$ the collection of representations compatible with N.

In $\mathrm{rep}((Q, R), N)$, we ask whether there exists a dense open set (for Zariski topology) such that all its representations are isomorphic. In such a case, we define $\mathrm{GenRep}(N)$ as a representation in the corresponding dense open set. Note that $\mathrm{GenRep}(N)$ only depends on its Jordan form. Therefore, we define $\mathrm{GenRep}(\boldsymbol{\lambda})$ the ***generic representation*** of $\boldsymbol{\lambda}$ to be one of the $\mathrm{GenRep}(N)$ where N is a collection of nilpotent linear maps compatible with a nilpotent endomorphism $N' \in \mathsf{NEnd}(E)$ such that $\mathrm{JF}(N') = \boldsymbol{\lambda}$.

Definition 4.3. A subcategory $\mathscr{C} \subseteq \mathrm{rep}(Q, R)$ is said to be ***canonically Jordan recoverable*** if, for any $X \in \mathscr{C}$, $\mathrm{GenRep}(\mathrm{GenJF}(X))$ is well-defined and is isomorphic to X.

Note that, in general, given a $\#Q_0$-integer partition, it could happen that $\mathrm{GenRep}(\boldsymbol{\lambda})$ cannot be defined. We refer to [Deq23b, Examples 2.21 and 2.23] for examples in the case where (Q, R) is a gentle quiver.

The following result shows that, at least, under some restrictions, we can ensure that $\mathrm{GenRep}(\boldsymbol{\lambda})$ is well-defined for any $\#Q_0$-integer partition.

Proposition 4.4 ([GPT23, Deq23a])**.** *Let Q be an ADE-type Dynkin quiver. Then, for any $\#Q_0$-integer partition $\boldsymbol{\lambda}$,* $\mathrm{GenRep}(\boldsymbol{\lambda})$ *is well-defined.*

Now, we restrict ourselves to type A_n quivers for some $n \in \mathbb{N}^*$, and we recall useful results. Fix $n \in \mathbb{N}^*$. Let us first recall the combinatorial statement that characterizes all the canonically Jordan recoverable subcategories of type A_n.

Two intervals $K, L \in \mathcal{I}_n$ are said to be ***adjacent*** if either $b(K) = e(L) + 1$ or $b(L) = e(K)+1$. Note that K and L are not adjacent if $K \cap L \neq \varnothing$, $b(K) > e(L)+1$ or $b(L) > e(K)+1$. We say that an interval subset $\mathscr{J} \subseteq \mathcal{I}_n$ is ***adjacency-avoiding*** if, for all $(K, L) \in \mathscr{J}^2$, the intervals K and L are not adjacent.

Theorem 4.5. *Let Q be an A_n type quiver. A subcategory $\mathscr{C} \subset \mathrm{rep}(Q)$ is canonically Jordan recoverable if and only if* $\mathrm{Int}(\mathscr{C})$ *is adjacency-avoiding.*

Determining all the maximal adjacency-avoiding subsets for the inclusion of $\mathcal{I}_n$ was the primary key to proving this result. Let us recall the exact statement. Given $\mathbf{B}, \mathbf{E} \subseteq \{1, \dots n\}$, we define the interval subset

$$\mathscr{J}(\mathbf{B}, \mathbf{E}) = \{K \in \mathcal{I}_n \mid b(K) \in \mathbf{B},\ e(K) \in \mathbf{E}\}.$$

Given $\mathbf{A} \subset \{1, \dots, n\}$, we set $\mathbf{A}[1] = \{a + 1 \mid a \in \mathbf{A}\}$.

Proposition 4.6. *The following assertions hold:*

(i) *For all* $\mathbf{B}, \mathbf{E} \subseteq \{1, \dots, n\}$*, if* $(\mathbf{B}, \mathbf{E}[1])$ *is a set partition of* $\{1, \dots, n+1\}$*, then* $\mathscr{J}(\mathbf{B}, \mathbf{E})$ *is a maximal adjacency-avoiding subset of* $\mathcal{I}_n$*; and,*

(ii) *Any maximal adjacency-avoiding subset of* $\mathcal{I}$ *is equal to* $\mathscr{J}(\mathbf{B}, \mathbf{E})$ *for some* $\mathbf{B}, \mathbf{E} \subseteq \{1, \dots, n\}$ *such that* $(\mathbf{B}, \mathbf{E}[1])$ *is a set partition of* $\{1, \dots, n+1\}$*.*

Another relevant key to proving Theorem 4.5 is to show that the notion of canonical Jordan recoverability on type A quivers does not depend on the orientation of the quiver. This, indeed, is a consequence of the techniques we will not detail here.

Define the subcategory $\mathscr{C}(\mathbf{B}, \mathbf{E}) = \mathrm{Cat}(\mathscr{J}(\mathbf{B}, \mathbf{E}))$ for any $\mathbf{B}, \mathbf{E} \subseteq \{1, \dots, n\}$. We can state the following result.

Theorem 4.7. *Let* Q *be an* A_n *type quiver. A subcategory* $\mathscr{C} \subseteq \mathrm{rep}(Q)$ *is a maximal canonically Jordan recoverable subcategory if and only if* $\mathscr{C} = \mathscr{C}(\mathbf{B}, \mathbf{E})$ *for some* $\mathbf{B}, \mathbf{E} \subseteq \{1, \dots, n\}$ *such that* $(\mathbf{B}, \mathbf{E}[1])$ *is a set partition of* $\{1, \dots, n+1\}$*.*

Example 4.8. By taking back the A_7 type quiver Q in Example 3.13, we explicit, in Figure 5, a maximal canonically Jordan recoverable subcategory of $\mathrm{rep}(Q)$ given by the pair $(\mathbf{B}, \mathbf{E})$ with $\mathbf{B} = \{1, 2, 3, 5\}$ and $\mathbf{E} = \{3, 5, 6, 7\}$.

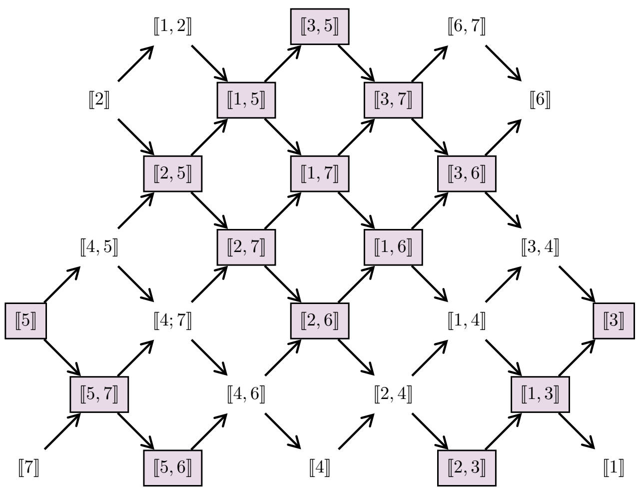

Figure 5. Example of a maximal canonically Jordan recoverable subcategory $\mathscr{C} = \mathrm{add}\,(\square)$ in $\mathrm{rep}(Q)$, for Q given in Figure 1.

There is also a representation-theoretic interpretation of Theorem 4.5, and it starts by noticing the following result, an obvious consequence from Theorem 2.7.

Lemma 4.9. *Let* Q *be an* A_n *type quiver. Then two intervals* $K, L \in \mathcal{I}_n$ *are adjacent if and only if one of the following assertions holds:*

- *there exists a short exact sequence*

$$0 \longrightarrow X_K \rightarrowtail E \twoheadrightarrow X_L \longrightarrow 0,$$

  such that $E \in \mathrm{Ind}(Q)$*; or,*
- *there exists a short exact sequence*

$$0 \longrightarrow X_L \rightarrowtail E \twoheadrightarrow X_K \longrightarrow 0,$$

  such that $E \in \mathrm{Ind}(Q)$*.*

This allows us to state the following result.

Theorem 4.10. *Let Q be an A_n type quiver. A subcategory $\mathscr{C} \subset \mathrm{rep}(Q)$ is canonically Jordan recoverable if and only if for any nonsplit short exact sequence*

$$0 \longrightarrow E \rightarrowtail F \twoheadrightarrow G \longrightarrow 0,$$

we have $F \notin \mathrm{Ind}(Q)$ whenever $E, G \in \mathscr{C}$.

Remark 4.11. Two things:

- One can restate the result by saying that a subcategory $\mathscr{C} \subset \mathrm{rep}(Q)$ is canonically Jordan recoverable if and only if all the short exact sequences whose extremities are in $\mathscr{C}$ must be in $\mathcal{E}_\diamond$.
- One can do a parallel with the definition of *maximal almost rigid* representations [BGMS23]. A representation $M \in \mathrm{rep}(Q)$ is said to be ***maximal almost rigid*** if for any nonsplit short exact sequence

$$0 \longrightarrow E \rightarrowtail F \twoheadrightarrow G \longrightarrow 0,$$

  where E and G are indecomposable summands of M, we have $F \in \mathrm{ind}(Q)$.

In the following section, we exhibit a strong link between GS operators on $\mathrm{rep}(Q)$ for $\mathcal{E}_\diamond$ and maximal canonically Jordan recoverable subcategories.

5. Type ADE algebras case

Convention 5.1. We fix Q an ADE type quiver. We endow $\mathrm{rep}(Q)$ with an exact structure $\mathcal{E}$.

Denote by $\mathrm{Tilt}(Q)$ the collection $\mathrm{Tilt}(\mathrm{rep}(Q))$. Given any finite collection of additive full subcategories $\mathscr{C}_i \subseteq \mathrm{rep}\, Q$, we write $\bigoplus_i \mathscr{C}_i$ for the additive full subcategory of $\mathrm{rep}(Q)$ generated by all the subcategories $\mathscr{C}_i$.

In this section, our aim is to describe, for any $T \in \mathrm{Tilt}(Q)$, that the subcategory $\mathrm{GS}_{\mathcal{E}}(T)$ in terms of a full and additive subcategory generated by tilting objects that are related to T via so-called $\mathcal{E}$*-mutations*. We apply this description to canonically Jordan recoverable subcategories, by restricting ourselves to type A quivers.

5.1. Tilting $\mathcal{E}$-mutations.

In this subsection, we recall the notion of $\mathcal{E}$*-mutations* on $\mathrm{Tilt}(Q)$, and we enumerate crucial and well-known results on left and right approximations, stated in our type ADE quivers framework.

Definition 5.2 ([AS93])**.** Let $\mathscr{C} \subseteq \mathrm{rep}(Q)$ be a subcategory, and $E \in \mathrm{rep}(Q)$.

- A ***left $\mathscr{C}$-approximation*** of E is a morphism $f : E \longrightarrow C$ with $C \in \mathscr{C}$ such that every morphism $\varphi : E \longrightarrow C'$ where $C' \in \mathscr{C}$ factors through f.

$$\begin{tikzcd} E \arrow[rr, "\forall\varphi"] \arrow[dr, "f"'] & & C' \\ & C \arrow[ur, dashed, "\exists\psi"'] & \end{tikzcd}$$

- Such a left $\mathscr{C}$-approximation of E is said to be ***minimal*** if any map $\alpha : C \longrightarrow C$ such that $\alpha f = f$ is bijective. Moreover, if f is an $\mathcal{E}$-monomorphism, it is a ***left $(\mathscr{C}, \mathcal{E})$-approximation*** of E.
- A ***right $\mathscr{C}$-approximation*** of E is a morphism $g : C \longrightarrow E$ with $C \in \mathscr{C}$ such that every morphism $\varphi : C' \longrightarrow E$ where $C' \in \mathscr{C}$ factors through g.

$$\begin{tikzcd} C' \arrow[rr, "\forall\varphi"] \arrow[dr, dashed, "\exists\psi"'] & & E \\ & C \arrow[ur, "g"'] & \end{tikzcd}$$

- Such a right $\mathscr{C}$-approximation of E is said to be ***minimal*** if any map $\beta : C \longrightarrow C$ such that $g\beta = g$ is bijective. Moreover, if g is an $\mathcal{E}$-epimorphsim, it is a ***right $(\mathscr{C}, \mathcal{E})$-approximation*** of E.

This first proposition states that, given a subcategory $\mathscr{C} \subset \mathrm{rep}(Q)$, from any $\mathscr{C}$-approximation, we can build a minimal one.

Proposition 5.3. *Let $\mathscr{C} \subseteq \mathrm{rep}(Q)$ be a subcategory. Let $E \in \mathrm{rep}(Q)$. The following assertions hold:*

(i) *if E admits a left $\mathscr{C}$-approximation, then E admits a minimal left $\mathscr{C}$-approximation.*

(ii) *if E admits a right $\mathscr{C}$-approximation, then E admits a minimal right $\mathscr{C}$-approximation.*

The following statement asserts that, for objects chosen rightfully relative to $\mathscr{C}$, any $\mathscr{C}$-approximation is an $\mathcal{E}$-epimorphism or an $\mathcal{E}$-monomorphism.

Proposition 5.4. *Let $\mathscr{C} \subseteq \mathrm{rep}(Q)$ be a subcategory. Let $E \in \mathrm{rep}(Q)$. The following assertions hold:*

(i) *if $E \in \mathrm{Gen}_{\mathcal{E}}(\mathscr{C})$, then any minimal right $\mathscr{C}$-approximation $g : C \longrightarrow E$ is an $\mathcal{E}$-epimorphism.*

(ii) *if $E \in \mathrm{Sub}_{\mathcal{E}}(\mathscr{C})$, then any minimal left $\mathscr{C}$-approximation $f : E \longrightarrow C$ is a $\mathcal{E}$-monomorphism.*

Lemma 5.5. *Let $M \in \mathrm{rep}(Q)$. The following assertions hold:*

(i) *Any $E \in \mathrm{Gen}_{\mathcal{E}}(M)$ admits a right $(\mathrm{add}(M), \mathcal{E})$-approximation; and,*

(ii) *Any $E \in \mathrm{Sub}_{\mathcal{E}}(M)$ admits a left $(\mathrm{add}(M), \mathcal{E})$-approximation.*

Remark 5.6. In particular, for any given $M \in \mathrm{rep}(Q)$, any object $N \in \mathrm{rep}(Q)$ admits a right and a left $\mathrm{add}(M)$-approximation.

We can now define the notion of $\mathcal{E}$-mutations on tilting objects of $\mathrm{rep}(Q)$.

Definition 5.7. Let $T \in \mathrm{Tilt}(Q)$ and $U \in \mathrm{ind}(Q)$.

- We say that T admits a ***left $\mathcal{E}$-mutation*** at U if U is an indecomposable summand of T, and, by writing $T = \widetilde{T} \oplus U$, we have that U admits a left minimal $(\mathrm{add}(\widetilde{T}), \mathcal{E})$-approximation f. In such a case, we define the ***left $\mathcal{E}$-mutation*** of T at U as $\boldsymbol{\mu}^{-}_{U,\mathcal{E}}(T) = \widetilde{T} \oplus \mathrm{Coker}(f)$.
- We say that T admits a ***right $\mathcal{E}$-mutation*** at U if U is an indecomposable summand of T, and, by writing $T = \widetilde{T} \oplus U$, we have that U admit a right minimal $(\mathrm{add}(\widetilde{T}), \mathcal{E})$-approximation g. In such a case, we define the ***right $\mathcal{E}$-mutation*** of T at U as $\boldsymbol{\mu}^{+}_{U,\mathcal{E}}(T) = \widetilde{T} \oplus \mathrm{Ker}(g)$.

The following proposition allows us to talk about $\mathcal{E}$-mutations.

Proposition 5.8. *Let $T \in \mathrm{Tilt}(Q)$ and $U \in \mathrm{ind}(Q)$. Then T admits at most one left $\mathcal{E}$-mutation or at most one right $\mathcal{E}$-mutation at U. Moreover if T admits a left $\mathcal{E}$-mutation at U, then T does not admits a right $\mathcal{E}$-mutation at U.*

Remark 5.9. C. Riedtmann and A. Schofield [RS91] proved this result for $\mathcal{E} = \mathcal{E}_{\max}$, and thus this proposition follows.

Definition 5.10. Let $U \in \mathrm{rep}(Q)$. We define the $\mathcal{E}$-mutation operator on $\mathrm{Tilt}(Q)$ as it follows:

$$\forall T \in \mathrm{Tilt}(Q),\ \boldsymbol{\mu}_{U,\mathcal{E}}(T) = \begin{cases} \boldsymbol{\mu}^{-}_{U,\mathcal{E}}(T) & \text{if } T \text{ admits a left } \mathcal{E}\text{-mutation at } U; \\ \boldsymbol{\mu}^{+}_{U,\mathcal{E}}(T) & \text{if } T \text{ admits a right } \mathcal{E}\text{-mutation at } U; \\ T & \text{otherwise.} \end{cases}$$

Definition 5.11. Let $T, T' \in \mathrm{Tilt}(Q)$. We say that T' is ***$\mathcal{E}$-reachable from*** T if there exists a finite sequence $(U_1, \ldots, U_m)$ of indecomposable representations of Q, such that

$$T' \cong \boldsymbol{\mu}_{U_m,\mathcal{E}} \circ \cdots \circ \boldsymbol{\mu}_{U_1,\mathcal{E}}(T).$$

In the case where $T' \cong T$ or $\boldsymbol{\mu}_{U_i,\mathcal{E}} = \boldsymbol{\mu}^{-}_{U_i,\mathcal{E}}$ for all $i \in \{1, \ldots, m\}$, we say that T' is ***left $\mathcal{E}$-reachable*** from T. Dually, if $T' \cong T$ or $\boldsymbol{\mu}_{U_i,\mathcal{E}} = \boldsymbol{\mu}^{+}_{U_i,\mathcal{E}}$ for all $i \in \{1, \ldots, m\}$, we say that T' is ***right $\mathcal{E}$-reachable*** from T.

The relation of $\mathcal{E}$-reachability on $\mathrm{Tilt}(Q)$ defines an equivalence relation, as $\boldsymbol{\mu}_{U,\mathcal{E}}$ is an involution by Proposition 5.8. We denote this relation by $\approx_{\mathcal{E}}$. In the following, given $T \in \mathrm{Tilt}(Q)$, we denote by $[T]_{\approx_{\mathcal{E}}}$ the equivalence class of tilting representation under $\approx_{\mathcal{E}}$.

In our setting, [HU05] shows that the right (and the left) $\mathcal{E}$-reachability relation defines an order relation on $\mathrm{Tilt}(Q)$. In the following, for $T, T' \in \mathrm{Tilt}(Q)$, we write $T' \trianglelefteq_{\mathcal{E}} T$ whenever T' is right $\mathcal{E}$-reachable from T. In the case where $\mathcal{E} = \mathcal{E}_{\max}$, we simply write $\trianglelefteq$ this relation. We also write $T' \triangleleft_{\mathcal{E}} T$ whenever T *covers* T', meaning that there exists $U \in \mathrm{ind}(Q)$ such that $\boldsymbol{\mu}^{+}_{U,\mathcal{E}}(T) \cong T'$.

Example 5.12. In Figure 6, we represent the poset $(\mathrm{Tilt}(Q), \trianglelefteq)$ for Q the A_4 type quiver seen Example 3.29. We also gather together tilting representations with respect to the equivalence relation $\approx_{\mathcal{E}_\diamond}$.

Remark 5.13. By the work of [IRTT15], we know that this poset $(\mathrm{Tilt}(Q), \trianglelefteq)$ is isomorphic to the poset of the torsion classes in $\mathrm{rep}(Q)$, and, therefore, it can be

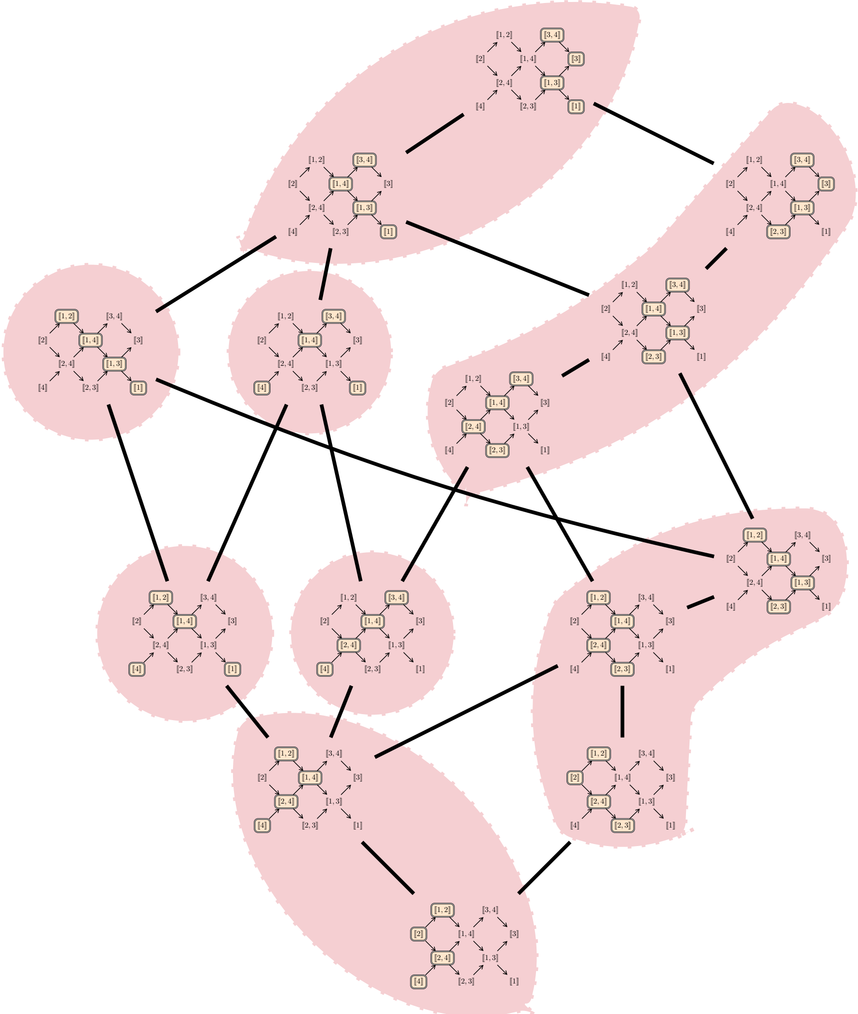

FIGURE 6. The poset $(\mathrm{Tilt}(Q), \trianglelefteq)$ for the A_4 type quiver Q seen in Example 3.29. We gather together tiltings of the same equivalence class for $\approx_{\mathcal{E}_\diamond}$.

endowed with a lattice structure. The obtained lattice is isomorphic to the positive part of the well-known *Cambrian lattice* in the sense of N. Reading [Rea06].

5.2. **Gen-Sub operators on tilting objects.** In this section, we establish that, for any $T \in \mathrm{Tilt}(Q)$, the subcategory $\mathrm{GS}_{\mathcal{E}}(T)$ is the category additively generated by tilting representations T' obtained from T by applying a finite sequence of $\mathcal{E}$-mutations. Let us consider first the following equivalence relation on $\mathrm{Tilt}(Q)$.

Definition 5.14. We define the $\mathrm{GS}_{\mathcal{E}}$***-relation***, denoted by $\sim_{\mathcal{E}}$, on $\mathrm{Tilt}(Q)$ as follows. Given $T, T' \in \mathrm{Tilt}(Q)$, write $T \sim_{\mathcal{E}} T'$ whenever we have $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}(T')$.

Corollary 5.15. *Let $T, T' \in \mathrm{Tilt}(Q)$. Then $T' \sim_{\mathcal{E}} T$ if and only if $T' \in \mathrm{GS}_{\mathcal{E}}(T)$.*

Given $T, T' \in \mathrm{Tilt}(Q)$, our aim is to prove that $T \sim_{\mathcal{E}} T'$ if and only if $T \approx_{\mathcal{E}} T'$.

Proposition 5.16. *Let $T, T' \in \mathrm{Tilt}(Q)$. If $T' \approx_{\mathcal{E}} T$, then $T' \sim_{\mathcal{E}} T$.*

Proof. To prove this proposition, it is enough to show that $\mathrm{GS}_{\mathcal{E}}(T') = \mathrm{GS}_{\mathcal{E}}(T)$ whenever $T' \cong \boldsymbol{\mu}_{U,\mathcal{E}}(T)$ for some $U \in \mathrm{ind}(Q)$. We obtain the desired result by induction on the length of the sequence of $\mathcal{E}$-mutations, which allows us to have $T' \approx_{\mathcal{E}} T$.

We have three cases to treat.

- If $\boldsymbol{\mu}_{U,\mathcal{E}}(T) \cong T$, then the result is obvious.
- If $\boldsymbol{\mu}_{U,\mathcal{E}}(T) = \boldsymbol{\mu}^{-}_{U,\mathcal{E}}(T)$, by writing $T = \widetilde{T} \oplus U$, we have the following short $\mathcal{E}$-exact sequence:

$$0 \longrightarrow U \overset{f}{\rightarrowtail} \Theta \longrightarrow\!\!\!\!\!\rightarrow \mathrm{Coker}(f) \longrightarrow 0,$$

 where f is a left minimal $(\mathrm{add}(\widetilde{T}), \mathcal{E})$-approximation of U. As $\Theta \in \mathrm{add}(T) \subset \mathrm{GS}_{\mathcal{E}}(T)$, and f is an $\mathcal{E}$-monomorphism, by Lemma 3.11, we have that $\mathrm{Coker}(f) \in \mathrm{GS}_{\mathcal{E}}(T)$. Therefore $T' \cong \widetilde{T} \oplus \mathrm{Coker}(f) \in \mathrm{GS}_{\mathcal{E}}(T)$.
- If $\boldsymbol{\mu}_{U,\mathcal{E}}(T) = \boldsymbol{\mu}^{+}_{U,\mathcal{E}}(X)$, by writing $T = \widetilde{T} \oplus U$, we have the following short $\mathcal{E}$-exact sequence:

$$0 \longrightarrow \mathrm{Ker}(f) \rightarrowtail \Theta' \overset{g}{\longrightarrow\!\!\!\!\!\rightarrow} U \longrightarrow 0,$$

 where g is a right minimal $(\mathrm{add}(\widetilde{T}), \mathcal{E})$-approximation of U. As $\Theta' \in \mathrm{add}(\widetilde{T})$ and g is an $\mathcal{E}$-epimorphism, by Lemma 3.11, we have that $\mathrm{Ker}(f) \in \mathrm{GS}_{\mathcal{E}}(T)$. So $T' \cong \widetilde{T} \oplus \mathrm{Ker}(f) \in \mathrm{GS}_{\mathcal{E}}(T)$.

In either case, we got that $T' \sim_{\mathcal{E}} T$. ■

Besides establishing that $T \sim_{\mathcal{E}} T'$ implies $T \approx_{\mathcal{E}} T'$, we also show that the class $[T]_{\approx_{\mathcal{E}}}$ additively generates $\mathrm{GS}_{\mathcal{E}}(T)$. To do so, we will have to show a few more results. First, we recall the notion of the Jacobson radical of a category.

Definition 5.17. Let $\mathscr{A}$ be an abelian category. The *(Jacobson) radical* of $\mathscr{A}$ is the two-sided ideal $\mathrm{Rad}_{\mathscr{A}}$ defined by

$$\mathrm{Rad}_{\mathscr{A}}(X, Y) = \left\{ h \in \mathrm{Hom}(X, Y) \;\middle|\; \forall g \in \mathrm{Hom}(Y, X),\ \mathrm{id}_Y - hg \text{ is invertible} \right\}$$

for all objects $X, Y \in \mathscr{A}$.

Remark 5.18. Following the definition, if $h \in \mathrm{Rad}_{\mathscr{A}}(X, Y)$, then h is not an isomorhism.

Lemma 5.19 ([ASS06])**.** *Let $\mathscr{A}$ be an abelian category and let*

$$X = \bigoplus_{i=1}^{n} X_i, \qquad Y = \bigoplus_{j=1}^{m} Y_j.$$

For $f \in \mathrm{Hom}(X, Y)$, we write $f_{ji} = \pi_{Y_j} f \iota_{X_i} \in \mathrm{Hom}(X_i, Y_j)$ using the canonical inclusions $\iota_{X_i} : X_i \to X$ and projections $\pi_{Y_j} : Y \to Y_j$. Then the following are equivalent:

(*i*) $f \in \mathrm{Rad}_{\mathscr{A}}(X, Y)$.

(*ii*) $f_{ji} \in \mathrm{Rad}_{\mathscr{A}}(X_i, Y_j)$ *for all* $1 \leq i \leq n$ *and* $1 \leq j \leq m$.

Lemma 5.20. *Let $\mathscr{A}$ be an abelian category and $M \in \mathscr{A}$. Consider a right minimal* $\operatorname{add}(M)$*-approximation g of $X \in \operatorname{Gen}(M)$. Then we have the following short exact sequence*

$$\xi : 0 \longrightarrow K_M \overset{f}{\rightarrowtail} E_M \overset{g}{\twoheadrightarrow} X \longrightarrow 0,$$

with $f \in \operatorname{Rad}_{\mathscr{A}}(K_M, E_M)$. Dually, if $X \in \operatorname{Sub}(T)$ and $f : X \to E_M$ is a left minimal $\operatorname{add}(M)$*-approximation of X, then $g \in \operatorname{Rad}_{\mathscr{A}}(E_M, \operatorname{Coker}(f))$.*

Proof. By Proposition 5.4, the map g is an epimorphism and we get the short exact sequence

$$\xi : 0 \longrightarrow K_M \overset{f}{\rightarrowtail} E_M \overset{g}{\twoheadrightarrow} X \longrightarrow 0,$$

with $E_M \in \operatorname{add}(M)$.

Consider the morphism $\mathrm{id}_{E_M} - fh \in \operatorname{End}(E_M)$ with $h \in \operatorname{Hom}(E_M, K_M)$. We have $g(\mathrm{id}_{E_M} - fh) = g - gfh = g$. Since g is minimal, $\mathrm{id}_{E_M} - fh$ is invertible for any $h \in \operatorname{Hom}(E_M, K_M)$. Hence, $f \in \operatorname{Rad}_{\mathscr{A}}(K_M, E_M)$. ■

Lemma 5.21 ([APT15])**.** *Let $T \in \operatorname{Tilt}(Q)$ and $X \in \operatorname{Gen}(T)$. Consider a right minimal* $\operatorname{add}(T)$*-approximation g of X, which is an epimorphism by Proposition 5.4. Then we have the following short exact sequence*

$$\xi : 0 \longrightarrow K_T \rightarrowtail E_T \overset{g}{\twoheadrightarrow} X \longrightarrow 0,$$

with $K_T \in \operatorname{add}(T)$. Dually, if $X \in \operatorname{Sub}(T)$ and f is a left minimal $\operatorname{add}(T)$*-approximation of X, then* $\operatorname{Coker}(f) \in \operatorname{add}(T)$*.*

Lemma 5.22. *Let $T \in \operatorname{Tilt}(Q)$ and $X \notin \operatorname{add}(T)$ indecomposable. The following assertions hold:*

(*i*) *if $X \in \operatorname{Gen}_{\mathcal{E}}(T)$, then there is an indecomposable $U \in \operatorname{add}(T)$ such that $U \in \operatorname{Sub}_{\mathcal{E}}(M)$ and $X \in \operatorname{Gen}_{\mathcal{E}}(M)$ with $T = U \oplus M$; and,*

(*ii*) *if $X \in \operatorname{Sub}_{\mathcal{E}}(T)$, then there is an indecomposable $U \in \operatorname{add}(T)$ such that $U \in \operatorname{Gen}_{\mathcal{E}}(M)$ and $X \in \operatorname{Sub}_{\mathcal{E}}(M)$ with $T = U \oplus M$.*

Proof. Consider $X \in \operatorname{Gen}_{\mathcal{E}}(T)$. By Proposition 5.3 and Lemma 5.5, there exists a right minimal $(\operatorname{add}(T), \mathcal{E})$-approximation g of X :

$$\xi : 0 \longrightarrow K_T \overset{f}{\rightarrowtail} E_T \overset{g}{\twoheadrightarrow} X \longrightarrow 0$$

with $E_T \in \operatorname{add}(T)$. Given that $X \notin \operatorname{add}(T)$, ξ does not split and $K_T \neq 0$. Moreover, $K_T \in \operatorname{add}(T)$ by Lemma 5.21 and $f \in \operatorname{Rad}_{\mathscr{A}}(K_T, E_T)$ by Lemma 5.20.

We will show that $\operatorname{add}(K_T) \cap \operatorname{add}(E_T) = 0$.

Consider any morphism $\alpha \in \operatorname{Hom}(K_T, E_T)$. Given that X is indecomposable and Q is of ADE Dynkin type, then X is rigid. By applying the functor $\operatorname{Hom}(-, X)$, we get the short exact sequence

$$\xi : 0 \longrightarrow \operatorname{Hom}(X, X) \rightarrowtail \operatorname{Hom}(E_T, X) \overset{\operatorname{Hom}(f,X)}{\twoheadrightarrow} \operatorname{Hom}(K_T, X) \longrightarrow 0,$$

Therefore, there exist $\beta \in \operatorname{Hom}(E_T, X)$ such that $\beta f = g\alpha$.

We can construct the following commutative diagram:

$$\begin{tikzcd}
 & 0 \arrow[r] & K_T \arrow[r, tail, "f"] \arrow[d, "\alpha"] & E_T \arrow[r, two heads, "g"] \arrow[d, dashed, "\beta"] & X \arrow[r] & 0 \\
0 \arrow[r] & K_T \arrow[r, tail, "f"'] & E_T \arrow[r, two heads, "g"'] & X \arrow[r] & 0 &
\end{tikzcd}$$

Since g is a right $\mathrm{add}(T)$-approximation of X, there is a morphism $\beta' \in \mathrm{End}(E_T)$ such that $\beta = g\beta'$. We get that $g(\alpha - \beta' f) = g\alpha - g\beta' f = \beta f - \beta f = 0$. By the kernel universal property, there is a morphism $\alpha' \in \mathrm{End}(K_T)$ such that $(\alpha - \beta' f) = f\alpha'$. Therefore, $\alpha = f\alpha' + \beta' f$. By Lemma 5.20, $f \in \mathrm{Rad}_{\mathrm{rep}\, Q}(K_T, E_T)$, so is α because $\mathrm{Rad}_{\mathrm{rep}\, Q}$ is a two-sided ideal of $\mathrm{rep}\, Q$.

Given that any morphism $\alpha \in \mathrm{Hom}(K_T, E_T)$ is in $\mathrm{Rad}_{\mathrm{rep}\, Q}(K_T, E_T)$, by Lemma 5.19, we get that $\mathrm{add}(K_T) \cap \mathrm{add}(E_T) = 0$.

Now, consider an indecomposable $U \in \mathrm{add}(K_T) \subseteq \mathrm{add}(T)$. Since $U \notin \mathrm{add}(E_T)$, we have $E_T \in \mathrm{add}(M)$ with $T = U \oplus M$. Therefore, since ξ is an $\mathcal{E}$-admissible short exact sequence, then $K_T \in \mathrm{Sub}_{\mathcal{E}}(M)$ and $X \in \mathrm{Gen}_{\mathcal{E}}(M)$. Hence, $U \in \mathrm{Sub}_{\mathcal{E}}(M)$ and $X \in \mathrm{Gen}_{\mathcal{E}}(M)$. ■

Remark 5.23. The above proof was mainly inspired by the proof of [AI12, Lemma 2.25].

Proposition 5.24. *Let $T \in \mathrm{Tilt}(Q)$. If T admits no left $\mathcal{E}$-mutation at any indecomposable summands, then $\mathrm{Gen}_{\mathcal{E}}(T) = \mathrm{add}(T)$. Dually, If T admits no right $\mathcal{E}$-mutation at any indecomposable summands, then $\mathrm{Sub}_{\mathcal{E}}(T) = \mathrm{add}(T)$.*

Proof. Suppose that $\mathrm{add}(T) \subsetneq \mathrm{Gen}_{\mathcal{E}}(T)$. Given that, there exist $X \in \mathrm{Gen}_{\mathcal{E}}(T)$ such that $X \notin \mathrm{add}(T)$. By Lemma 5.22, there exist an indecomposable $U \in \mathrm{add}(T)$ such that $U \in \mathrm{Sub}_{\mathcal{E}}(\widetilde{T})$ with $T = \widetilde{T} \oplus U$. By Proposition 5.3 and Lemma 5.5, U admits a left minimal $(\mathrm{add}(\widetilde{T}), \mathcal{E})$-approximation f. In the sense of Definition 5.7, T admits a left $\mathcal{E}$-mutation at U which is a contradiction. Hence, $\mathrm{Gen}_{\mathcal{E}}(T) = \mathrm{add}(T)$. ■

The following results highlight the description of $\mathrm{Gen}_{\mathcal{E}}(T)$ via the tilting representations that cover T, and the description of $\mathrm{Sub}_{\mathcal{E}}(T)$ via the tilting representations that T covers for $\lhd_{\mathcal{E}}$.

Proposition 5.25. *Let $T \in \mathrm{Tilt}(Q)$. Then the following equalities hold:*

(*i*) $\mathrm{Gen}_{\mathcal{E}}(T) = \mathrm{add}(T) \oplus \Big(\bigoplus_{T \lhd_{\mathcal{E}} T'} \mathrm{Gen}_{\mathcal{E}}(T') \Big)$; *and,*

(*ii*) $\mathrm{Sub}_{\mathcal{E}}(T) = \mathrm{add}(T) \oplus \Big(\bigoplus_{T' \lhd_{\mathcal{E}} T} \mathrm{Sub}_{\mathcal{E}}(T') \Big)$.

Proof. The assertions (*i*) and (*ii*) are dual statements. It suffices for us to show (*i*).

First, we will show that $\mathrm{add}(T) \oplus \Big(\bigoplus_{T \lhd_{\mathcal{E}} T'} \mathrm{Gen}_{\mathcal{E}}(T') \Big) \subseteq \mathrm{Gen}_{\mathcal{E}}(T)$.

Let $X \in \mathrm{add}(T) \oplus \Big(\bigoplus_{T \lhd_{\mathcal{E}} T'} \mathrm{Gen}_{\mathcal{E}}(T') \Big)$ be an indecomposable representation. If $X \in \mathrm{add}(T)$, then obviously $X \in \mathrm{Gen}_{\mathcal{E}}(T)$. Assume that $X \in \mathrm{Gen}_{\mathcal{E}}(T')$ for some $T \lhd_{\mathcal{E}} T'$. Set $U \in \mathrm{ind}(Q)$ such that $T \ncong \boldsymbol{\mu}^{-}_{U,\mathcal{E}}(T) = T'$. We have the following $\mathcal{E}$-admissible short exact sequence

$$\xi : 0 \longrightarrow U \overset{f}{\rightarrowtail} \Theta \overset{g}{\twoheadrightarrow} \mathrm{Coker}(f) \longrightarrow 0,$$

with f a left minimal $(\mathrm{add}(\widetilde{T}), \mathcal{E})$-approximation of U with $T = \widetilde{T} \oplus U$. Since $\Theta \in \mathrm{add}(\widetilde{T}) \subset \mathrm{add}(T)$, then $\mathrm{Coker}(f) \in \mathrm{Gen}_{\mathcal{E}}(T)$. Given that $T' = \widetilde{T} \oplus \mathrm{Coker}(f)$, then $\mathrm{Gen}_{\mathcal{E}}(T') \subseteq \mathrm{Gen}_{\mathcal{E}}(T)$, so $X \in \mathrm{Gen}_{\mathcal{E}}(T)$.

Now, we will show that $\mathrm{Gen}_{\mathcal{E}}(T) \subseteq \mathrm{add}(T) \oplus \left(\bigoplus_{T \lhd_{\mathcal{E}} T'} \mathrm{Gen}_{\mathcal{E}}(T')\right)$.

Let $X \in \mathrm{Gen}_{\mathcal{E}}(T)$ be an indecomposable representation. If $X \in \mathrm{add}(T)$, we are done. Otherwise, by Lemma 5.22, there is an indecomposable representation $U \in \mathrm{add}(T)$ such that $X \in \mathrm{Gen}_{\mathcal{E}}(\widetilde{T})$ and $U \in \mathrm{Sub}_{\mathcal{E}}(\widetilde{T})$ with $T = \widetilde{T} \oplus U$. By Proposition 5.3 and Lemma 5.5, U admits a left minimal $(\mathrm{add}(\widetilde{T}), \mathcal{E})$-approximation f. In the sense of Definition 5.7, T admits a left $\mathcal{E}$-mutation at U, and we get $\boldsymbol{\mu}^{-}_{U,\mathcal{E}}(T) = T'$ for some tilting object $T \lhd_{\mathcal{E}} T'$ with $\mathrm{add}(\widetilde{T}) \subseteq \mathrm{add}(T')$. Therefore, $X \in \mathrm{Gen}_{\mathcal{E}}(\widetilde{T}) \subseteq \mathrm{Gen}_{\mathcal{E}}(T')$ implies $X \in \mathrm{Gen}_{\mathcal{E}}(T')$ for some tilting object $T \lhd_{\mathcal{E}} T'$. Hence $\mathrm{Gen}_{\mathcal{E}}(T) \subseteq \mathrm{add}(T) \oplus \left(\bigoplus_{T \lhd_{\mathcal{E}} T'} \mathrm{Gen}_{\mathcal{E}}(T')\right)$. ■

Proposition 5.26. *Let $T \in \mathrm{Tilt}(Q)$. Then the following equalities hold:*

(1) $\mathrm{Gen}_{\mathcal{E}}(T) = \displaystyle\bigoplus_{T \unlhd_{\mathcal{E}} T'} \mathrm{add}(T')$*; and,*

(2) $\mathrm{Sub}_{\mathcal{E}}(T) = \displaystyle\bigoplus_{T' \unlhd_{\mathcal{E}} T} \mathrm{add}(T')$.

Proof. Since Q is representation-finite, the result follows directly from Propositions 5.24 and 5.25 by induction. ■

Before we show the proof of the last theorem, we will show how to choose some special tilting object $T' \in [T]_{\approx_{\mathcal{E}}}$ to generate $\mathrm{GS}_{\mathcal{E}}(T)$.

Lemma 5.27. *Let $T_1, T_2 \in \mathrm{Tilt}(Q)$. If $\mathrm{Gen}_{\mathcal{E}}(T_1) = \mathrm{Gen}_{\mathcal{E}}(T_2)$, then $T_1 \cong T_2$. Dually, it holds if $\mathrm{Sub}_{\mathcal{E}}(T_1) = \mathrm{Sub}_{\mathcal{E}}(T_2)$.*

Proof. Since $\mathrm{Gen}_{\mathcal{E}}(T_1) = \mathrm{Gen}_{\mathcal{E}}(T_2)$, we have the two $\mathcal{E}$-admissible short exact sequences:

$$\xi_1 : 0 \longrightarrow K \rightarrowtail \Theta_1 \twoheadrightarrow T_2 \longrightarrow 0,$$

with $\Theta_1 \in \mathrm{add}(T_1)$, and

$$\xi_2 : 0 \longrightarrow K' \rightarrowtail \Theta_2 \twoheadrightarrow T_1 \longrightarrow 0,$$

with $\Theta_2 \in \mathrm{add}(T_2)$. Applying the functors $\mathrm{Hom}(T_1, -)$ to ξ_1 and $\mathrm{Hom}(T_2, -)$ to ξ_2, because $\mathrm{Ext}^1(T_1, \Theta_2) = \mathrm{Ext}^1(T_2, \Theta_2) = 0$ and because $\mathrm{rep}\, Q$ is hereditary, we obtain $\mathrm{Ext}^1(T_1, T_2) = \mathrm{Ext}^1(T_2, T_1) = 0$. Since T_1, T_2 are maximally rigid objects, then $T_1 \cong T_2$. ■

Proposition 5.28. *Let $T \in \mathrm{Tilt}(Q)$. Then $[T]_{\approx_{\mathcal{E}}}$ has a unique minimal element T' for $\unlhd_{\mathcal{E}}$ and $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{Gen}_{\mathcal{E}}(T')$. Moreover, $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}^1_{\mathcal{E}}(T')$.*

Proof. It is clear that $[T]_{\approx_{\mathcal{E}}}$ has minimal elements for $\unlhd_{\mathcal{E}}$. We must only show that there is indeed a unique one. Let $T' \in [T]_{\approx_{\mathcal{E}}}$ be a minimal element. By Proposition 5.16, we already have $\mathrm{Gen}_{\mathcal{E}}(T') \subseteq \mathrm{GS}_{\mathcal{E}}(T') = \mathrm{GS}_{\mathcal{E}}(T)$.

Let us show that $\mathrm{GS}_{\mathcal{E}}(T') \subseteq \mathrm{Gen}_{\mathcal{E}}(T')$ by induction. First, we have that $\mathrm{GS}^0_{\mathcal{E}}(T') = \mathrm{add}(T') \subseteq \mathrm{Gen}_{\mathcal{E}}(T')$. Assume that, for a fixed $n \in \mathbb{N}$, we have $\mathrm{GS}^n_{\mathcal{E}}(T') \subseteq \mathrm{Gen}_{\mathcal{E}}(T')$. Consider $X \in \mathrm{GS}^{n+1}_{\mathcal{E}}(T')$. We have two cases to treat.

If $X \in \mathrm{Gen}_{\mathcal{E}}(F)$ for some $F \in \mathrm{GS}^n_{\mathcal{E}}(T')$, by induction hypothesis, $F \in \mathrm{Gen}_{\mathcal{E}}(T')$. As $\mathcal{E}$-epimorphisms are closed under composition, so is X.

Now assume that $X \in \mathrm{Sub}_{\mathcal{E}}(F)$ for some $F \in \mathrm{GS}^n_{\mathcal{E}}(T')$. Then we have the following short $\mathcal{E}$-exact sequence

$$0 \longrightarrow X \rightarrowtail F \twoheadrightarrow C \longrightarrow 0.$$

By induction hypothesis, $F \in \mathrm{Gen}_{\mathcal{E}}(T')$, and therefore, there exists a short $\mathcal{E}$-exact sequence

$$0 \longrightarrow K \rightarrowtail \Theta \twoheadrightarrow F \longrightarrow 0,$$

with $\Theta \in \mathrm{add}(T')$. We obtain the following commutative pull-back diagram.

$$\begin{tikzcd}
 & 0 & 0 & 0 & \\
0 \arrow[r] & 0 \arrow[r, tail] \arrow[u] & C' \arrow[r, two heads] \arrow[u] & C \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & K \arrow[r, tail] \arrow[u, two heads] & \Theta \arrow[r, two heads] \arrow[u, two heads] & F \arrow[r] \arrow[u, two heads] & 0 \\
0 \arrow[r] & K \arrow[r, tail] \arrow[u, equal] & PB \arrow[r, two heads] \arrow[u, tail] & X \arrow[r] \arrow[u, tail] & 0 \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

The short sequences given by the three columns and those given by the last two rows are exact. By Lemma 3.6, the short sequence given by the first row is also exact. Thus $C' \cong C$. Moreover, the short exact sequences given by the three rows and those given by the first and third columns are in $\mathcal{E}$. By Lemma 3.6, the short sequence given by the middle column is also $\mathcal{E}$-exact. So $PB \in \mathrm{Sub}_{\mathcal{E}}(\Theta)$.

Since $\Theta \in \mathrm{add}(T')$ and T' admits no right $\mathcal{E}$-mutation (as T' is minimal for $\trianglelefteq_{\mathcal{E}}$), then $\mathrm{Sub}_{\mathcal{E}}(\Theta) \subseteq \mathrm{Sub}_{\mathcal{E}}(T') = \mathrm{add}(T')$ by Proposition 5.24. Hence, $PB \in \mathrm{add}(T')$. Given that the last row is $\mathcal{E}$-exact, $X \in \mathrm{Gen}_{\mathcal{E}}(PB) \subseteq \mathrm{Gen}_{\mathcal{E}}(T')$.

Thus, this completes the induction proof and we get $\mathrm{GS}_{\mathcal{E}}(T') \subseteq \mathrm{Gen}_{\mathcal{E}}(T')$. Thus $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}(T') = \mathrm{Gen}_{\mathcal{E}}(T')$. The uniqueness directly follows from Lemma 5.27. Recall that $\mathrm{Gen}_{\mathcal{E}}(T') \subseteq \mathrm{GS}_{\mathcal{E}}(T)$ and $\mathrm{Sub}_{\mathcal{E}}(T') = \mathrm{add}(T') \subseteq \mathrm{GS}_{\mathcal{E}}(T)$ from Propositions 5.24 and 5.16. So $\mathrm{GS}^1_{\mathcal{E}}(T') \subseteq \mathrm{GS}_{\mathcal{E}}(T) = \mathrm{Gen}_{\mathcal{E}}(T') \subseteq \mathrm{GS}^1_{\mathcal{E}}(T')$. ■

By dual statements, we get the following corollary.

Corollary 5.29. *Let $T \in \mathrm{Tilt}(Q)$. Then $[T]_{\approx_{\mathcal{E}}}$ has a unique maximal element T'' of $[T]_{\approx_{\mathcal{E}}}$ for $\trianglelefteq_{\mathcal{E}}$ such that $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{Sub}_{\mathcal{E}}(T')$. Moreover, $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}^1_{\mathcal{E}}(T'')$.*

We can now provide proof of the results we aimed for.

Proposition 5.30. *Let $T, T' \in \mathrm{Tilt}(Q)$. Then $T \sim_{\mathcal{E}} T'$ if and only if $T \approx_{\mathcal{E}} T'$.*

Proof. By Proposition 5.16, we have that if $T' \approx_{\mathcal{E}} T$, then $T' \sim_{\mathcal{E}} T$. Consider $T, T' \in \mathrm{Tilt}(Q)$ such that $T' \sim_{\mathcal{E}} T$. By definition, $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}(T')$. Due to Proposition 5.28, there exist $S \in [T]_{\approx_{\mathcal{E}}}$ and $S' \in [T']_{\approx_{\mathcal{E}}}$ such that $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{Gen}_{\mathcal{E}}(S)$ and $\mathrm{GS}_{\mathcal{E}}(T') = \mathrm{Gen}_{\mathcal{E}}(S')$. We obtain $\mathrm{Gen}_{\mathcal{E}}(S) = \mathrm{Gen}_{\mathcal{E}}(S')$ and, by Lemma 5.27, $S \cong S'$. We conclude that $[T]_{\approx_{\mathcal{E}}} = [T']_{\approx_{\mathcal{E}}}$. ■

We can finally prove one of our main results.

Theorem 5.31. *Let $T \in \mathrm{Tilt}(Q)$. Then*

$$\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{add}\left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}}}} T'\right)$$

Proof. First of all,

$$\mathrm{add}\left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}}}} T'\right) \subseteq \mathrm{GS}_{\mathcal{E}}(T)$$

by Proposition 5.16. By Proposition 5.28, there exists (a unique) $S \in [T]_{\approx_{\mathcal{E}}}$ such that $\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{Gen}_{\mathcal{E}}(S)$. By Proposition 5.26, we can write

$$\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{Gen}_{\mathcal{E}}(S) \; = \bigoplus_{S \trianglelefteq_{\mathcal{E}} T'} \mathrm{add}(T') \subseteq \mathrm{add}\left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}}}} T'\right).$$

■

Example 5.32. The following figure illustrates an example of Theorem 5.31, as well as Corollary 5.15 and Proposition 5.16, for a specific equivalence class of tilting representations selected from those shown in Figure 6.

$$[T_1]_{\approx_{\mathcal{E}_\diamond}} = [T_2]_{\approx_{\mathcal{E}_\diamond}} = [T_3]_{\approx_{\mathcal{E}_\diamond}}$$

$$\mathrm{GS}_{\mathcal{E}_\diamond}(T_1) = \mathrm{add}\,(T_1 \oplus T_2 \oplus T_3)$$

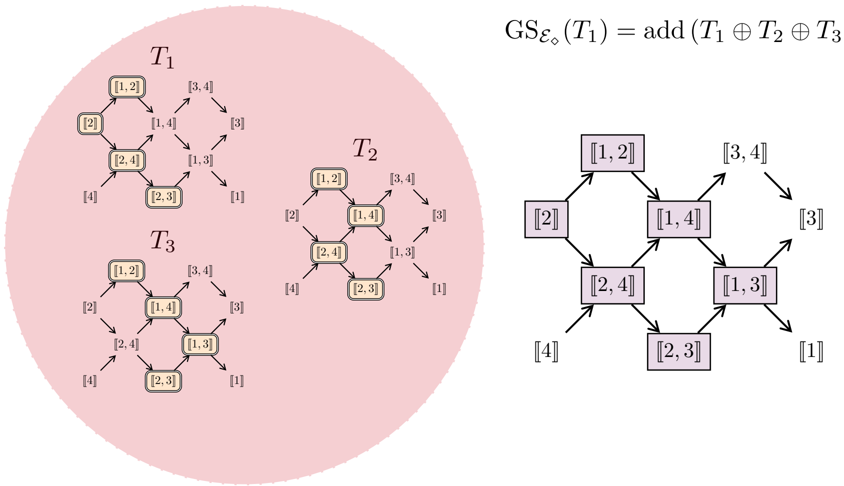

Figure 7. Illustration of Theorem 5.31 via one equivalence class in $\mathrm{Tilt}(Q)$ for $\approx_{\mathcal{E}_\diamond}$ and for Q the A_4 type quiver in Example 3.29.

5.3. **Canonically Jordan recoverable subcategories for type A algebras.** In the following, we prepare the ground for the proof of our main result, which will be an application of the results in Sections 3.5, 5.2 and 5.3. To do so, we show that any maximal canonically Jordan recoverable subcategory of $\mathrm{rep}(Q)$ is extension-closed and contains a tilting representation.

Proposition 5.33. *Let Q be an A_n type quiver for some $n \in \mathbb{N}^*$. Then all the maximal canonically Jordan recoverable subcategories of* $\mathrm{rep}(Q)$ *are extension-closed.*

Proof. Let $\mathscr{C} \subseteq \text{rep}(Q)$ be a maximal canonically Jordan recoverable subcategory. By Theorem 4.7, consider $\mathbf{B}, \mathbf{E} \subseteq \{1, \dots, n\}$ such that:

- the pair $(\mathbf{B}, \mathbf{E}[1])$ is a set partition of $\{1, \dots, n+1\}$; and,
- we have $\mathscr{C} = \boldsymbol{\mathscr{C}}(\mathbf{B}, \mathbf{E})$.

To check that $\mathscr{C}$ is extension closed, by the fact that Ext^1 is a bilinear functor, it is enough to prove that for any pair (D, F) of indecomposable representations in $\mathscr{C}$, and for any nonsplit short exact sequence

$$0 \longrightarrow D \rightarrowtail E \twoheadrightarrow F \longrightarrow 0,$$

we have $E \in \mathscr{C}$.

Given such a short exact sequence, by Theorems 2.7 and 4.10, we have that $E \cong E_1 \oplus E_2$ and there exist $K, L \in \mathcal{I}_n$ with $K \cap L \neq \varnothing$ such that:

- $K, L, K \cap L$ and $K \cup L$ are four different intervals in $\mathcal{I}_n$; and,
- either $\{D, F\} = \{X_K, X_L\}$ and $\{E_1, E_2\} = \{X_{K\cap L}, X_{K \cup L}\}$, or $\{D, F\} = \{X_{K\cap L}, X_{K\cup L}\}$ and $\{E_1, E_2\} = \{X_K, X_L\}$.

Set $K = [\![b_1, e_1]\!]$ and $L = [\![b_2, e_2]\!]$. Up to exchanging the role of K and L, assume that $e_1 \leqslant e_2$. To make sure that $K, L, K \cap L$ and $K \cup L$ are four distinct intervals nonempty in $\mathcal{I}_n$, we must have either

(1) $b_1 < b_2 \leqslant e_1 < e_2$; or,

(2) $b_2 < b_1 \leqslant e_1 < e_2$.

Assume that we are in the case (1). If $\{D, F\} = \{X_K, X_L\}$, then $b_1, b_2 \in \mathbf{B}$ and $e_1, e_2 \in \mathbf{E}$. Therefore, $K \cap L = [\![b_2; e_1]\!] \in \text{Int}(\mathscr{C})$ and $K \cup L = [\![b_1, e_2]\!] \in \text{Int}(\mathscr{C})$. So $E \in \mathscr{C}$. The case where $\{D, F\} = \{X_{K\cap L}, X_{K\cup L}\}$ can be treated similarly. These arguments yield the same result if we start with case (2).

In either case, we get $E \in \mathscr{C}$, and so the desired result. ∎

Remark 5.34. In contrast to Example 3.35, every maximal $\mathcal{E}_\diamond$-adapted subcategory is extension-closed. This is a direct consequence of Proposition 5.33 and Theorem 4.10.

Proposition 5.35. *Let Q be an A_n type quiver. Let $T \in \text{rep}(Q)$ be a rigid representation. Then $\text{add}(T)$ is canonically Jordan recoverable.*

Proof. It follows from Theorem 4.10. ∎

Proposition 5.36. *Let Q be an A_n type quiver. Consider $\mathscr{C} \subset \text{rep}(Q)$ a maximal canonically Jordan recoverable subcategory. Then $\mathscr{C}$ contains a tilting representation.*

Proof. By Theorem 4.7, consider $\mathbf{B}, \mathbf{E} \subseteq \{1, \dots, n\}$ such that:

- the pair $(\mathbf{B}, \mathbf{E}[1])$ is a set partition of $\{1, \dots, n+1\}$; and,
- we have $\mathscr{C} = \boldsymbol{\mathscr{C}}(\mathbf{B}, \mathbf{E})$.

Let $\boldsymbol{\mathscr{T}}(\mathbf{B}, \mathbf{E}) \subseteq \boldsymbol{\mathscr{J}}(\mathbf{B}, \mathbf{E}) = \text{Int}(\mathscr{C})$ be the set such that:

- for any $e \in \mathbf{E}$, $[\![1, e]\!] \in \boldsymbol{\mathscr{T}}(\mathbf{B}, \mathbf{E})$; and,
- for any $b \in \mathbf{B} \setminus \{1\}$,
  - if, for any $e \in E$, the interval $[\![b, e]\!]$ is either on the top or at the bottom of $[\![1, n]\!]$, then $[\![b, n]\!] \in \boldsymbol{\mathscr{T}}(\mathbf{B}, \mathbf{E})$;

- otherwise, by setting $e_{(b)} \in \mathbf{E}$ the minimal element in $\mathbf{E}$ such that $[\![b, e_{(b)}]\!]$ is neither on the top and at the bottom of $[\![1, n]\!]$, then $[\![b, e_{(b)}]\!] \in \mathscr{T}(\mathbf{B}, \mathbf{E})$.

Set

$$T = T_{(\mathbf{B},\mathbf{E})} = \bigoplus_{K \in \mathscr{T}(\mathbf{B},\mathbf{E})} X_K.$$

Let us show that T is tilting.

First $\mathrm{rep}(Q)$ is a hereditary category. Therefore, to show that T is a tilting representation, it suffices to show that:

- $\#\mathscr{T}(\mathbf{B}, \mathbf{E}) = n$; and,
- $\mathrm{Ext}^1(X_K, X_L) = 0$ for all $K, L \in \mathscr{T}(\mathbf{B}, \mathbf{E})$.

The first point is obvious, as one can check that

$$\#\mathscr{T}(\mathbf{B}, \mathbf{E}) = \#\mathbf{E} + \#\mathbf{B} - 1 = n.$$

By Theorem 2.7, the short exact sequences

$$0 \longrightarrow D \rightarrowtail E \twoheadrightarrow F \longrightarrow 0,$$

where $D, F \in \mathrm{ind}(Q)$, are parametrized by only two kind of pair of intervals $(K, L) \in \mathcal{I}_n$, namely:

(1) the case where $K \cap L = \varnothing$ and $K \cup L \in \mathcal{I}_n$; or,

(2) the case where $K \cap L \neq \varnothing$, and $K, L, K \cap L$ and $K \cup L$ are four distinct intervals.

The case (1) is equivalent to say that K and L are adjacent, and as $\mathscr{T}(\mathbf{B}, \mathbf{E}) \subset \mathscr{J}(\mathbf{B}, \mathbf{E})$ which is adjacency-avoinding, this cannot happen, with $D, F \in \mathrm{add}(T)$ two indecomposable representations.

Assume the case (2) holds with $D, F \in \mathrm{add}(T)$ two indecomposable representations. To exchange the role of K and L, assume that $e(K) < e(L)$. Then $b(K) < b(L) \leqslant e(K) < e(L)$. By construction of $\mathscr{T}(\mathbf{B}, \mathbf{E})$, both of the following assertions holds:

- $b(L) \neq 1$;
- if $e(L) \neq n$, then L is neither on the top nor at the bottom of $[\![1, n]\!]$; and,
- $K \cap L \notin \mathscr{T}(\mathbf{B}, \mathbf{E})$; and,
- $K \cap L$ is either on the top or at the bottom of $[\![1, n]\!]$, and, therefore, of $K \cup L$.

So, in such case, we have $\{D, F\} = \{X_K, X_L\}$. However, by the last point, we cannot have a non-zero morphism between X_K and X_L which factors through $X_{K \cap L} \oplus X_{K \cup L}$. We sought a contradiction.

Therefore, any short exact sequence

$$0 \longrightarrow D \rightarrowtail E \twoheadrightarrow F \longrightarrow 0,$$

where $D, F \in \mathrm{add}(T)$ are indecomposable, must split, and so $\mathrm{Ext}^1(F, D) = 0$. We proved that $T \in \mathscr{C}$ is tilting. ■

Example 5.37. We continue with Example 4.8, and we exhibit a tilting representation in this maximal canonically Jordan recoverable following the procedure given in the proof of Proposition 5.36.

$$Q = 1 \longrightarrow 2 \longleftarrow 3 \longrightarrow 4 \longrightarrow 5 \longleftarrow 6 \longrightarrow 7$$

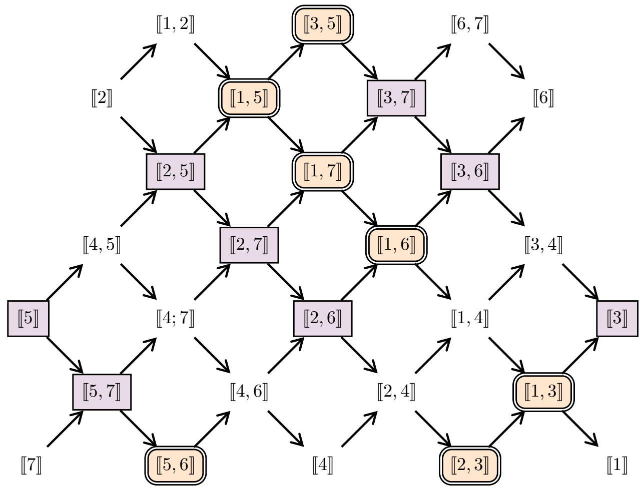

FIGURE 8. Example of a tilting representation $T = \oplus$ in the maximal canonically Jordan recoverable subcategory given in Figure 5.

Proposition 5.38. *Let $n \in \mathbb{N}^*$ and Q be an A_n type quiver. Consider $T \in \operatorname{rep}(Q)$ a rigid representation. Then $\mathrm{GS}_{\mathcal{E}_\diamond}(T)$ is canonically Jordan recoverable.*

Proof. It follows from Proposition 3.22 and Theorem 4.10. ∎

We can finally prove our main results.

Theorem 5.39. *Let Q be an A_n type quiver for some $n \in \mathbb{N}^*$. A subcategory $\mathscr{C} \subset \operatorname{rep}(Q)$ is a maximal canonically Jordan recoverable subcategory if, and only if, there exists $T \in \operatorname{Tilt}(Q)$ such that $\mathscr{C} = \mathrm{GS}_{\mathcal{E}_\diamond}(T)$.*

Proof. Let $\mathscr{C}$ be a maximal canonically Jordan recoverable subcategory of $\operatorname{rep}(Q)$. By Proposition 5.36, there exists a tilting representation $T \in \operatorname{rep}(Q)$ such that $T \in \mathscr{C}$. By Theorem 4.10, the category $\mathscr{C}$ is $\mathcal{E}_\diamond$-adapted. Moreover, $\mathscr{C}$ is closed under extensions by Proposition 5.33. Therefore, we have $\mathrm{GS}_{\mathcal{E}_\diamond}(T) \subseteq \mathscr{C}$. By Theorem 3.33, we get $\mathrm{GS}_{\mathcal{E}_\diamond}(T) = \mathscr{C}$. The converse is obvious by Theorem 3.33 and Theorem 4.10. ∎

Corollary 5.40. *Let Q be an A_n type quiver for some $n \in \mathbb{N}^*$. Let $\mathscr{C} \subseteq \operatorname{rep}(Q)$ be a maximal canonically Jordan recoverable subcategory. Then there exists $T \in \operatorname{Tilt}(Q)$ such that*

$$\mathscr{C} = \operatorname{add}\left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}_\diamond}}} T' \right).$$

Proof. It follows directly from Theorems 5.31 and 5.39. ∎

6. To go further

This section provides an overview of a few directions we are interested in investigating in the near future. The reader is invited to have a look at these various or related problems.

6.1. **Cokernel-Kernel operators.** This section introduces a closure operator on subcategories of $(\mathscr{A}, \mathcal{E})$ that is slightly different from the Gen-Sub operator. More precisely, given a subcategory $\mathscr{C}$, we will construct the smallest subcategory $\mathscr{D} \supseteq \mathscr{C}$ such that:

- $\mathscr{D}$ is closed under cokernels of $\mathcal{E}$-monomorphisms in $\mathscr{D}$; and,
- $\mathscr{D}$ is closed under kernels of $\mathcal{E}$-epimorphisms in $\mathscr{D}$.

The $\mathrm{Coker}_{\mathcal{E}}$***-operator*** on subcategories of $\mathscr{A}$ is defined as follows: given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define $\mathrm{Coker}_{\mathcal{E}}(\mathscr{C})$ as the full subcategory, closed under sums and summands, of $\mathscr{A}$ containing the cokernel of any $\mathcal{E}$-monomorphism $f : X \longrightarrow Y$ such that $X, Y \in \mathscr{C}$. The $\mathrm{Ker}_{\mathcal{E}}$***-operator*** on subcategories of $\mathscr{A}$ is defined analogously: given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define $\mathrm{Ker}_{\mathcal{E}}(\mathscr{C})$ as the full subcategory, closed by sums and summands, of $\mathscr{A}$ containing the kernel of any $\mathcal{E}$-epimorphism $g : Y \longrightarrow X$ such that $Y, X \in \mathscr{C}$. Given an object $M \in \mathscr{C}$, by abusing of notation, we set $\mathrm{Coker}_{\mathcal{E}}(M) = \mathrm{Coker}_{\mathcal{E}}(\mathrm{add}(M))$ and $\mathrm{Ker}_{\mathcal{E}}(M) = \mathrm{Ker}_{\mathcal{E}}(\mathrm{add}(M))$.

Given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define a sequence of subcategories $(\mathrm{CK}^i_{\mathcal{E}}(\mathscr{C}))_{i \in \mathbb{N}}$ as it follows:

- we set $\mathrm{CK}^0_{\mathcal{E}}(\mathscr{C}) = \mathscr{C}$; and,
- for all $i \geqslant 1$, we define $\mathrm{CK}^i_{\mathcal{E}}(\mathscr{C})$ to be the subcategory of $\mathscr{A}$ additively generated by objects in both $\mathrm{Coker}_{\mathcal{E}}(\mathrm{CK}^{i-1}_{\mathcal{E}}(\mathscr{C}))$ and $\mathrm{Ker}_{\mathcal{E}}(\mathrm{CK}^{i-1}_{\mathcal{E}}(\mathscr{C}))$.

By the previous remark, $(\mathrm{CK}^i_{\mathcal{E}}(\mathscr{C}))_{i \in \mathbb{N}}$ is an increasing sequence of additive subcategories of $\mathscr{A}$. Therefore, as $\mathscr{A}$ is abelian, this sequence of subcategories admits a colimit.

Definition 6.1. The $\mathrm{CK}_{\mathcal{E}}$***-operator*** on subcategories of $\mathscr{A}$ is defined as follows: given a subcategory $\mathscr{C} \subseteq \mathscr{A}$, we define $\mathrm{CK}_{\mathcal{E}}(\mathscr{C})$ as the colimit of the sequence of subcategories $(\mathrm{CK}^i_{\mathcal{E}}(\mathscr{C}))_{i \in \mathbb{N}}$. As before, we set $\mathrm{CK}_{\mathcal{E}}(M) = \mathrm{CK}_{\mathcal{E}}(\mathrm{add}(M))$ by abuse of notations.

Remark 6.2. It is clear that $\mathrm{CK}_{\mathcal{E}}(\mathscr{C}) \subseteq \mathrm{GS}_{\mathcal{E}}(\mathscr{C})$. Therefore, by Proposition 3.22, $\mathrm{CK}_{\mathcal{E}}(T)$ is $\mathcal{E}$-adapted for any rigid object T.

Example 6.3. In Figure 9, we calculate $\mathrm{CK}_{\mathcal{E}_\diamond}(\mathscr{C})$ for a subcategory $\mathscr{C}$ of $\mathrm{rep}(Q)$ and a fixed quiver Q of A_7 type. This example illustrates the slight difference between $\mathrm{CK}_{\mathcal{E}_\diamond}(\mathscr{C})$ and $\mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C})$.

For the same subcategory $\mathscr{C}$, $\mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C})$ still corresponds to the subcategory shown in Figure 1. By adding the object $[\![1, 3]\!]$ to $\mathscr{C}$, thereby making it maximally rigid, we obtain $\mathrm{GS}_{\mathcal{E}_\diamond}(\mathscr{C}) = \mathrm{CK}_{\mathcal{E}_\diamond}(\mathscr{C})$. We observe that, in general, a minimal basic object T for which $\mathrm{GS}_{\mathcal{E}}(T)$ is a maximally $\mathcal{E}$-adapted extension-closed subcategory does not necessarily have to be tilting. This suggests that studying the $\mathrm{CK}_{\mathcal{E}}$-operator could be valuable.

Remark 6.4. By minimal, we mean that no direct summand can be removed from T basic without losing the property that $\mathrm{GS}_{\mathcal{E}}(T)$ is a maximally $\mathcal{E}$-adapted extension-closed subcategory.

$$Q = \ 1 \longrightarrow 2 \longleftarrow 3 \longrightarrow 4 \longrightarrow 5 \longleftarrow 6 \longrightarrow 7$$

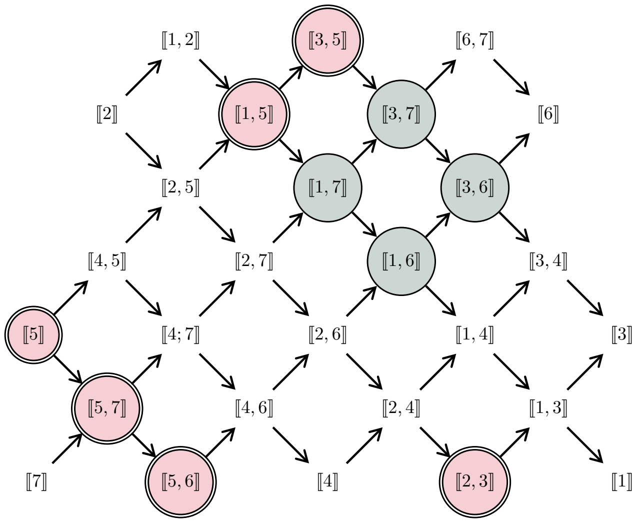

FIGURE 9. Calculation of $\mathrm{CK}_{\mathcal{E}_\diamond}(\mathscr{C})$ where $\mathscr{C} = \mathrm{add}\,(\ )$. We get $\mathrm{CK}^1_{\mathcal{E}_\diamond}(\mathscr{C}) = \mathrm{add}\,(\ + \ )$. Here, we obtain that $\mathrm{CK}_{\mathcal{E}_\diamond}(\mathscr{C}) = \mathrm{CK}^1_{\mathcal{E}_\diamond}(\mathscr{C})$.

Definition 6.5. We recall that a subcategory $\mathscr{D}$ is called ***$\mathcal{E}$-wide*** if, for any short $\mathcal{E}$-exact sequence

$$\xi : \quad 0 \longrightarrow E \rightarrowtail F \twoheadrightarrow G \longrightarrow 0,$$

whenever two of the three objects E, F, or G belong to $\mathscr{D}$, then so does the third.

Proposition 6.6. *Let $(\mathscr{A}, \mathcal{E})$ be an exact category with $\mathscr{A}$ hereditary. Then the following statements hold:*

(a) *if $T \in \mathscr{A}$ is rigid, then $\mathrm{CK}_{\mathcal{E}}(T)$ is an $\mathcal{E}$-adapted and $\mathcal{E}$-wide subcategory;*
(b) *if $T \in \mathscr{A}$ is tilting, then $\mathrm{CK}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}(T)$.*

Proof. The proof of (a) follows as in Proposition 3.24, noting that the subcategory is no longer necessarily closed under arbitrary $\mathcal{E}$-subobjects and $\mathcal{E}$-quotients, but it remains closed under taking kernels of $\mathcal{E}$-epimorphisms and cokernels of $\mathcal{E}$-monomorphisms within itself.

The proof of Theorem 3.33 can be adapted to the setting of $\mathrm{CK}_{\mathcal{E}}(T)$ for showing (b). Together with the result from (a), this implies that both $\mathrm{CK}_{\mathcal{E}}(T)$ and $\mathrm{GS}_{\mathcal{E}}(T)$ are maximal $\mathcal{E}$-adapted extension-closed subcategories with $\mathrm{CK}_{\mathcal{E}}(T) \subseteq \mathrm{GS}_{\mathcal{E}}(T)$, which yields the desired equality. ∎

Lemma 6.7. *Let $(\mathscr{A}, \mathcal{E})$ be an exact category with $\mathscr{A}$ hereditary. Then, $T \in \mathscr{A}$ is minimal such that $\mathrm{CK}_{\mathcal{E}}(T)$ forms a maximally $\mathcal{E}$-adapted extension-closed subcategory if and only if T is tilting.*

6.2. **Congruences on the tilting lattice.** The equivalence relation $\approx_{\mathcal{E}}$ seems to hide other interesting properties in the lattice $(\mathrm{Tilt}(Q), \trianglelefteq)$. Considering the equivalence relation $\approx_{\mathcal{E}}$, we can ask ourselves if it defines well-behaved lattice quotients, usually called *lattice congruences.*

Definition 6.8. Let $(L, \leqslant, \vee, \wedge)$. A ***congruence*** of L is an equivalence relation $\sim$ on L such that, for any $a, b, x, y \in L$, if $a \sim x$ and $b \sim y$, then both $a \vee b \sim x \vee y$ and $a \wedge b \sim x \wedge y$. The ***quotient lattice*** of L on $\sim$ is the lattice $(L/\sim, \leqslant, \vee, \wedge)$ induced by L.

N. Reading gives a criterion on equivalence relations that are congruences on a lattice without using join and meet operations [Rea04].

For any $n \in \mathbb{N}$, denote by $\mathcal{B}_n$ the Boolean lattice of order n. In Figure 6, we can check that $\approx_{\mathcal{E}_\diamond}$ defines a congruence on $(\mathrm{Tilt}(Q), \trianglelefteq)$, and the lattice congruence obtained is isomorphic to the boolean lattice $\mathcal{B}_3$. In light of this example, we made a stronger conjecture that encapsulates the general behavior of $\approx_{\mathcal{E}_\diamond}$.

Lemma 6.9. *Let Q be an A_n type quiver for some $n \in \mathbb{N}^*$. The equivalence relation $\approx_{\mathcal{E}_\diamond}$ defines a congruence on $(\mathrm{Tilt}(Q), \trianglelefteq)$, and the obtained lattice congruence quotient is isomorphic to the Boolean lattice $\mathcal{B}_{n-1}$*

Example 6.10. In Figure 6, the equivalence relation $\approx_{\mathcal{E}_\diamond}$ gives a lattice congruence, given by identifying gathered tilting representations, and the quotient is isomorphic to $\mathcal{B}_3$.

A rough plan to prove this conjecture is as follows. First, we show that $\approx_{\mathcal{E}_\diamond}$ defines a congruence on $(\mathrm{Tilt}(Q), \trianglelefteq)$ by using Reading's criterion. Then, we can easily count the number of equivalence classes of $\approx_{\mathcal{E}_\diamond}$ in $\mathrm{Tilt}(Q)$.

Lemma 6.11. *Let Q be an A_n type quiver for some $n \in \mathbb{N}^*$. There are exactly 2^{n-1} equivalence classes for $\approx_{\mathcal{E}_\diamond}$ in $\mathrm{Tilt}(Q)$.*

Proof. By Theorems 4.7 and 5.39, those equivalence classes are parametrized by pairs $(\mathbf{B}, \mathbf{E})$ such that $\{\mathbf{B}, \mathbf{E}[1]\}$ is a bipartition of $\{1, \dots, n+1\}$. Then there is a one-to-one correspondence ϕ from subsets of $\{2, \dots, n\}$ to such pairs given by

$$\forall \mathbf{A} \subseteq \{2, \dots, n\},\ \phi(\mathbf{A}) = (\{1\} \cup \mathbf{A}, \{2, \dots, n+1\}) \setminus \mathbf{A}.$$

We got the desired result. ■

Finally, we can construct the lattice isomorphism explicitly. Note that the order of subsets of $\{2, \dots, n\}$ induced by $(\mathrm{Tilt}(Q)/\approx_{\mathcal{E}_\diamond}, \trianglelefteq)$ does not coincide with the usual inclusion order on those subsets.

Remark 6.12. If Q is a linearly-oriented A_n type quiver, then $(\mathrm{Tilt}(Q), \trianglelefteq)$ is isomorphic to the classical order on *binary trees* with n internal nodes, and $\approx_{\mathcal{E}_\diamond}$ identifies trees by their *canopy* [Vie09]. In this special case, the conjecture was proved by J.-L. Loday and M. O. Racia [LR98, Lod04].

We aim to prove this conjecture soon. We can also try to characterize exact structures $\mathcal{E}$ on a type ADE quiver Q such that $\approx_{\mathcal{E}}$ defines a congruence on $\mathrm{Tilt}(Q)$. It could provide a deeper understanding of a subfamily of well-known lattice congruences within the Cambrian lattice.

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